QUALITATIVE-BEHAVIOUR CONTROL WITH UNCONTROLLABLE LINEARISATION: BRANCHING FEEDBACK AND THE GEOMETRY OF ITS SINGULARITY LOCUS

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Dedicated, with deep gratitude, to the memory of Vladimir Arnold, Roger Brockett, Mark Krasnosel'skii, Alexander Krasnosel'skii, Aleksei Pokrovskii, Aleksei Kleptsyn and Nikolai Bobylev

Abstract. We study feedback synthesis at a working point where the linearisation of the plant is *uncontrollable*. We recast the qualitative-behaviour control machinery of [1–4] in the language of geometric control. The organising object is the integral manifold M of the plant–reference product system; solvability of the *qualitative-behaviour control (QBC) problem*, choosing feedback so that the closed loop reproduces the dynamics of a prescribed reference system, is equivalent to M being the graph $x = F(z)$ of a diffeomorphism F, and uncontrollability of the linearisation forces F to be singular whenever the spectrum of the reference does not contain some uncontrollable eigenvalue of the plant. When the inverse of this projection is multivalued, as in our examples, the feedback obtained by inverting it is multivalued, ramifying along the critical value set $\pi(\Sigma)$ of that projection. This set depends on the chosen coupling, and the controllability degeneracy forces it to contain the working point. We give the geometry of Σ (Section 2.5), apply the construction to the nonholonomic (Brockett) integrator, where a weighted blow-up resolves the plant into a family of surfaces on which the synthesis has a half-attraction time (the time to halve the size of the state) that does not depend on the initial point, and where the ramification over the excluded axis is a circle of directions, and where rotating the branch chosen on the axis turns the synthesis into a continuous time-periodic stabiliser of Coron's type (Section 4), and use the branching feedback to synthesise a periodic motion of *prescribed frequency* when the critical mode $\pm i\omega_0$ is uncontrollable (Section 5). No smooth feedback can change the frequency ω_0 of this mode. A branching feedback can: for non-degenerate quadratic control terms it produces an attracting cycle with a prescribed frequency from an explicit range, apart from a few resonances with ω_0. Note that a similar branching feedback can also make the working point asymptotically stable for a plant whose linearisation there is uncontrollable and unstable, when its quadratic terms satisfy an explicit sign condition (Section 3).

1. Introduction

Let a plant $\dot{x} = v(x, u)$ be given, with state x in the *state space* X and control u in the *control space* $\mathcal{U}$; near the working point $x = 0$, $u = 0$ it reads

$$\dot{x} = Ax + Cu + b(x, u), \qquad x \in \mathbb{R}^N,\ u \in \mathbb{R}^\ell, \quad b = o(x, u). \tag{1}$$

Let also a *reference system* $\dot{z} = w(z)$ be given on the *reference state space* Z; near $z = 0$ it reads

$$\dot{z} = Dz + d(z), \qquad z \in \mathbb{R}^N, \quad d = o(z). \tag{2}$$

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We wish the closed loop to reproduce the qualitative behaviour of the reference. The *qualitative-behaviour control (QBC) problem* asks for a feedback $u = s(x)$ making the closed loop C^k-conjugate to (2). When (A, C) is controllable, the linear problem is classical: it is solved by pole placement or by Rosenbrock's structural theorem, and the nonlinear problem reduces to the linear one [1, Thm. 2.5]. The paper shows how to deal with this problem, especially in the *linearly uncontrollable* case — (A, C) uncontrollable.

The paper [1]. The method comes from the author's paper [1] of 1990. It has two parts: control (§2–§5) and observation (§6). It is cited almost only for the second part: French researchers name it, together with later works, as the starting point of the Kazantzis–Kravaris/Luenberger (KKL) observer theory [5]; see also [6] (for a survey of that theory see [7]). The control part, which treats the QBC problem, is practically unknown, although the QBC problem contains stabilisation, the synthesis of bifurcations and the control of frequency as particular cases. The present preprint restates the control part in the language of geometric control and extends it.

How to read this paper. The main text, Sections 1–6 (pp. 1–29), states the results; proofs of a few lines stay in it, and the longer proofs are only outlined there. The extended proofs are in the appendices (pp. 29–46), and each appendix opens with an outline of its proof. The references are on pp. 46–49. A short version of this paper, with the results and their mechanism but without the proofs, is the preprint [8].

Where linear controllability is lost. Engineering design prefers plants with controllable linearisation, but when a plant depends on parameters, the loss of linear controllability is inevitable as soon as the family of plants is transversal to a surface of degenerate linear controllability in the space of systems with control [1, §4.11]. Several cases in the literature show this situation: a quadrocopter that has lost propellers, a spacecraft with two control torques, a surface vessel without lateral thrust, a horizontal arm with a passive joint, the unicycle. For each, the practical task has been solved, and always by abandoning smooth static feedback. The quadrocopter hovers by spinning about its thrust axis [9, 10]; the others are stabilised by laws that switch or depend on time [11–15]. Loss of controllability of the linearisation also happens in practice in the areas of fluid control [16], digital (sampled-data) control [17], turbomachinery [18], robotics [19], networks [20], quantum control [21], chemistry [22] and biology [23]. A nonlinear feedback generically stabilises a Hopf bifurcation even when the critical modes of the linearisation are uncontrollable [24, 25]; such feedback changes the criticality, stability and amplitude of the cycle, but its onset frequency stays ω_0. Asymptotic stabilisation of a position, asymptotic stabilisation of a trajectory and the synthesis of a cycle of prescribed frequency are instances of one problem, the QBC problem with different references: a trajectory is stabilised by stabilising the zero of the error between state and trajectory, a system that may depend on time. In all the cases above this problem is of practical importance. The area of applications becomes even wider when one considers the stabilisation of a desired trajectory (the author started his research in this area from stable following of a glide path in landing). This paper gives a method, branching feedback, which generalises regular static feedback and produces the required control (Sections 3–4) even when no regular feedback solving the problem exists, and also considers more general equilibrium points for asymptotic stabilisation (Section 3) and a case which, as far as we know, has not been addressed in the literature: a cycle of prescribed frequency (Section 5).

The obstruction, and what is known about it. That no continuous static feedback can stabilise certain systems with uncontrollable linearisation is well established. Brockett's necessary condition [26] is the topological source; it was shown by Ryan [27] to remain necessary when the feedback is discontinuous and the solutions are understood in Filippov's sense, so researchers turned to other kinds of feedback: sampled proximal-aiming feedback [28], smooth time-varying

feedback and the return method [29], patchy and hybrid constructions [30]. Recent accounts of these topological obstructions, including their extension from points to sets and to safety constraints, are [31,32]. Each of these constructions establishes the obstruction to feedback stabilising the plant and then *constructs a substitute*. For the Brockett integrator, Zeng [33] locates the discontinuity of a time-invariant law on the x_3-axis and observes the branching there, but does not describe the axis as the critical value set of a projection, which is how it appears in this paper. In parallel, the control-bifurcation programme [34,35] classifies systems with uncontrollable linearisation by smooth normal forms and analyses the low-codimension bifurcations (fold, transcritical, Hopf); it proves — for the fold control bifurcation — that no continuous static feedback *stabilises* the equilibria on both sides of the set in state–control space where controllability is lost, and then passes to smooth controlled-centre dynamics; the geometry of the feedback itself is not its subject.

What this paper adds. The machinery of [1] for the QBC problem is as follows. Its central object is not the plant alone but the integral manifold M of the *plant–reference product system* $\dot{x} = v\big(x, h(x,z)\big)$, $\dot{z} = w(z)$, where $u = h(x,z)$ is a *coupling* of plant and reference $\dot{z} = w(z)$; the problem is solvable iff M is the graph of a diffeomorphism F from reference space to plant space (the Theorem of §2.3 below). *Uncontrollability forces F to be singular at the working point* whenever the reference spectrum does not contain some uncontrollable eigenvalue (the Proposition of §2.4 below), so F is not a diffeomorphism. (We usually assume that F is finite, i.e. every state has only finitely many preimages under F (see, however, Section 4)). Then the QBC problem has no C^1 solution. When F^{-1} is multivalued, as at a fold, a ramification, z^n or the Brockett circle, the feedback $s(x) = h\big(x, F^{-1}(x)\big)$ has one value for each branch of F^{-1} (values on different branches may coincide), and its *ramification locus* (or, in the usual terminology, *caustic*) — where it loses smoothness — is the critical value set $\pi(\Sigma)$, the image of the critical set

$$\Sigma \;=\; \{\, z : \det \partial_z F(z) = 0 \,\}$$

of the projection $(x,z) \mapsto x$ restricted to M — its *bifurcation set* in the sense of [36], of which the fold is only the generic codimension-one stratum (cusps and deeper singularities occur in general). The locus depends on the chosen reference and coupling; what the controllability degeneracy forces is that it contains the working point. Multivaluedness from a singular projection is familiar from optimal-control synthesis, where the projection of a Lagrangian manifold is singular along its caustic; here it arises from the stabilisation obstruction itself, with no cost functional. The resulting feedback is realised as a *branching closed/open-loop control* (Section 2.5).

We then put the machinery to work. Section 4 applies it to the nonholonomic (Brockett) integrator. A weighted blow-up resolves the plant into a family of surfaces indexed by a cylinder, and a single reference is carried on the blow-up itself, with the graph of the blow-down as the coupling manifold: the control is bounded by the size of the state and the half-attraction time (the time to halve the size of the state) does not depend on the initial point. The ramification locus is the axis and carries a circle of branches. In the second approach we keep the surfaces of the family invariant instead, with the coupling the blow-up suggests. This makes the attraction non-uniform; prefixing an initial phase restores uniformity, and then a constant frequency, fixed by the designer, survives the motion. Finally, in the third approach, rotating the branch used on the axis, in opposite directions on its two halves, and merging the branches into one value near the axis gives a continuous time-periodic stabiliser of Coron's type with uniform exponential convergence (Section 4.4). Section 5 is another example of our main result, the method of qualitative-behaviour branching control: a synthesis of periodic motion of *prescribed frequency* whose critical mode is an uncontrollable imaginary pair $\pm i\omega_0$. Smooth feedback leaves $\pm i\omega_0$ in the closed-loop linearisation, so the onset frequency of a Hopf bifurcation from that pair is fixed

by the plant. We give conditions on the quadratic part of the control under which a branching feedback is constructed that installs an attracting cycle on which $x_<$ moves at any prescribed angular velocity $2\omega \neq \omega_0$, or -2ω, by transporting a reference cycle of frequency ω through the projection $z \mapsto z^2$ of the coupling manifold.

Main results. The results are stated in detail in Sections 2–5.

(1) *The singularity is forced* (Proposition of §2.4 below). If the linearisation (A, C) is uncontrollable and the spectrum of the reference does not contain some uncontrollable eigenvalue of A, then the map F of every coupling manifold is singular at the working point. Then the QBC problem has no C^1 solution. When F^{-1} is multivalued, as at a fold, a ramification, z^n or the Brockett circle, the feedback obtained by inverting F has one value for each branch of F^{-1}, and its branches meet over the ramification locus $\pi(\Sigma)$ (§2.5 below).

(2) *Stabilisation* (Section 3). For the corank-one normal form the sign of one number Δ, computed from the linear and quadratic coefficients, decides: for $\Delta > 0$ no bounded control stabilises the working point, and for $\Delta < 0$ a branching feedback does. The model $\dot{x} = u^n$, $x, u \in \mathbb{C}$, has no continuous stabiliser, but an explicit n-branch one; under small deformations its branch point splits into at most $n-1$ nearby branch points, and the same reference and coupling serve the whole family of deformations with a control bounded uniformly, where every smooth stabiliser needs gains that grow without bound.

(3) *The Brockett integrator* (Section 4). A branching feedback in closed form with $|u| \leq c\nu$, $\nu = \left((x_1^2 + x_2^2)^2 + x_3^2\right)^{1/4}$ the size of the state, and with a time after which ν has halved that does not depend on the initial point. Over each point of the x_3-axis the branches form a circle. Rotating the branch used on the axis, in opposite directions on its two halves, gives a continuous time-periodic stabiliser with uniform exponential convergence (Theorem of §4.4 below).

(4) *Prescribed frequency* (Section 5). Let the critical pair $\pm i\omega_0$ be uncontrollable and let the control enter it through $Q(u) = Gu^2 + H|u|^2 + K\bar{u}^2$. Smooth feedback cannot change the frequency ω_0. The position of the complex number H relative to the ellipse $E(G,K) = \{\vartheta \in \mathbb{C} : \ |\vartheta|^2 + |\vartheta^2 - 4GK| = 2(|G|^2 + |K|^2)\}$ (see (41) in §5.2 below) decides what a branching feedback can do; H is inside $E(G,K)$ when $|H|^2 + |H^2 - 4GK| < 2(|G|^2 + |K|^2)$. If H is strictly inside the ellipse, every angular velocity $\pm 2\omega$ of $x_<$ is realised on an attracting cycle, except $2\omega = \omega_0$. First consider $|G| \neq |K|$. If H is on the ellipse, every angular velocity 2ω is realised, except at the resonances $\omega/\omega_0 \in \{\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, 1, 2\}$. If H is outside the ellipse, the same holds for all ω with $\omega/\omega_0 < U$, where $U \geq 1/\sqrt{2}$ is given by an explicit formula in G, H, K (45), and $U \sim \operatorname{dist}(H, E(G,K))^{-1/2}$, so U grows without bound as H approaches the ellipse (Corollary of §5.5 below). The case $|G| = |K|$, where the ellipse degenerates to a segment, is considered separately (§5.2).

Engineering aspects. In engineering practice the branching feedback may be attractive for several reasons. It is given in closed form, with explicit bounds; the controller needs no clock, only the current branch, which suits computer control; and it allows one approach to different problems, similar to those solved by a switch with hysteresis [37,38], by hybrid controllers [30,39], and by Coron's continuous time-varying stabilisers [29]. Laws with memory, similar in this respect, are standard wherever an actuator has discrete states: a thermostat or a relay has two branches, an automatic gearbox with p gears has p branches, since it shifts up and down at different speeds, and staged heaters or compressors have one branch per stage. In all these cases memory prevents chattering. A further advantage is expected: insensitivity to small errors in measuring the state. A discontinuous stabiliser that depends only on the measured state is in

general sensitive to such errors near its discontinuities, since an error can push the control back and forth across the switching surface. Insensitivity is obtained by a controller with an internal state [1, 40] (in [1] this insensitivity is discussed in §6 in the context of observation problems); in the branching feedback this state is the current branch. It is not re-decided from the measured state at each instant: the branch changes only near the points of Σ where branches collide, inside a small ball around such a point in Z (§2.5). In branching closed/open-loop control [1, §3] the ball is not removed: while the lifted state is inside it, the control is a short programmed control $u(t) = h\big(F(z(t)), z(t)\big)$, the coupling along the reference trajectory, which brings the state to the next branch without using the measured state; the control stays continuous and the trajectory $x(t)$ smooth, although it may intersect itself. In branching feedback alone the ball is cut out of Z: on its boundary the state jumps to another branch over the same x, and the control jumps there. In both cases, under additional conditions of transversality of the trajectories on M to the singular set Σ, a small measurement error cannot flip the branch. Such balls are not explicitly built in this paper; they will be constructed elsewhere.

On language. We have deliberately dropped the original "Ashby" terminology, which never found readers and collides with unrelated usages; the objects are renamed in the vocabulary of geometric control. However, the name of Ashby hinted at a connection between the stabilisation of equilibria by branching feedback and the notion of ultrastability, which Ashby developed in the framework of cybernetics [41].

2. Control of qualitative behaviour: setup

This section restates the framework of [1, §1–§2] in modern language. The point is to make it legible and to fix the notation the later sections use.

2.1. **Plant, reference, feedback.** The plant $\dot x = v(x,u)$, $x \in X$, $u \in \mathcal{U}$, and the reference system $\dot z = w(z)$, $z \in Z$, are those of the Introduction, with the germs (1) and (2) at the working point. (In [1] the admissible controls at x form the fibre of a bundle over X; locally this is the product $X \times \mathcal{U}$.) A *feedback* is a map $s\colon X \to \mathcal{U}$; it gives the *closed loop* $\dot x = v\big(x, s(x)\big)$.

2.2. **The qualitative-behaviour control problem.**

Definition. The *qualitative-behaviour control (QBC) problem* of class C^k ($0 \le k \le \infty$) for the plant v and the reference w asks for a feedback $s\colon X \to \mathcal{U}$ of class C^k such that the closed loop $\dot x = v\big(x, s(x)\big)$ is C^k-conjugate to $\dot z = w(z)$, that is, a C^k-diffeomorphism carries the flow of the reference to the flow of the closed loop. Its local variant (germs at $x = 0, u = 0, z = 0$) is the *local* QBC problem.

Modal synthesis, Rosenbrock's structural theorem, equilibrium stabilisation, and the synthesis of stable periodic and quasi-periodic motions are all instances, for a suitably chosen reference w [1, §2].

2.3. **The product system and its coupling manifold.** We need only the local situation (for the global definition see [1, §2]).

Definition. A *coupling* of the plant and the reference is a germ $u = h(x,z)$ at $x = 0$, $z = 0$, with $h(0,0) = 0$. A *coupling manifold* of h is a germ $M \subset X \times Z$ of an integral manifold of the *product system*

$$\dot x = v\big(x, h(x,z)\big), \qquad \dot z = w(z),$$

that is a graph over Z: $M = \operatorname{graph}(x = F(z)) = \{(x,z) : x = F(z)\}$, $F(0) = 0$. We call the septet $(X, \mathcal{U}, v;\ Z, w;\ h, M)$ a *coupling scheme*.[1] Its members are the *state space* X, the *control*

[1] This is the object called an "Ashby relation" in [1].

space $\mathcal{U}$, the *plant field* v, the *reference state space* Z, the *reference field* w, the *coupling* h and the *coupling manifold* M. Derived from them are the map F, whose graph is M; the critical set Σ of the projection $\pi\colon M \to X$, $\pi(x,z) = x$ (§2.5); and its image $\pi(\Sigma) \subset X$, the *ramification locus*, or *caustic*.

Remark (coupling manifold and Lagrangian manifold). The construction parallels optimal control. There the state space is doubled by the cotangent bundle: to the state x one adjoins the adjoint variable ψ, the extremals sweep out a Lagrangian manifold in the space of pairs (x,ψ), and the optimal feedback is read off this manifold through the projection $(x,\psi) \mapsto x$. Here the state space is doubled by the reference system: to the state x one adjoins the reference state z, the coupling manifold M lies in the space of pairs (x,z), and the feedback is read off M through the projection $(x,z) \mapsto x$. So z plays the role of ψ and the coupling manifold that of the Lagrangian manifold. In both cases the feedback is single-valued where the projection is a local diffeomorphism and branches over its critical value set: the caustic in optimal control, the ramification locus of §2.5 below here. The comparison is continued in §6.1 below.

Theorem ([1, Thm. 2.2]). *Let the state space X, the control space $\mathcal{U}$ and the reference state space Z be C^k manifolds, and let the plant field v, the reference field w and the couplings below be of class C^k, $k \geq 1$. The local QBC problem of class C^k is solvable iff there exist a coupling h and a coupling manifold $M = \operatorname{graph}(x = F(z))$ of h with F a germ of a C^k-diffeomorphism. In that case $s(x) = h\big(x, F^{-1}(x)\big)$ solves it, and F conjugates the closed loop to the reference.*

Proof. As in [1, §2]: given a solving feedback s and a conjugacy F, set $h(x,z) = s(x)$; then $M = \operatorname{graph}(x = F(z))$ is invariant. Conversely, if F is a diffeomorphism, the closed loop under $s(x) = h(x, F^{-1}(x))$ is the push-forward of the reference by F, since M is invariant. □

2.4. **When is F a diffeomorphism?** The map F is a diffeomorphism near 0 iff its linear part $F_0 = \partial_z F(0)$ is invertible. With the linearised coupling $u = Kx + \Psi z$, the linearised product system is $\dot{x} = (A+CK)x + C\Psi z$, $\dot{z} = Dz$, and invariance of M gives $(A+CK)F_0 - F_0 D = -C\Psi$.

Proposition. *If (A,C) is controllable and the plant field v and the reference field w are of class C^2, then the local problem is C^1-solvable iff its linearisation is solvable ([1, Thm. 2.5], via Rosenbrock). If (A,C) is uncontrollable and λ is an uncontrollable eigenvalue of A that is not an eigenvalue of D, then for every coupling the map F is not a diffeomorphism [1, Cor. 2.7].*

Proof. Let (A,C) be controllable and let the linearisation be solvable: $(A+CG)\Phi = \Phi D$ with Φ invertible. Choose K so that the spectra of $A+CK$ and D satisfy the conditions of [1, Thm. 1.2], which is possible by controllability, and put $\Psi = (G-K)\Phi$. Then the coupling manifold exists, its linear part is the unique solution of the equation above, and this solution is $F_0 = \Phi$; so F is a diffeomorphism, and by the Theorem of §2.3 (with $k = 1$) the local problem is C^1-solvable. Conversely, let a feedback s of class C^1 solve the local problem, with a conjugacy H of class C^1 carrying the reference to the closed loop. Differentiating the conjugacy at 0 gives $\big(A + C\,ds(0)\big)DH(0) = DH(0)\,D$, so $G = ds(0)$ and $\Phi = DH(0)$ solve the linearisation. For the second assertion: the image of C lies in the controllable subspace, so $A+CK$ and A induce the same map on the quotient by it, and an uncontrollable eigenvalue λ of A is an eigenvalue of $A+CK$ for every K. If F were a diffeomorphism, the closed loop with the feedback of the Theorem of §2.3 would be conjugate by F to the reference, so its linearisation, of the form $A+CK'$, would be similar to D, and λ would be an eigenvalue of D. □

Where F cannot be inverted, the feedback has several branches; this is the subject of §2.5 below.

2.5. **Branching feedback and its ramification locus.** Fix a coupling h with coupling manifold $M = \mathrm{graph}(x = F(z))$, and let

$$\Sigma \;=\; \{\, z : \det \partial_z F(z) = 0 \,\} \tag{3}$$

be the critical set of the projection $\pi\colon M \to X$, $\pi(x,z) = x$ (we identify M with Z by $z \mapsto (F(z), z)$). Off Σ the map F is a local diffeomorphism. By the Proposition of §2.4 above, the working point lies in Σ when (A, C) is uncontrollable and the spectrum of the reference does not contain some uncontrollable eigenvalue of A. The *ramification locus* $\pi(\Sigma) \subset X$ depends on the reference and the coupling as well as on the plant.

Branches. Assume F is *finite*: proper near the fibre in question, with finite fibres. Then off $\pi(\Sigma)$ the equation $x = F(z)$ has, locally, finitely many smooth solution branches $z = \zeta_1(x), \dots, \zeta_\mu(x)$, their number constant on each component of $X \setminus \pi(\Sigma)$; a global labelling of the branches may be obstructed by monodromy, which is why the label is changed across cuts. On $\pi(\Sigma)$ branches meet. In the two simplest cases two branches meet. At a *fold*, as for $x = z^2$ on $\mathbb{R}$, there are two branches on one side and none on the other: they meet and end. At a *ramification point*, as for $x = z^2$ on $\mathbb{C}$, there are two branches at every nearby point, and going once around the point exchanges them. More branches can meet at one point: for $x = z^n$ on $\mathbb{C}$, n branches meet at 0, and going once around it permutes them cyclically; at the point of a cusp, three branches of a real map meet. Finite branching is the case in §3 and §5 below: the tuck of §3.1, the map $z \mapsto z^n$ and its small deformations, and $(z, x_>) \mapsto (z^2, x_>)$. For the Brockett integrator the branches over each axis point form a circle (§4.2 below).

The branching feedback. Along M the coupling h of the Definition of §2.3 above takes the value

$$p(z) \;=\; h\big(F(z), z\big),$$

the *coupling control* carried by the reference. It is a function of the reference state z, not of x, so on its own it is not a feedback. Inverting the projection gives the *branching feedback*: the values $u(x,j) = p(\zeta_j(x))$, $j = 1, \dots, \mu$, one for each branch, together with a switching rule α for the branch index j. The closed loop is the plant together with a discrete part [1, §3]:

$$\dot{x} = v\big(x, u(x,j)\big), \qquad j(t+0) = \alpha\big(x(t), j(t-0)\big),$$

where $j(t)$ is piecewise constant, with finitely many jumps on each finite time interval (this holds, for example, when the trajectories of the reference on Z cross Σ transversally). When F is not a diffeomorphism, the QBC problem has no C^1 solution (§2.4 above), and one asks instead for a branching feedback whose closed loop is the image of the reference under F: the *branching QBC problem* (the modified problem of [1, §3]).

Proposition (local uniqueness off Σ; [1, Prop. 3.1])**.** *Solutions of the closed loop that follow the branch j are locally unique at every (x,j) with $\zeta_j(x) \notin \Sigma$. Uniqueness can fail only where the active branch meets Σ.*

Parasitic solutions. Choosing the branch does not make the solutions unique. Near $\pi(\Sigma)$, where branches meet, uniqueness fails, and a solution may continue along the caustic and stop following the reference: on the caustic the closed-loop field is the projection $\partial_z F \cdot w$ of the reference field, which is tangent to the caustic (and zero if w lies in the kernel of $\partial_z F$), and the non-Lipschitz closed loop admits solutions running along it. There are two ways to avoid these parasitic solutions; we do not go into the details of either.

(i) *Branching closed/open-loop control.* Near $\pi(\Sigma)$ the feedback is replaced by the programmed control $u(t) = p(z(t))$ along the reference trajectory, and it is resumed on the branch reached. No cuts and no jumps are needed: the branch label simply changes

along $u(t)$, and the control stays continuous in time. This is the sense of the name *branching closed/open-loop control* [1, §3].

(ii) *Branching feedback alone.* Without programmed control, the state is switched from one branch to another before it reaches $\pi(\Sigma)$. In Z, a small ball is cut out around each point of Σ where branches collide and end (two, or any even number of them). We assume that over the boundary of each cut ball there is a branch that does not take part in the collision. When the lifted state reaches the boundary of such a ball, j jumps to another branch over the same x that does not take part in the collision, and the feedback continues on it. Nothing is cut out of X or around the new branch. The control jumps in time at these switches, and the trajectories are continuous there but not smooth.

In (i) each solution is the projection of one reference trajectory, and the closed loop reproduces, projected by π, the portrait of the reference. In (ii) each solution is the projection of a concatenation of reference trajectories, with a reset of z at each switch; stability and finitely many switches must then be checked for the chosen reset rule.

3. Branching stabilisation with uncontrollable linearisation

3.1. **The generic second-order degeneracy.** Before the model, the general corank-one situation, where the second-order term is an arbitrary quadratic. Let the linearisation of (1) at the origin have corank one, with $x \in \mathbb{R}^2$ and scalar u. By a linear change of variables and a feedback substitution absorbing the $\dot{x}_2$-component, the system reduces to the normal form [1, §4.2]

$$\dot{x}_1 = ax_1 + b_{20}x_2^2 + b_{11}x_2u + b_{02}u^2 + b_3(x_1, x_2, u), \qquad \dot{x}_2 = u, \tag{4}$$

with $a > 0$, constants b_{20}, b_{11}, b_{02}, and a remainder b_3 that is merely *differentiable*, $b_3 = O(|x_1|^2 + |x_1||x_2| + |x_1||u| + |x_2|^3 + |u|^3)$. Set

$$\Delta \;=\; a^2b_{02}^2 + 2a\,b_{02}b_{11} + 4\,b_{02}b_{20}. \tag{5}$$

Theorem (dichotomy at a corank-one degeneracy; [1, Thm. 4.2, Prop. 4.4])**.** *For* (4) *the sign of the invariant* Δ *decides the QBC problem.*

(1) *If* $\Delta > 0$*, the origin cannot be stabilised at all: there are* $c, \delta > 0$ *and initial values arbitrarily close to the origin whose trajectory leaves the* δ*-neighbourhood for* every *control* $u(t)$*, however discontinuous, with* $|u| \le c$*. Moreover no small periodic motion with nonzero rotation about the origin exists.*

(2) *If* $\Delta < 0$*, the stabilisation problem is solvable by branching control: for every* $\Lambda > 0$ *one can write out explicitly a coupling with a Hurwitz reference, of decay* $\|z(t)\| \sim e^{-kt}$*,* $k > \Lambda$*, whose coupling manifold is the graph of a map* $F(z_1, z_2) = (f(z_1, z_2), z_2)$ *equivalent to the tuck (or cusp) [36].*

Remark (the sign of Δ and the case $\Delta = 0$). The sign of Δ is an invariant of (4) [1, §4.2]. The excluded case $\Delta = 0$ has codimension two in the space of plants: one condition for the uncontrollable linearisation and one for $\Delta = 0$. There the quadratic terms do not decide, and the verdict depends on the third- and fourth-order terms of b_3, which must then be smooth enough; this case is not treated here.

If $\Delta < 0$, more is possible: the same normal form also admits the synthesis of a whole Hopf bifurcation, at a frequency fixed by the plant.

Theorem (synthesis of the Andronov–Hopf bifurcation; [1, Thms. 4.5, 4.6, Rem. 4.7])**.** *For almost all plants* (4) *with* $a^2b_{02}^2 + ab_{11}b_{02} + 2b_{20}b_{02} < 0$ *(a condition that implies* $\Delta < 0$*), a branching feedback depending on a parameter* ε *reproduces the stable Andronov–Hopf bifurcation:*

for small $\varepsilon > 0$ the closed loop has an asymptotically stable cycle of diameter of order $\sqrt{\varepsilon}$, with nonzero rotation about the origin and angular frequency ω,

$$\omega^2 = -\frac{ab_{11} + 2b_{20} + a^2 b_{02}}{2b_{02}}.$$

The projection of the coupling manifold is equivalent to the tuck, and near the cycle the closed loop is topologically orbitally equivalent to a standard model. With a second scalar control any prescribed frequency can be reached.

3.2. **The model $\dot{x} = u^n$.** The model is *more* degenerate than (4): its linearisation is totally uncontrollable and it carries two real controls, so it is not a special case of the dichotomy above but several strata down. Fix $n \geq 2$ and consider

$$\dot{x} = u^n, \qquad x \in \mathbb{C} \cong \mathbb{R}^2,\ u \in \mathbb{C}, \tag{6}$$

whose linearisation at the origin vanishes identically in u. For $n = 2$ this is the system of [1, §4].

Proposition (no continuous stabiliser; [1, Prop. 4.1])**.** *For (6) with any continuous $u = u(x)$, $u(0) = 0$, the origin is not asymptotically stable. (Degree argument: an asymptotically stable equilibrium is isolated and has index 1; on a small circle around the origin free of zeros of u, the closed-loop field $u(x)^n$ has rotation $n\nu$, ν the rotation of u, never 1 for $n \geq 2$.)*

Theorem (branching stabilisation; [1, §4] for $n = 2$)**.** *The stabilisation problem for (6) is solvable in the class of branching controls. In complex notation, $x = x_1 + ix_2$, $u = u_1 + iu_2$, take the stable reference $\dot{z} = \lambda z$, $\Re\lambda < 0$, the coupling $u = z$ and the coupling manifold $x = F(z) = z^n/(n\lambda)$. Let $(n\lambda x)^{1/n}$ be a branch of the root continuous on the x-plane cut along a ray $x_2 = kx_1$, $x_1 > 0$. Then the n-branch control*

$$u(x, j) = \omega^j (n\lambda x)^{1/n}, \qquad \omega = e^{2\pi i/n}, \quad j = 0, \ldots, n-1,$$

gives the closed loop $\dot{x} = n\lambda x$ on every branch, so $x(t) \to 0$. The ramification locus is $\pi(\Sigma) = \{0\}$; when x crosses the ray $x_2 = kx_1$, $x_1 > 0$, with k chosen by the designer, the branch is switched by the law $j(t+0) = \alpha\big(x(t), j(t-0)\big)$ of §2.5, where $\alpha(x, j) = j + \operatorname{sgn}(\Im\lambda) \bmod n$ on the ray and $\alpha(x, j) = j$ off it, and the control $u(t) = z(t)$ stays continuous.

For real λ the trajectories are rays into 0 and the branch never changes.

3.3. **Deformations of the model.**

Definition. Let a degeneracy of controllability have codimension k in the space of plants. If a k-parameter family of plants crosses the set of degenerate plants transversally [36], then every family close to it crosses that set too: a small change of the family only moves the parameter value at which the degenerate plant occurs, and does not remove it. The QBC problem of stabilising every plant of such a family, with one reference and one coupling for all of them, is called the *QBC ultrastabilisation problem*, after Ashby's ultrastability [41].

Deform the model by switching on the remaining low-order terms. This subsection has two purposes. It shows, in this example, the branching completely: how the branch point of order n splits, where the new branch points lie, and how the n branches are permuted around them. And it shows that the same reference and coupling serve the whole family of deformations, across the degeneracy, with one bound on the control, which is what plants whose control enters through a power require (see the end of the versal paragraph below). The family below is holomorphic in x and u. A generic real perturbation also contains terms in $\bar{u}$; then the critical set is a union of curves, with folds and cusps, not a set of isolated points, and this case is not treated here. For $n = 2$ the quadratic part u^2 stays a double covering under the small real perturbations

$Gu^2 + H|u|^2 + K\bar{u}^2$, $G \approx 1$, $H, K \approx 0$, since such H lies inside the ellipse of §5.2 below. The plant is

$$\dot{x} = P(x, u, \varepsilon), \qquad x, u \in \mathbb{C}, \tag{7}$$

reducing at $\varepsilon = 0$ to the model (6) of §3.2 above,

$$P(x, u, 0) = u^n, \qquad n \geq 2. \tag{8}$$

The versal deformation. The simplest such family does not contain x:

$$P(u) = u^n + c_{n-1}u^{n-1} + \cdots + c_1 u, \qquad c_k \in \mathbb{C}, \tag{9}$$

the versal deformation of u^n among polynomials with $P(0) = 0$; the constant term is zero because the working point $x = 0$, $u = 0$ is an equilibrium. With the reference $\dot{z} = \lambda z$, $\Re\lambda < 0$, and the coupling $u = z$, invariance of $M = \mathrm{graph}(x = F(z))$ reads $\lambda z F'(z) = P(z)$, so

$$F(z) = \frac{1}{\lambda}\int_0^z \frac{P(s)}{s}\,ds = \frac{1}{\lambda}\Big(\frac{z^n}{n} + \sum_{k=1}^{n-1}\frac{c_k}{k}\,z^k\Big), \qquad F'(z) = \frac{P(z)}{\lambda z}.$$

As c varies, F runs over the versal deformation of $z^n/(n\lambda)$. The critical set Σ_c is the set of roots of $P(z)/z = z^{n-1} + c_{n-1}z^{n-2} + \cdots + c_1$, at most $n-1$ points (it contains 0 when $c_1 = 0$ and is $\{0\}$ when $c = 0$), and the ramification locus $\pi(\Sigma_c) = F(\Sigma_c)$ is the set of critical values of this versal deformation. The closed loop is $x(t) = F(z_0 e^{\lambda t}) \to 0$ and the control is $u(t) = z_0 e^{\lambda t}$. So $|u(t)| \leq |z_0| \leq R$ for every c, including $c = 0$, where no continuous stabiliser exists: the lifted state stays in the disc $|z| \leq R$ of M. By Rouché's theorem F maps this disc onto a neighbourhood of $x = 0$ that contains a fixed disc for all small c. For $c_1 \neq 0$ the linearisation is controllable, but every C^1 feedback s with $s(0) = 0$ that stabilises the origin with decay rate $|\lambda|$ needs the gain $\|Ds(0)\| \geq |\lambda|/|c_1|$, since the closed-loop linearisation is $c_1 Ds(0)$; this gain grows without bound as $c_1 \to 0$. The same reference and coupling thus serve the whole family, across the degeneracy, with a bound that does not depend on c: they solve the QBC ultrastabilisation problem (Definition 3.3) for the versal family, an $(n-1)$-parameter family through the degenerate plant $c = 0$, whose degeneracy has complex codimension $n-1$. The plant origin is stable when the branch is started at a lift z_0 of $x(0)$ of smallest modulus; from another lift the trajectory still tends to 0 but may first move away. Indeed the product of the n roots of $F(z) = x$ has modulus $n|\lambda||x|$, so the smallest one satisfies $|z_0| \leq (n|\lambda||x(0)|)^{1/n}$, and $|z(t)| \leq |z_0|$.

An example: braking a resonator driven at half frequency. Electrostatic actuators produce a force proportional to the square of the voltage. A resonator of frequency ω_0 is therefore often driven by an alternating voltage of frequency $\omega_0/2$ without DC bias: the force then has a component at ω_0, while the electrical feedthrough stays away from ω_0 [42,43]. For the resonator $\ddot{y} + 2\gamma\dot{y} + \omega_0^2 y = cV^2$ with $V(t) = \mathrm{Re}\big(ue^{i\omega_0 t/2}\big)$, where the complex amplitude u of the voltage carries two real controls, averaging gives for the slowly varying amplitude a of $y = \mathrm{Re}\big(ae^{i\omega_0 t}\big)$ the equation

$$\dot{a} = -\gamma a + \kappa u^2, \qquad \kappa = -\frac{ic}{4\omega_0},$$

our model with $n = 2$ and a weak damping γ (a high quality factor). To stop the resonator quickly, at a rate $R \gg \gamma$, a continuous law $u = s(a)$ would need $\kappa s(a)^2 \approx -Ra$, that is, a continuous square root of $-a$, which does not exist. The construction above solves the problem as a particular case: the reference $\dot{z} = -\frac{\gamma+R}{2}z$, the coupling $u = z$ and the coupling manifold $a = -\kappa z^2/R$ give the two branches $u = \pm\sqrt{-Ra/\kappa}$ and the closed loop $\dot{a} = -(\gamma + R)a$. In a simulation of the full resonator with quality factor 10^4 and $R = 0.02\,\omega_0$, this feedback reduces the amplitude tenfold in about 18 periods instead of about 7000, from every initial phase, while

a continuous law $u = \beta a/|a|^{1/2}$ first amplifies the oscillation by a factor that grows without bound as the initial phase approaches one direction. A drive at $\omega_0/4$ gives the model with $n = 4$. For a parallel-plate actuator, normalised as $\ddot{y} + 2\gamma\dot{y} + y = \beta V^2/(1-y)^2$ with $\omega_0 = 1$ and $V(t) = \mathrm{Re}\big(ue^{it/4}\big)$, the force βV^2 makes the plate oscillate at $\omega_0/2$ with the complex amplitude $\frac{2\beta}{3}u^2$, and the gap term $2\beta V^2 y$ multiplies this by V^2 again; the resonant part is $\frac{\beta^2}{3}\mathrm{Re}\big(u^4 e^{it}\big)$, and averaging gives

$$\dot{a} = -\gamma a - \tfrac{i}{2}\beta|u|^2 a + \kappa_4 u^4, \qquad \kappa_4 = -\tfrac{i}{6}\beta^2,$$

where the middle term only rotates a. The four-branch feedback $u = (-Ra/\kappa_4)^{1/4}$, with the branch carried along, brakes this resonator too: in a simulation of the full nonlinear resonator with quality factor 10^4, $\beta = 1$ and $R = 0.01$, the amplitude falls from 0.05 to 0.005 (gap 1) in 17 to 28 periods, depending on the initial phase, instead of about 7000. In both cases the branches meet only at the caustic $a = 0$, which the trajectory approaches but never crosses; when a winds around it, the branch label changes on a cut chosen by design, a ray issuing from $a = 0$, and the feedback needs no open-loop segment. Plants $\dot{a} \approx \kappa u^n$ with $n > 2$ arise in the same way whenever an oscillation is driven, through a nonlinearity of order n, by a signal at $1/n$ of its frequency, as in frequency multipliers (a tripler gives $n = 3$) and harmonic generation; no continuous feedback $u = s(a)$ then brakes it from every phase. Other physical deviations of the resonator add lower powers of u, which gives the versal deformation above; the same reference and coupling then still give a branching feedback that brakes the resonator. More generally, whenever a part of the state that does not stabilise by itself is reached by two controls only through a homogeneous quadratic map, no continuous feedback stabilises it: such a map is even, so its degree is even and never 1.

The general family. Take

$$P(x,u,\varepsilon) \;=\; u^n + \sum_{a+b\le n} \varepsilon_{ab}\, x^a u^b, \qquad \varepsilon_{ab} \in \mathbb{C}, \tag{10}$$

a polynomial in $x, u \in \mathbb{C}$ with complex coefficients; it contains (9) as the terms without x. (The polynomiality here is only local normal-form (jet) data.) Here $\varepsilon = (\varepsilon_{ab})$ is the vector of all coefficients of (10) other than that of u^n, and ranges over a neighbourhood of 0 in $\mathbb{C}^K$ with

$$K = \tfrac{1}{2}(n+1)(n+2) - 1.$$

We assume

$$\varepsilon_{00} = 0, \tag{11}$$

since the working point $x = 0$, $u = 0$ is an equilibrium.

Invariance of $M = \mathrm{graph}(x = F(z))$ over a reference $\dot{z} = w(z)$ carrying the control $u = p(z)$ reads

$$F'(z)\, w(z) \;=\; P\big(F(z), p(z), \varepsilon\big).$$

Take $\dot{z} = \lambda z$ with $\Re\lambda < 0$ and choose $u = p(z) = z$; invariance becomes

$$\lambda z\, \partial_z F(z,\varepsilon) \;=\; P\big(F(z,\varepsilon), z, \varepsilon\big). \tag{12}$$

Theorem (the branch point splits)**.** *Assume* (10) *and* (11)*, and choose* λ *with* $\Re\lambda < 0$ *and*

$$m\lambda \neq \varepsilon_{10} \qquad \textit{for every integer } m \ge 1, \tag{13}$$

which excludes only the points $\lambda = \varepsilon_{10}/m$ *and is therefore always possible. Then*:

(1) (12) *has a unique solution* $F(\cdot,\varepsilon)$ *holomorphic near* 0 *with* $F(0,\varepsilon) = 0$*, its coefficients determined recursively by the divisors* $m\lambda - \varepsilon_{10}$*; and* $F(z,0) = z^n/(n\lambda)$.

(2) *$\partial_z F(z,0) = z^{n-1}/\lambda$ has a zero of order $n-1$ at the origin, so by the Weierstrass preparation theorem $\partial_z F(\cdot,\varepsilon)$ has, for small ε, exactly $n-1$ zeros near 0 counted with multiplicity. The branch point of order n therefore* splits*: Σ_ε consists of up to $n-1$ points, generically simple, which coalesce at the origin as $\varepsilon \to 0$.*

(3) *The ramification locus in the plant is their image*

$$\pi(\Sigma_\varepsilon) \;=\; F(\Sigma_\varepsilon, \varepsilon),$$

at most $n-1$ points, all tending to $x = 0$ with ε. The feedback remains n-valued, $u = p(\zeta_j(x)) = \zeta_j(x)$, holomorphic off $\pi(\Sigma_\varepsilon)$, and its branches are permuted by the monodromy of loops encircling those points.

(4) *For $n = 2$ and $P = u^2 + \varepsilon_{01} u + \varepsilon_{10} x$ everything is explicit:*

$$\Sigma_\varepsilon = \left\{ -\tfrac{\varepsilon_{01}(\varepsilon_{10}-2\lambda)}{2(\varepsilon_{10}-\lambda)} \right\}, \qquad \pi(\Sigma_\varepsilon) = \left\{ \tfrac{\varepsilon_{01}^2(\varepsilon_{10}-2\lambda)}{4(\varepsilon_{10}-\lambda)^2} \right\}.$$

Thus, for $\varepsilon_{01} \neq 0$, the branches no longer meet at the working point: they meet at a nearby point, at a distance from the origin proportional to ε_{01}^2.

Proof sketch. The coefficients of F are found recursively from (12). By the Weierstrass preparation theorem, $F(z,\varepsilon) = x$ is near 0 a polynomial equation of degree n in z. Its n roots are the branches ζ_j, and the theorem describes how they branch as ε varies. □

Condition (13) excludes only countably many λ, so λ may be taken real; then $z(t) = z_0 e^{\lambda t}$ moves along a segment of a ray, its image $x(t) = F(z(t))$ winds around $\pi(\Sigma_\varepsilon)$ only finitely often, and the branch label changes finitely often. Trajectories through a point of Σ_ε ($n-1$ curves of initial states) would stop for an instant at a branch point, where the closed loop is not Lipschitz. With jumps on cuts around the branch points the state never reaches them; with open-loop segments it passes them along the reference (§2.5).

4. The Brockett integrator

The nonholonomic (Brockett) integrator [26]

$$\dot{x}_1 = u_1, \qquad \dot{x}_2 = u_2, \qquad \dot{x}_3 = x_1 u_2 - x_2 u_1, \tag{14}$$

$x \in \mathbb{R}^3$, $u \in \mathbb{R}^2$, fails Brockett's necessary condition at the origin: no continuous $s(x)$ stabilises it. The unicycle is brought to this form by a feedback transformation [44], so the syntheses below apply to it. It is the corank-one instance of the present framework — (x_1, x_2) controllable, x_3 uncontrollable with the bracket entry $x_1 u_2 - x_2 u_1$ (no linear part). A weighted blow-up resolves the plant into a family of surfaces indexed by a cylinder (§4.1 below). In §4.2 below we use the construction of §2.3, on each side of the surface where the reference switches, with the blow-up Z as the reference state space and the blow-down

$$\beta \colon Z \to \mathbb{R}^3, \qquad (r, \theta, \rho) \mapsto (r\cos\theta,\; r\sin\theta,\; \rho r^2),$$

as the map F. The reference is a field on Z, and the coupling $u = h(z)$ depends on $z \in Z$ alone: it is the control under which β carries the reference to the plant. The feedback is

$$s(x) = h\big(\beta^{-1}(x)\big).$$

Off the x_3-axis $\beta^{-1}(x)$ is one point, and $s(x)$ is one vector. Over a point $(0, 0, x_3)$ of the axis, β^{-1} is the circle $\{r = 0,\ \theta \in S^1\}$ (in the coordinates (r, θ, x_3) of (16)), so there the branches of s form a circle: at the point θ the value of h is $\eta |x_3|^{1/2} (\cos\theta, \sin\theta)$, a push away from the axis in the direction θ, with $\eta > 0$ a constant chosen in §4.2. Choosing the branch means choosing θ, that is, the point of the circle from which the reference trajectory starts. Every choice is admissible, and once it is made the motion is unique. Let $r^2 = x_1^2 + x_2^2$; we call $\nu = (r^4 + x_3^2)^{1/4}$ the *size* of

the state. The control satisfies $|u| \le c\,\nu$ for a constant c, and the half-attraction time, the time after which ν has halved, does not depend on the initial point. The feedback s does not depend on time. Over the x_3-axis $\pi(\Sigma)$, where β is not a diffeomorphism, it has a circle of branches. It jumps only on the surface $|x_3| = \rho_0(x_1^2 + x_2^2)$, where the reference changes and whose position $\rho_0 > 0$ the designer chooses. This jump could be removed (Remark at the end of §4.2), but we keep it: it gives the exact duration of the outward motion, and in §4.3 the position ρ_0 of this surface is how the designer chooses the frequency. Write $\rho = x_3/r^2$; the surfaces $\{x_3 = \rho r^2\}$, ρ constant, are paraboloids. §4.3 below keeps each of them invariant instead; the half-attraction time on the paraboloid labelled ρ then grows like $(2\ln 2/c)\,|\rho|^{1/2}$, which tends to ∞ as the axis is approached, $\rho \to \pm\infty$. Finally, §4.4 below makes the feedback depend on time and obtains a law $u(x,t)$ that is continuous everywhere.

4.1. **The blow-up and the family of horns.** *The blow-up.* Blow-up is a standard tool for dynamics near degenerate points; for a survey see [45]. Consider the weighted $(1,1,2)$ blow-up of $\mathbb{R}^3$ at the origin,

$$x_1 = r\cos\theta, \qquad x_2 = r\sin\theta, \qquad x_3 = \rho\, r^2, \tag{15}$$

with $r \in [0,\infty)$, $\theta \in S^1$ and $\rho \in \mathbb{R}$. Its set $\{r = 0\}$ is the cylinder over the origin, with coordinates (θ,ρ). In (15), $r = 0$ forces $x_3 = 0$, so (15) never reaches a point $(0,0,x_3)$ with $x_3 \neq 0$. That is why near such points we use the cylindrical coordinates (r,θ,x_3),

$$x_1 = r\cos\theta, \qquad x_2 = r\sin\theta, \qquad r \in [0,\infty),\ \theta \in S^1,\ x_3 \neq 0. \tag{16}$$

The blow-up Z is the manifold covered by two charts, (r,θ,ρ) of (15) and (r,θ,x_3) of (16). Where both apply, $r > 0$ and $x_3 \neq 0$, the coordinates change by

$$(r,\theta,\rho) \mapsto (r,\theta,x_3) = (r,\theta,\rho r^2), \qquad (r,\theta,x_3) \mapsto (r,\theta,\rho) = (r,\theta,x_3/r^2).$$

The blow-down $\beta\colon Z \to \mathbb{R}^3$ is (15) in the first chart and (16) in the second. Over each point $(0,0,x_3)$, $x_3 \neq 0$, of the axis it has the circle which in the coordinates of the second chart is $\beta^{-1}(0,0,x_3) = \{r = 0,\ \theta \in S^1\}$ (which was used in the introduction to §4). The *exceptional set* $E = \beta^{-1}(x_3\text{-axis}) = \{r = 0\}$ is the cylinder over the origin (chart (15)) together with the circles over the other axis points (chart (16)). Off E, β is a diffeomorphism onto $\mathbb{R}^3$ minus the x_3-axis. The set E is not compact: the two end circles $\rho = \pm\infty$ are missing. To compactify it one would need a third chart, $x_1 = ts\cos\theta$, $x_2 = ts\sin\theta$, $x_3 = \pm t^2$ with $t,s \in [0,\infty)$, in which $\rho = \pm 1/s^2$ and the end circles are $\{t = s = 0\}$. We do not need it: no trajectory of the closed loop reaches these circles.

A horn for each point of the cylinder. The coordinate $\rho = x_3/r^2$ is constant on the paraboloids

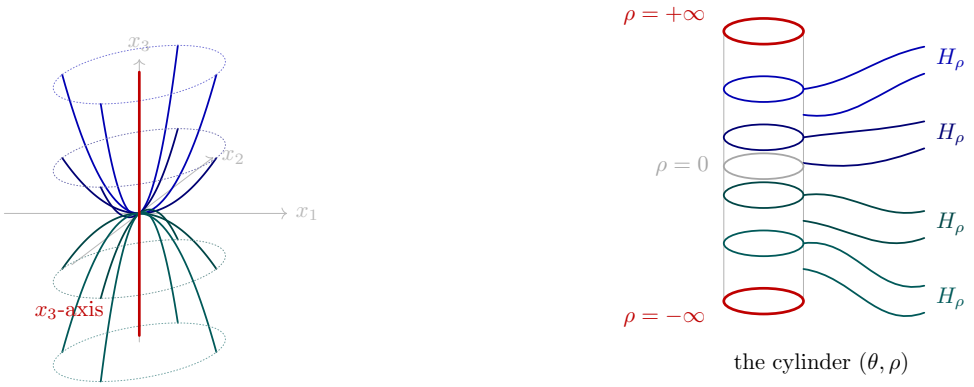
$$H_\rho = \{\, x_3 = \rho\, r^2 \,\}, \qquad r^2 = x_1^2 + x_2^2,$$

which foliate $\mathbb{R}^3$ minus the x_3-axis; each is a two-dimensional *horn* with apex at the origin. Downstairs all the horns share that apex; in the formula (15) they are disjoint, each meeting the cylinder in the circle $\{r = 0,\ \rho = \text{const}\}$. That cylinder therefore *indexes* the family: a point of it names one horn.

Area is the only way to move x_3. Writing $x_1 = r\cos\theta$, $x_2 = r\sin\theta$, a control is the velocity of that point, and

$$\dot{x}_3 = x_1 u_2 - x_2 u_1 = r^2\,\dot{\theta}. \tag{17}$$

The right-hand side is twice the areal velocity of the projected motion: $x_1\dot{x}_2 - x_2\dot{x}_1 = r^2\dot{\theta}$.



(a) (b)

FIGURE 1. (a) The horns $H_\rho = \{x_3 = \rho r^2\}$ in x-space: paraboloids of revolution with a common apex at the origin, tangent there to the plane $x_3 = 0$. They foliate $\mathbb{R}^3$ minus the x_3-axis (thick), which is their common limit as $\rho \to \pm\infty$. (b) After the blow-up: in the formula (15) the set $\{r = 0\}$ is the cylinder with coordinates (θ, ρ), and the whole cylinder maps to the original origin; each horn is attached along one circle $\rho = \text{const}$ (the circle $\rho = 0$ along the plane $x_3 = 0$). The two end circles $\rho = \pm\infty$ are not part of the cylinder. The circles over the points $(0, 0, x_3)$ of the x_3-axis lie in the chart (16), not on the cylinder; as $x_3 \to 0^\pm$ they approach the ends $\rho = \pm\infty$. The synthesis of §4.2 below drives ρ from the two ends towards the circle $\rho = 0$.

4.2. **The synthesis: outward motion, then descent.** The construction has two regimes, and the horn label ρ decides between them: where $|\rho|$ is large the state runs outward at constant height (the *outward motion*), which is what brings $|\rho|$ down, and where $|\rho|$ is small it descends (the *descent*) to the working point with the radius and the height decaying at rates of their own.

The plant on the blow-up. In the coordinates (ν, θ, ρ) of the first chart (15), with $\nu = r(1 + \rho^2)^{1/4}$, write the control as

$$u = \nu A\, e_r + \nu(1 + \rho^2)^{1/4} B\, e_\theta, \qquad e_r = (\cos\theta, \sin\theta), \quad e_\theta = (-\sin\theta, \cos\theta), \tag{18}$$

so that $\dot r = \nu A$ and $\dot x_3 = \nu^2 B$. The new controls are $A, B \in \mathbb{R}$. The plant (14) lifted to the blow-up is

$$\dot\rho = B\,(1 + \rho^2)^{1/2} - 2\rho A\,(1 + \rho^2)^{1/4}, \tag{19}$$

$$\frac{\dot\nu}{\nu} = A\,(1 + \rho^2)^{-3/4} + \tfrac{1}{2}\,\rho B\,(1 + \rho^2)^{-1/2}, \tag{20}$$

$$\dot\theta = B\,(1 + \rho^2)^{1/2},$$

and the size of the control is

$$|u|^2 = \nu^2\big[A^2 + B^2(1 + \rho^2)^{1/2}\big]. \tag{21}$$

The right-hand sides do not contain ν or θ, so the lifted system extends to the cylinder $\nu = 0$ over the origin, which is invariant.

The control bound. We require $|u| \le c\,\nu$ with a constant $c > 0$. By (21) it reads

$$A^2 + B^2(1 + \rho^2)^{1/2} \le c^2. \tag{22}$$

This bound decides how far from the axis the state must go before it can descend. Near the axis, moving outward is cheap and moving the height is expensive, so the state first moves away from the axis and only then descends (Appendix A).

The reference. Let Z denote the blow-up of §4.1 above, in the two formulas (15) and (16), and let $\beta\colon Z \to \mathbb{R}^3$ be the blow-down. Here the reference, the coupling h and the map $F = \beta$ are defined globally, on all of Z, not only near one point as in the Definition of §2.3. The reference $\dot z$ on Z is as follows, with constants $b > 2a > 0$, $\eta > 0$ and $\rho_0 > 0$ chosen by the designer.

Outward motion, on $|\rho| > \rho_0$, in the coordinates (r, θ, x_3) of (16):

$$\dot r = \eta\,\nu, \qquad \dot\theta = 0, \qquad \dot x_3 = 0, \tag{23}$$

with $\nu = (r^4 + x_3^2)^{1/4}$. In the coordinates (ν, θ, ρ) it reads $\dot\rho = -2\eta\rho(1+\rho^2)^{1/4}$, $\dot\theta = 0$, $\dot\nu/\nu = \eta(1+\rho^2)^{-3/4}$, which also covers the cylinder $\nu = 0$.

Descent, on $|\rho| \le \rho_0$, in the coordinates (ν, θ, ρ) of (15):

$$\dot\rho = -(b-2a)\rho, \qquad \dot\theta = -b\rho, \qquad \frac{\dot\nu}{\nu} = -\,\frac{a + \frac12 b\rho^2}{1+\rho^2}. \tag{24}$$

The coupling $u = h(z)$ is

$$u = \eta\,\nu\,e_r \quad \text{on } \{|\rho| > \rho_0\}, \qquad u = -r\,(a\,e_r + b\rho\,e_\theta) \quad \text{on } \{|\rho| \le \rho_0\}, \tag{25}$$

with $r = \nu(1+\rho^2)^{-1/4}$. It is the control (18) with $A = \eta$, $B = 0$ in the outward motion and $A = -a(1+\rho^2)^{-1/4}$, $B = -b\rho(1+\rho^2)^{-1/2}$ in the descent, and under it the lifted plant (19)–(20) is the reference. Pushed down by β it gives the feedback in x. Write $r = (x_1^2 + x_2^2)^{1/2}$, $\nu(x) = (r^4 + x_3^2)^{1/4}$ and $\rho(x) = x_3/r^2$. Then

$$u = \begin{cases} \eta\,\nu(x)\,\dfrac{(x_1, x_2)}{r} & \text{where } |x_3| > \rho_0\, r^2 \quad \text{(outward motion)}, \\[2mm] -a\,(x_1, x_2) - b\,\rho(x)\,(-x_2, x_1) & \text{where } |x_3| \le \rho_0\, r^2 \quad \text{(descent)}, \end{cases} \tag{26}$$

and $u(0) = 0$. The two regions are separated by the horns $|x_3| = \rho_0 r^2$.

The checks that both formulas are bounded, that the switch is crossed once, and the largest admissible ρ_0 are given in Appendix A.

The coupling manifold is the graph of the blow-down. It is

$$M = \operatorname{graph}\bigl(x = \beta(z)\bigr) \subset X \times Z,$$

invariant under the product system because the closed loop is by construction the push-forward of the reference. Here β is a local diffeomorphism off the exceptional set E, so

$$\Sigma = E, \qquad \pi(\Sigma) = \{\, x_1 = x_2 = 0 \,\},$$

the ramification locus is the x_3-axis. The fibre of π over a point of the axis other than the origin is the fibre of β, a circle; over the origin it is the cylinder $\{r = 0\}$ of (15). So β fails to be a diffeomorphism exactly on E.

Proposition (the branching feedback)**.** *Let $a, b, \eta, \rho_0 > 0$ with $b > 2a$, and write $r = (x_1^2 + x_2^2)^{1/2}$, $\nu(x) = (r^4 + x_3^2)^{1/4}$, $\rho(x) = x_3/r^2$. The branching feedback of the coupling (25) is*

$$u(x,\theta) = \begin{cases} \eta\,\nu(x)\,(x_1, x_2)/r & \text{if } r > 0,\ |x_3| > \rho_0 r^2, \\ \eta\,|x_3|^{1/2}(\cos\theta, \sin\theta) & \text{if } r = 0,\ x_3 \neq 0, \\ -a\,(x_1, x_2) - b\,\rho(x)\,(-x_2, x_1) & \text{if } r > 0,\ |x_3| \le \rho_0 r^2, \\ 0 & \text{if } x = 0, \end{cases} \tag{27}$$

where the branch $\theta \in S^1$ is the point of the circle $\beta^{-1}(x)$ over an axis point; off the axis there is one branch and u does not depend on θ. Thus $u(x,\theta) = h(z)$, with z the point of $\beta^{-1}(x)$ of angle θ, and u is the coupling on Z; it extends continuously to the axis within the outward-motion region and is continuous within each regime.

(i) *For every initial state $x(0)$ and branch $\theta(0)$ the closed loop has a unique lifted solution, $x(t) = \beta(z(t))$, where $z(t)$ is the solution of the reference from the point of $\beta^{-1}(x(0))$ of angle $\theta(0)$. The branch matters only if $x(0)$ is on the axis.*

(ii) *A solution that starts on the axis leaves it at once in the direction $\theta(0)$, and no solution reaches the axis at a later time. So the branch is chosen once, at $t = 0$, and never switched.*

The choice of branch. Every branch gives the same motion of r and x_3, the same times and the same estimates; the branch fixes only the direction in which the state leaves the axis. One may also fix one branch θ_0 for all axis points; this gives a feedback $u(x) = u(x,\theta_0)$ without a branch argument, with the same estimates. Uniqueness in (i) refers to the closed loop with its branch, that is, to solutions with a continuous lift to Z. For the static $u(x)$, a solution starting on the axis may leave it in any direction, with the same estimates. By Brockett's condition [26] no continuous $u(x)$ stabilises the origin, and indeed $u(x)$ is discontinuous at every point $x^* = (0,0,x_3)$, $x_3 \neq 0$, of the x_3-axis: $u(x^*) = \eta|x_3|^{1/2}(\cos\theta_0, \sin\theta_0)$, while at the points $x = (-\varepsilon\cos\theta_0, -\varepsilon\sin\theta_0, x_3)$, which tend to x^* as $\varepsilon \to 0$, $u(x)$ tends to $-u(x^*)$. The discontinuity at the axis is not seen along any solution.

Proposition (the estimates)**.** *In the setting of the previous Proposition, let $x(t)$ be a solution of the closed loop with the branching feedback (27), with $x(0) \neq 0$ and any branch $\theta(0)$, and write $u(t) = u(x(t),\theta(0))$. Put*

$$C = (1+\rho_0^{-2})^{1/4}, \qquad c = \max\Big\{\eta,\ \max_{|\rho|\le\rho_0} \sqrt{a^2+b^2\rho^2}\,(1+\rho^2)^{-1/4}\Big\}.$$

Then:

(i) *$|u(t)| \le c\,\nu(x(t)) \le cC\,\nu(x(0))$ — the control is bounded by the size ν, hence bounded on each neighbourhood of the origin and vanishing at it;*

(ii) *the region $\{|\rho| \le \rho_0\}$ is reached by the time*

$$T_{\mathrm{out}} = \frac{\tau(\rho_0)}{\eta}, \qquad \tau(\rho_0) = \int_0^{\rho_0^{-1/2}} \frac{d\xi}{(1+\xi^4)^{1/4}}, \tag{28}$$

which does not depend on $x(0)$, and is not left again;

(iii) *afterwards $\rho(t) \to 0$ exponentially at rate $b - 2a$, and (24) gives*

$$\nu(x(t)) \le C\,\nu(x(0))\,e^{-a(t-T_{\mathrm{out}})};$$

(iv) *consequently the attraction is uniform: given $\varepsilon > 0$, one has $\nu(x(t)) \le \varepsilon\,\nu(x(0))$ for all $t \ge T(\varepsilon) = \tau(\rho_0)/\eta + a^{-1}\log(C/\varepsilon)$, with $T(\varepsilon)$ independent of the initial point; in particular the* half-attraction time *$T(1/2) = \tau(\rho_0)/\eta + a^{-1}\log(2C)$, after which $\nu(x(t)) \le \frac{1}{2}\nu(x(0))$, is the same for every initial point, and each further halving takes at most $T(1/2)$;*

(v) *the path length $\int_0^\infty |u|\,dt$ and the energy $\int_0^\infty |u|^2\,dt$ are at most constant multiples of $\nu(x(0))$ and $\nu(x(0))^2$.*

The horns have played no part in this beyond labelling the two regimes; they are crossed, not preserved. What preserving them would give instead is recorded in §4.3 below.

The switching surface $|\rho| = \rho_0$ can be removed (Appendix A).

4.3. **The horns as invariant leaves.** The synthesis of §4.2 above crosses the horns. Here they are kept invariant instead, as the blow-up suggests: one reference for each horn. This gives two things the synthesis of §4.2 does not have, a rotation at a prescribed frequency and the clock used in §4.4 below.

A reference preserves every horn exactly when $\dot\rho = 0$, that is by (19) when $B = 2\rho A(1+\rho^2)^{-1/4}$. Write $\dot r = -\frac12 k\, r$ and $\dot x_3 = -k\, x_3$, which is this condition with $k = -2A(1+\rho^2)^{1/4}$. On the horn labelled ρ the state then turns and shrinks at the rates

$$\omega = \dot\theta = -k\rho, \qquad \frac{\dot\nu}{\nu} = -\frac{k}{2}, \qquad |u| = k\, r\sqrt{\tfrac14 + \rho^2}. \tag{29}$$

Since ρ is constant along the motion, k may be any positive function of ρ; the fastest one allowed by $|u| \le c\nu$ is

$$k(\rho) = \frac{2c\,(1+\rho^2)^{1/4}}{\sqrt{1+4\rho^2}}, \qquad k(\rho) \sim \frac{c}{|\rho|^{1/2}} \quad (\rho \to \pm\infty). \tag{30}$$

The horns alone do not give uniform attraction: the rate $k/2$ tends to 0 near the axis, and no other choice of k helps, because $|\dot x_3| = |x_1u_2 - x_2u_1| \le r|u|$ while on an invariant horn the radius r only decreases, so lowering the height of a state close to the axis takes a long time.

The outward motion or a rotation, then a horn. If $|\rho(0)| > \rho_0$, run the outward motion (23) until $|\rho| = \rho_0$; it lasts at most $\tau(\rho_0)/\eta$. If $|\rho(0)| < \rho_0$, rotate the state about the x_3-axis instead, with $u = \sigma c\nu\, e_\theta$, $\sigma = \operatorname{sign} x_3(0)$ (either sign if $x_3(0) = 0$: two branches). Then r is constant, $\dot x_3 = \sigma c\nu r$ and $\frac{d}{dt}|\rho| = c(1+\rho^2)^{1/4} \ge c$, so $|\rho| = \rho_0$ is reached within the time ρ_0/c, and ν grows at most by the factor $(1+\rho_0^2)^{1/4}$. At the first arrival at $|\rho| = \rho_0$ switch to the horn-preserving law: keep the state on the horn it has reached, with the rate (30), so $k = k(\rho_0)$ and the attraction is uniform:

$$\nu\big(x(t)\big) \le C\,\nu\big(x(0)\big)\, e^{-\frac12 k(\rho_0)\,(t-T_0)},$$

with

$$C = \max\big\{(1+\rho_0^{-2})^{1/4}, (1+\rho_0^2)^{1/4}\big\}, \quad T_0 = \max\big\{\tau(\rho_0)/\eta,\ \rho_0/c\big\}.$$

The law as a function of the state, the branch and a mode. Put $e_r = (x_1,x_2)/r$, $e_\theta = (-x_2,x_1)/r$, $\rho(x) = x_3/r^2$, $k = k(\rho_0)$ and, with a constant $\gamma > 0$, $q(x) = \rho(x) + (\gamma/k)\big(\rho(x) - \sigma\rho_0\big)$. The closed loop carries, besides the branch, a mode $m \in \{0,1\}$, which is switched by the switching rule $\beta_{\rm sw}$: from the present state $x(t)$ and the mode $m(t-0)$ just before the present moment, it gives the mode $m(t+0)$ just after it,

$$m(t+0) = \beta_{\rm sw}\big(x(t), m(t-0)\big), \qquad \beta_{\rm sw}(x,1) = 1, \qquad \beta_{\rm sw}(x,0) = \begin{cases} 1 & \text{if } r>0,\ |x_3| = \rho_0 r^2,\\ 0 & \text{otherwise,}\end{cases}$$

and $m(t_0) = \beta_{\rm sw}\big(x(t_0),0\big)$: m becomes 1 when the state first reaches $|\rho| = \rho_0$, which a continuous trajectory cannot jump over, and never returns to 0. The law is

$$u(x,\theta,\sigma,m) = \begin{cases} \eta\,\nu\, e_r & \text{if } m=0,\ r>0,\ |x_3| > \rho_0 r^2,\\ \eta\,|x_3|^{1/2}(\cos\theta, \sin\theta) & \text{if } m=0,\ r=0,\ x_3 \ne 0,\\ \sigma c\,\nu\, e_\theta & \text{if } m=0,\ r>0,\ |x_3| < \rho_0 r^2,\\ -k\big(\frac12(x_1,x_2) + q(x)\,(-x_2,x_1)\big) & \text{if } m=1,\\ 0 & \text{if } x = 0,\end{cases}$$

where the branch $\theta \in S^1$ is used over an axis point and $\sigma = \operatorname{sign} x_3$, with the two branches $\sigma = \pm1$ if $x_3 = 0$. Both are chosen at t_0 and never switched, $\theta(t+0) = \theta(t-0)$, $\sigma(t+0) = \sigma(t-0)$: a trajectory starting on the axis leaves it at once and does not return, and $x_3 \ne 0$ with the sign σ

for $t > t_0$. So m is the only switch. The horn law cannot be used on the surface $|x_3| = \rho_0 r^2$ alone. The outward motion and the rotation both push the state onto this surface from its two sides. An exact solution could follow the horn law on the surface, but with any error in measuring the state or in the switching, and for solutions in Filippov's sense, a law depending on x alone would alternate between the two near the surface (jitter), and the state would slide along the surface under a combination of the two motions. Both increase ν, since $\frac{d}{dt}\nu^4 = 4r^3\dot r + 2x_3\dot x_3$ is positive for each of them, so the sliding state moves away from the origin. The mode prevents this. It is not necessary for that: since the law for $m = 1$ makes the horn $\rho = \sigma\rho_0$ attracting (see below), it could be used without a mode on a layer $\rho_0 - \delta \le |\rho| \le \rho_0 + \delta$, whose edges the laws on both sides cross only inwards; with errors of size E in the measured ρ this needs $\gamma\delta > (k+\gamma)E$. The mode is nevertheless beneficial: the law changes only once, at the first arrival, and after that a disturbance never returns the state to the outward motion or the rotation. With errors in measuring the state, the switch takes place when the measured state first reaches or crosses the horn, and this ends the approach. Later crossings of the horn caused by errors do not change the law: the law for $m = 1$ is the same on both sides of the horn, depends continuously on x there and pulls the state back towards the horn, so the state only wobbles around it. Without the mode, on a layer, errors near its edges could switch the law back and forth during the entry. The mode, like the branch, is a part of the state of the closed loop and is constant between switches (§2.5); with it there is no jitter, since after the switch the law for $m = 1$ is the only one in use and the switching law no longer depends on x.

For $m = 1$ one has $\dot r = -\frac12 k\, r$ and $\dot\rho = -\gamma(\rho - \sigma\rho_0)$. Without errors the state stays on the horn $\rho = \sigma\rho_0$, where $q = \rho$, and moves there as in (29) with $k = k(\rho_0)$. The term with γ makes this horn attracting. Let ρ be measured with an error $e(t)$, $|e| \le E$; this is an error model for ρ itself, since bounded errors in x_3 and r need not give bounded errors in x_3/r^2 as $r \to 0$. Without the term ($\gamma = 0$) every horn is kept and none is preferred: $\dot\rho = -k\, e$, and the errors can accumulate. With $\gamma > 0$, $h = \rho - \sigma\rho_0$ obeys $\dot h = -\gamma h - (k+\gamma)e$, so, with t counted from the switch,

$$|h(t)| \le e^{-\gamma t}|h(0)| + \frac{k+\gamma}{\gamma}\, E\left(1 - e^{-\gamma t}\right) :$$

ρ stays in a band around $\sigma\rho_0$ of width of the order of E that does not grow with time, and the angular velocity $k|q|$ stays correspondingly close to $|\omega_*|$ below. Without errors $|u| \le \max\{\eta, c\}\,\nu$ along the whole motion, with $|u| = c\nu$ on the target horn; with errors the constant for $m = 1$ is enlarged according to the band.

Every trajectory ends on a horn $|\rho| = \rho_0$ and spirals into the origin with the constant angular velocity

$$|\omega_*| = k(\rho_0)\rho_0 = \frac{2c\,\rho_0\,(1+\rho_0^2)^{1/4}}{\sqrt{1+4\rho_0^2}}, \tag{31}$$

which increases from 0 to ∞ with ρ_0: every frequency is available, and the designer chooses it through ρ_0. In §4.2, by contrast, $\rho \to 0$ and the rotation dies out. The result is a decaying spiral of prescribed frequency; §5 below obtains a periodic motion of prescribed frequency by the same mechanism.

4.4. **A continuous time-periodic law from a rotating branch.** The branching feedback (27) has a circle of branches over the axis. A continuous feedback that does not depend on time cannot stabilise the plant, by Brockett's condition [26]; a time-dependent one is not subject to that condition. Here the branch used on the axis is made to rotate, in opposite directions on the upper and the lower half-axis, and inside a thin tube around the axis the branches are merged into one value. The result is a continuous time-periodic stabiliser. Outside the tube it is the

smooth field (54), so §4.2–§4.3 above supply most of the proof; only the analysis near the axis is new.

The sphere of shapes. The dilations $\delta_\lambda(x_1,x_2,x_3)=(\lambda x_1,\lambda x_2,\lambda^2 x_3)$, $\lambda>0$, carry each state $x\neq 0$ to exactly one point of the surface $S=\{\nu=1\}$, its *shape*. The surface S is a sphere with the poles $(0,0,\pm1)$ on the axis and the equator in the plane $x_3=0$. The laws of this section are homogeneous, $u(\delta_\lambda x)=\lambda u(x)$, so the closed loop commutes with the dilations and moves the shapes along a vector field on S, the *shape field*. At a zero of the shape field off the equator a state keeps θ and ρ; since $\dot{x}_3=r^2\dot{\theta}$, it then keeps x_3 and r, so it does not move. Take a continuous static law that equals (54) away from the axis. Its shape field is continuous on a cap $\{\rho\geq\rho_c\}$ around the north pole and, by $\dot{\rho}<0$ for (54), points out of the cap on its boundary circle. A vector field on a closed disk that points outward on the boundary vanishes inside, so the law has a state that does not move, as Brockett's condition requires. The blow-up replaces each pole by the circle of branches and turns S into a cylinder; on the cylinder the shape field of (54) vanishes only on the equator, where the states shrink in the plane $x_3=0$. This is why the branching feedback has no state that does not move. The law below is continuous, so the zero in the cap is there again, but now as a zero of the shape field in the frame that rotates with the branch: a state there keeps its shape and turns with the branch, $\dot{\theta}=\omega$, and (29) gives $\dot{\nu}/\nu=\omega/(2\rho)$. Turning against the sign of x_3 makes it shrink. The chart y in the proof (Appendix B below) is the chart of S at the north pole in the rotating frame; step (b) of the proof gives the boundary circle of the cap, step (c) treats the zeros inside it, and the Remark after the Theorem below is the case of the wrong direction of rotation.

The law. Let $A(\rho)$, $B(\rho)$ be the pair (54), with $\eta,a>0$ and $b>2a$. Put $w=r/\nu=(1+\rho^2)^{-1/4}\in[0,1]$, so that $w=0$ on the axis and $w=1$ on the plane $x_3=0$. Fix a tube width $w_1\in(0,1)$, a frequency $\omega_0>0$ and a phase φ_0, and let

$$\sigma(w)=3(w/w_1)^2-2(w/w_1)^3\ (w\leq w_1),\qquad \sigma(w)=1\ (w\geq w_1),$$

$$e_\pm(t)=\big(\cos(\mp\omega_0 t+\varphi_0),\,\sin(\mp\omega_0 t+\varphi_0)\big).$$

The *tube* is $\{w<w_1\}$. Define $u(0,t)=0$ and, for $x\neq 0$,

$$u(x,t)=\nu A(\rho)\Big[\sigma(w)\frac{(x_1,x_2)}{r}+\big(1-\sigma(w)\big)e_{\operatorname{sgn}x_3}(t)\Big]+\nu(1+\rho^2)^{1/4}B(\rho)\,\frac{(-x_2,x_1)}{r}. \tag{32}$$

Outside the tube $\sigma=1$ and (32) is the smooth field (54). On the axis it is $u=\eta\nu\,e_\pm(t)$: the branch of (27) used at time t is $\theta=\mp\omega_0 t+\varphi_0$, which rotates with period $P=2\pi/\omega_0$. The law is continuous: $e_\pm$ enters only inside the tube, where $x_3\neq 0$ keeps its sign; near the axis $\sigma(w)(x_1,x_2)/r=3w_1^{-2}r(x_1,x_2)/\nu^2-2w_1^{-3}r^2(x_1,x_2)/\nu^3$ tends to 0; and the last term equals $-b\,x_3r^2(-x_2,x_1)/\nu^4$. It is C^1 in $x\neq 0$, satisfies $|u|\leq(\eta+a+b)\nu$, and is homogeneous: $u(\lambda x_1,\lambda x_2,\lambda^2x_3,t)=\lambda u(x,t)$.

Theorem (continuous stabilisation by a rotating branch)**.** *For all $\eta,a>0$, $b>2a$, $w_1\in(0,1)$ and $\omega_0>0$, the closed loop (14), (32) has unique solutions defined for all $t\geq t_0$, and there are $K,\gamma>0$, independent of t_0 and $x(t_0)$, such that $\nu(x(t))\leq Ke^{-\gamma(t-t_0)}\nu(x(t_0))$ for $t\geq t_0$.*

The proof is in Appendix B below.

Remark (the opposite rotation is necessary). If both half-axes rotate the same way, $\varphi(t)=\omega t+\varphi_0$, the circle of step (b) of the proof in Appendix B is unchanged, since ℓ does not involve ω. The field has index one on it, so it has an equilibrium inside, and there $\dot{\nu}/\nu=\omega/(2\rho)$. In the half-space where $\omega x_3>0$ this is positive: the corresponding solution grows exponentially, and by the dilations such solutions start arbitrarily close to the origin, which is therefore neither stable nor attractive. The sign rule "rotate against the sign of ρ" is the clock $\omega=-k\rho$ of (29).

For a frozen time the law (32) is a continuous selection of the branching feedback away from the axis, smoothed across it; the time dependence serves only to move the selected branch around the circle of branches. With $\omega_0 = 0$ the smoothed law is static and continuous, and it acquires the spurious equilibria that Brockett's condition predicts.

5. Control of frequency

We consider the plant (1) at a working point where the linearisation (A, C) is uncontrollable, with a critical oscillatory pair $\pm i\omega_0$ ($\omega_0 \geq 0$) among the uncontrollable modes. Smooth feedback leaves the uncontrollable eigenvalues $\pm i\omega_0$ in the spectrum of the closed-loop linearisation, so a Hopf cycle born from that pair under smooth feedback has onset frequency ω_0. The *control-of-frequency* problem is to install by feedback an attracting cycle on which $x_<$ moves at a prescribed angular velocity. Three frequencies must be kept apart. The reference cycle has angular frequency ω. The coupling manifold is $x_< = z^2$, so $x_< = z^2$ moves at angular velocity 2ω (at -2ω for $x_< = \bar{z}^2$). The whole closed-loop orbit, control included, has period $2\pi/\omega$. The Theorem of §5.3 below assigns the angular velocity $\pm 2\omega$ of $x_<$, for every $\omega > 0$ except $2\omega = \omega_0$ in the rotation $+2$, when the quadratic part Q of the control has no folds. §5.2 below records what a *linear* coupling gives at second order, and the Theorems of §5.4 and §5.5 below show that a linear coupling with a cubic friction term realises every frequency that passes that test, whether or not Q has folds, for the plane system and for the full plant, up to a few resonances.

5.1. **The construction.** Consider the plant (1) at a working point $x = 0$, $u = 0$ where the linearisation (A, C) is uncontrollable at a critical oscillatory pair $\pm i\omega_0$ ($\omega_0 \geq 0$). The Kalman decomposition of (A, C) splits the state, $x = (x_<, x_>)$, into the uncontrollable linearly uncontrollable part $x_< \in \mathbb{C} \cong \mathbb{R}^2$, carrying that pair, and a controllable complement $x_>$, bringing (1) to the normal form

$$\dot{x}_< = i\omega_0 x_< + Q(u, x), \qquad \dot{x}_> = A_> x_> + A_{><} x_< + C_> u + R(u, x), \tag{33}$$

in which $u \in \mathbb{R}^\ell$ is the *full* control, and $Q(u, x)$, $R(u, x)$ collect the terms of order ≥ 2 in the two equations — Q two-dimensional, valued in the critical plane. (The Kalman form allows the linear coupling $A_{><} x_<$ in the controllable block; the critical block receives nothing linear.) The linearly uncontrollable part $x_<$ turns at $\pm i\omega_0$ under *no* linear control — so the control reaches the critical mode only through Q — while $(A_>, C_>)$ is controllable. We assume

- (H1) the uncontrollable subspace is two-dimensional with critical pair $\pm i\omega_0$ (for $\omega_0 = 0$ the uncontrollable block is assumed to be 0), and the quadratic-in-u part of Q not identically zero;
- (H2) the complement $(A_>, C_>)$ is controllable, and $\dim_{\mathbb{R}} \ker C_> = 2$.

Remark (the dimensions here need not be minimal). Both dimensions above are fixed at their smallest values, not their only possible ones. The uncontrollable subspace of (H1) is taken to be exactly two-dimensional, carrying just the pair $\pm i\omega_0$; it could instead be higher-dimensional, with $\pm i\omega_0$ appearing as one invariant block among others. Likewise $\dim_{\mathbb{R}} \ker C_>$ in (H2) is fixed at 2; it could instead exceed 2. Neither possibility is pursued here.

Assume $x_> = 0$ and let u range over $\ker C_>$, identified with $\mathbb{C}$ by (H2) — chosen so as not to disturb the complement once it is switched back on in §5.3 — in the first equation $\dot{x}_< = i\omega_0 x_< + Q(u, x)$ of (33); such u leaves the complement equation untouched, since $C_> u = 0$. Redefine $x_<$ as x. Then Q, in the control alone, is the general $\mathbb{C}$-valued real quadratic

$$Q(u) = G\, u^2 + H\, |u|^2 + K\, \bar{u}^2 \qquad (G, H, K \in \mathbb{C}).$$

The terms $u\cdot x$, x^2 and those of order three are of higher order in the reference variable below ($u\sim z$, $x\sim z^2$), and the *plane system* is

$$\dot x = i\omega_0 x + Q(u), \qquad x,u\in\mathbb{C}, \tag{34}$$

uncontrollable at its linearisation ($\partial_u Q|_0=0$). For $\omega_0=0$ and $Q(u)=u^2$ it is the model (6) with $n=2$; for $\omega_0=0$ and other Q it need not resemble that model (for $Q(u)=|u|^2$, $\operatorname{Re}x$ never decreases and $\operatorname{Im}x$ is constant). The full plant is taken up again in §5.3.

Fix $\omega>0$ and take for reference the Andronov–Hopf system

$$\dot z = w(z) := (\varepsilon+i\omega)z - |z|^2 z, \qquad z\in\mathbb{C}\cong\mathbb{R}^2,\quad \varepsilon>0, \tag{35}$$

whose attracting cycle $|z|^2=\varepsilon$ carries $\dot z = i\omega z$. The first coupling to try is the real-linear $p(z)=bz+b'\bar z$. With $Q(p(z))=d_{20}z^2+d_{11}z\bar z+d_{02}\bar z^2$,

$$d_{20}=Gb^2+Hb\bar b'+K\bar b'^2,\quad d_{11}=2Gbb'+H(|b|^2+|b'|^2)+2K\bar b\bar b',\quad d_{02}=Gb'^2+H\bar b b'+K\bar b^2, \tag{36}$$

the invariance equation $i\omega_0 F+Q(p(z))=F_z\dot z+F_{\bar z}\dot{\bar z}$ for a graph $x=F(z)=\sum_{m+n\ge2}c_{mn}z^m\bar z^n$ is solved order by order; at second order

$$c_{20}=\frac{d_{20}}{2\varepsilon+i(2\omega-\omega_0)},\qquad c_{11}=\frac{d_{11}}{2\varepsilon-i\omega_0},\qquad c_{02}=\frac{d_{02}}{2\varepsilon-i(2\omega+\omega_0)}, \tag{37}$$

and §5.2 below uses their limits at $\varepsilon=0$,

$$c_{20}=\frac{d_{20}}{i(2\omega-\omega_0)},\qquad c_{11}=\frac{i\,d_{11}}{\omega_0},\qquad c_{02}=\frac{d_{02}}{-i(2\omega+\omega_0)}, \tag{38}$$

which require $2\omega\neq\omega_0$ and $\omega_0>0$.

Remark (a linear coupling does not give an invariant graph). The order-by-order solution is formal: the graph it defines at the origin need not extend to the reference cycle. Take $\dot x = i\omega_0 x+u^2$ and $u=z$, with $\omega_0>0$, $\omega_0/\omega\notin\mathbb{N}$ and $\delta:=2\omega-\omega_0\neq0$. Then there is no continuous invariant graph $x=F(z)$ with $F(0)=0$ over a disc containing the reference cycle (Appendix C.1 below). The reason is seen in the product system $(z,x)\in\mathbb{C}^2$. Over the reference cycle $\{|z|^2=\varepsilon\}$ it has the closed orbit Γ_0, the periodic solution $x=(\varepsilon/i\delta)e^{2i\omega t}$, surrounded by nested invariant tori, none of which attracts. The trajectory leaving the origin does not arrive on Γ_0: it arrives on the torus of radius $\pi/(2|\sinh(\pi\delta/(2\varepsilon))|)$ around it and winds on it forever with the two frequencies ω and ω_0. This defect is exponentially small in $1/\varepsilon$ and invisible at every finite order. The same holds for every linear coupling.

The tori exist because the critical mode of the plant has no friction: the term $i\omega_0 x$ only rotates, and a coupling that depends on z alone adds none. A coupling that also depends on x can supply it. Two such constructions follow. In §5.3 below the graph is prescribed and the coupling is solved for pointwise; the closed loop $\dot x_<=[2(\varepsilon-|x_<|)+2i\omega]x_<$ has the friction towards the circle built in. In §5.4 below the linear coupling is kept and a cubic term multiplying x is added; it turns the family of tori into the basin of one attracting cycle.

5.2. **The second-order test for a linear coupling.** This subsection asks at which frequencies the quadratic part of the formal graph of a linear coupling is a nondegenerate double point. By the second Remark of §5.1 above this is a test on second-order data, not the existence of a synthesis; the existence theorems are in §5.3 and §5.4 below. Throughout, $\omega_0>0$ and the coefficients are those of (38).

Let $F_2(z)=c_{20}z^2+c_{11}z\bar z+c_{02}\bar z^2$. Its branch point at $z=0$ is nondegenerate exactly where $|\partial_z F_2|^2-|\partial_{\bar z}F_2|^2$ is sign-definite on $z\neq0$. Writing $z=re^{i\theta}$ and expanding $\partial_z F_2 = r(2c_{20}e^{i\theta}+c_{11}e^{-i\theta})$, $\partial_{\bar z}F_2=r(c_{11}e^{i\theta}+2c_{02}e^{-i\theta})$ gives $|\partial_z F_2|^2-|\partial_{\bar z}F_2|^2=r^2\varphi(\theta)$ with

$$\varphi(\theta)=4\big(|c_{20}|^2-|c_{02}|^2\big)+4\operatorname{Re}\big(Ae^{2i\theta}\big),\qquad A:=c_{20}\bar c_{11}-c_{11}\bar c_{02}, \tag{39}$$

so φ is sign-definite over all θ exactly when

$$|c_{20}|^2 - |c_{02}|^2 > |A| \tag{40}$$

(rotation $+2$; the reverse inequality with the roles of c_{20}, c_{02} exchanged gives rotation -2, and reduces to the simpler $|c_{20}| > |c_{02}|$ when $c_{11} = 0$). We call such ω *reachable at second order* and write Ω for their set.

Remark (three outcomes at any order of approximation). At any order of Taylor approximation of the plant, a system falls into one of three classes: those for which qualitative-behaviour branching control can already be built from the data available at that order; those for which it can already be shown impossible at that order (by some genuine obstruction; for instance, for a real positive-definite pair $Q_1(u_1, u_2), Q_2(u_1, u_2)$ (or in complex notation, $Q(u)$), no circular cycle at any frequency other than ω_0 can be built this way — not pursued further here); and a remaining border class, where the data at that order decides nothing either way, and only a higher-order approximation could. (40) is a test at second order: if it holds, F_2 is a branched double cover of a neighbourhood of $x = 0$.

The quantities c_{20}, c_{02}, A in (40) all depend on the coupling b, b', which is ours to choose; the plant's own coefficients G, H, K and the target frequency ω do not. So the question is: for given G, H, K, ω, does *some* choice of b, b' satisfy (40)? Eliminating b, b' turns this into a condition on G, H, K alone, and that condition takes the shape of a region in the complex plane where H (as a complex number) can sit — an ellipse built from G, K:

$$\frac{\xi^2}{(|G| + |K|)^2} + \frac{\eta^2}{(|G| - |K|)^2} \le 1, \qquad \xi + i\eta := H\, e^{-i \arg(GK)/2}, \tag{41}$$

centred at 0 with semi-axes $|G| + |K|$ and $\big||G| - |K|\big|$ along and across the direction $\arg(GK)/2$, and foci $\pm 2\sqrt{GK}$. The formula (41) applies when $GK \neq 0$ and $|G| \neq |K|$. If $GK = 0$, the angle $\arg(GK)$ is undefined, the foci coincide at 0, and the region is the disc $|H| \le |G| + |K|$. If $|G| = |K| \neq 0$, the second denominator vanishes and the region is the segment joining the foci, $\{\pm 2t\sqrt{GK} : 0 \le t \le 1\}$.

With $\Pi := |H^2 - 4GK|$ and $D := |H|^2 + \Pi - 2(|G|^2 + |K|^2)$, the point H lies strictly inside the ellipse exactly when $D < 0$, on it when $D = 0$, and strictly outside it when $D > 0$.

Strictly inside the ellipse both rotations occur: every frequency but the resonance is reachable at second order with rotation $+2$, and also with rotation -2. On or outside the ellipse only rotation $+2$ occurs: no linear coupling reverses the rotation inherited from the uncontrollable plant mode $+i\omega_0$. So for ω/ω_0 (the resonance at $\frac{1}{2}$), the set Ω is as follows.

Inside or on the ellipse: no ceiling. If H lies inside (41) or on it, every frequency but the resonance is reachable at second order,

$$\Omega = \mathbb{R}_{>0} \setminus \{\tfrac{1}{2}\}, \tag{42}$$

with one exception. When $|G| = |K|$ the ellipse degenerates to the segment joining its foci $\pm 2\sqrt{GK}$, and at these two foci themselves Ω is only the floor (43) below (example: $G = K = 1$, $H = -2$, that is, $Q(u) = -4(\operatorname{Im} u)^2$). Everywhere else on the closed ellipse there is no ceiling, meaning the ceiling U of (45) below is $+\infty$. Strictly inside the ellipse there is moreover a linear coupling with $d_{11} = 0$ and $|d_{02}| < |d_{20}|$; outside it no linear coupling has $d_{11} = 0$ and $|d_{20}| \neq |d_{02}|$.

The universal floor. For every nontrivial Q, whatever the position of H, the band

$$\big(0, \tfrac{1}{\sqrt{2}}\big) \setminus \{\tfrac{1}{2}\} \subseteq \Omega \tag{43}$$

is always reachable at second order. The floor is the whole of Ω exactly when $|H|^2 - 2(|G|^2 + |K|^2) = |H^2 - 4GK|$: this holds for $G = K = 0$, at the two foci of a degenerate ellipse, and at the

points strictly outside the ellipse where the ceiling below equals $1/\sqrt{2}$ (for example $G = K = 1$, $H = -\frac{5}{2}$). In every other case Ω is larger than the floor.

Strictly outside the ellipse: the ceiling. If H lies strictly outside the ellipse, Ω is a bounded segment whose only interior gap is the resonance,

$$\Omega = (0, U) \setminus \{\tfrac{1}{2}\}, \tag{44}$$

with ceiling U given, in the coefficients G, H, K, by

$$U = \frac{1}{2}\sqrt{\frac{|H|^2 + 3\,\Pi - 2\big(|G|^2 + |K|^2\big)}{|H|^2 + \Pi - 2\big(|G|^2 + |K|^2\big)}}, \qquad \Pi = |H^2 - 4GK|. \tag{45}$$

On the ellipse itself the denominator $|H|^2 + \Pi - 2(|G|^2 + |K|^2)$ vanishes identically while the numerator equals 2Π there. Π vanishes only at the foci, which lie strictly inside the ellipse when $|G| \neq |K|$ and are the two ends of the segment when $|G| = |K|$. So $U \to +\infty$ approaching the ellipse from outside, independently of direction, at every point of the ellipse other than these two ends: there is no ceiling on the ellipse itself, exactly as inside it, apart from the exception noted above.

The proof is in Appendix C.2 below.

Strictly inside the ellipse the Theorem of §5.3 below realises every frequency with both rotations by a nonlinear coupling, except $2\omega = \omega_0$ in rotation $+2$. On and outside the ellipse the Jacobian of Q vanishes on some line through 0 (folds in general, degenerate critical points in special cases) and that Theorem does not apply; there the Theorem of §5.4 below realises every frequency of Ω, except $\omega/\omega_0 \in \{\frac{1}{4}, \frac{1}{2}\}$, by a linear coupling with a cubic friction term. So, for the plane system, Ω is the set of realised frequencies up to these resonances. The three cases — inside, on and outside the ellipse — are stated together in the Corollary of §5.5 below, where the Theorem of §5.4 is also extended to the full plant.

5.3. **Prescribed frequency across uncontrollability.** We now prescribe the graph and solve for the coupling. The graph is $x_< = z^2$, the order-two branch point; the graph $x_< = \bar{z}^2$ reverses the orientation and is treated in the same way. For the reference (35), $x_< = z^2$ obeys $\frac{d}{dt} z^2 = 2zw(z) = 2(\varepsilon - |z|^2 + i\omega)z^2$, that is,

$$\dot{x}_< = V(x_<) := \big[2(\varepsilon - |x_<|) + 2i\omega\big]x_<. \tag{46}$$

This field has a repelling equilibrium at $x_< = 0$ and an attracting circle $|x_<| = \varepsilon$ traversed at angular velocity 2ω. The coupling must make the critical equation of (33) equal to (46). The requirement is a pointwise equation for the control, solvable when Q has no folds:

(H3) $|G| \neq |K|$ and $\big|K\bar{H} - \bar{G}H\big| < \big||G|^2 - |K|^2\big|$.

The Jacobian of Q is

$$|Q_u|^2 - |Q_{\bar{u}}|^2 = 4\Big[\big(|G|^2 - |K|^2\big)|u|^2 + \mathrm{Re}\big((G\bar{H} - H\bar{K})u^2\big)\Big], \tag{47}$$

so (H3) says exactly that the Jacobian is strictly positive everywhere or strictly negative everywhere on $u \neq 0$: Q has no critical rays. (H3) holds exactly when H lies strictly inside the ellipse (41), that is, when $D < 0$. Indeed, expanding both sides gives

$$\big(|G|^2 - |K|^2\big)^2 - \big|K\bar{H} - \bar{G}H\big|^2 = \tfrac{1}{4}\,D\,(D - 2\Pi),$$

and $2\Pi - D \geq 2(|G| - |K|)^2 \geq 0$, because $\Pi \geq |H|^2 - 4|GK|$; so the left side is positive exactly when $D < 0$. (H3) excludes, for instance, $Q(u) = |u|^2$.

Lemma (a quadratic without folds is a double covering)**.** *Under* (H3), Q *maps* $\mathbb{C}\setminus\{0\}$ *onto* $\mathbb{C}\setminus\{0\}$, *every* $\vartheta\neq 0$ *has exactly the two preimages* $\pm u$, *the curve* $\theta\mapsto Q(e^{i\theta})$ *winds twice around* 0, *and* $\|dQ(u)^{-1}\|\le C_Q/|u|$ *for a constant* C_Q.

The proof is in Appendix C.3 below.

Close the complement as before: $u = L\cdot x_> + p$ with $p\in\ker C_>\cong\mathbb{C}$ and the gain $L\colon x_>\to W$, W a complement to $\ker C_>$ in $\mathbb{R}^\ell$, chosen with $A_L := A_> + C_>L$ Hurwitz (possible by (H2)). Write the critical equation of (33) as

$$\dot x_< = i\omega_0 x_< + Q(p) + T(p,x), \qquad T(p,x) := Q(Lx_> + p,\, x) - Q(p). \tag{48}$$

Since the plant is of class C^3 near 0, $Q(u,x)$ has order ≥ 2 and its part quadratic in u alone is, on $\ker C_>$, the form $Q(p)$,

$$|T| \le C_T\big(|x|(|p|+|x|)+|p|^3\big), \quad |\partial_p T| \le C_T\big(|x|+|p|^2\big), \quad |\partial_x T| \le C_T\big(|p|+|x|\big) \tag{49}$$

for small (p,x). With $\delta := 2\omega-\omega_0$, the critical equation equals (46) exactly when

$$Q(p) + T(p,x) = \big[2(\varepsilon - |x_<|) + i\delta\big]x_<. \tag{50}$$

This is an equation for the value p at each state x, not a differential equation, and no homological divisors enter it.

Theorem (prescribed frequency, H strictly inside the ellipse)**.** *This Theorem is the case of* H *strictly inside the ellipse* (41), *which is the case that* Q *has no folds, i.e.* (H3). *Assume that the plant is of class* $C^{\max(k,3)}$, $k\ge 1$, *and* (H1)–(H3), *and let* $\omega>0$ *with* $\delta = 2\omega-\omega_0\neq 0$. *There are* $\varepsilon_0>0$ *and* $K_0>0$ *such that for* $0<\varepsilon<\varepsilon_0$ *the following hold on the region*

$$\mathcal{N}_\varepsilon = \big\{x:\ |x_<|\le 2\varepsilon,\ |x_>|\le K_0|x_<|\big\}.$$

(i) *For* $x\in\mathcal{N}_\varepsilon$ *with* $x_<\neq 0$, (50) *has exactly two solutions* $p_0(x), p_1(x)$ *with* $|p|\le R_0|x_<|^{1/2}$ (R_0 *a constant*). *They are* C^k *in* x, *tend to* 0 *as* $x\to 0$, *and are exchanged by continuation once around* $x_<=0$. *The branching feedback is* $u(x,j) = Lx_> + p_j(x)$, $j\in\{0,1\}$; *its ramification locus in* $\mathcal{N}_\varepsilon$ *is the origin.*

(ii) *Label the branch by* z *with* $z^2 = x_<$. *On the lifted region* $\tilde{\mathcal{N}}_\varepsilon = \{(z,x_>) : |z|^2\le 2\varepsilon,\ |x_>|\le K_0|z|^2\}$, *which is positively invariant, the closed loop is the skew product*

$$\dot z = w(z), \qquad \dot x_> = A_L x_> + A_{><}z^2 + R\big(Lx_> + p(z,x_>),\, z^2,\, x_>\big). \tag{51}$$

So along every closed-loop solution in $\mathcal{N}_\varepsilon$ $x_<$ *is exactly* $x_<(t) = z(t)^2$, *with* $z(t)$ *a solution of the reference* (35) *and time preserved. The solution from* $x=0$ *stays at* 0.

(iii) *The lifted closed loop* (51) *has a periodic solution* γ *of period* $2\pi/\omega$ *on which* $x_<(t) = \varepsilon e^{2i(\omega t+\theta_0)}$: $x_<$ *traverses the circle* $|x_<|=\varepsilon$ *at angular velocity* 2ω, *twice per period of* γ. *The orbit* γ *is hyperbolic and orbitally asymptotically stable with asymptotic phase, lies in the interior of* $\mathcal{N}_\varepsilon$, *and every solution in* $\mathcal{N}_\varepsilon$ *with* $x_<\neq 0$ *tends to it.*

(iv) *Along* γ *the control* $u(t)$ *and the complement* $x_>(t)$ *are periodic with period* $2\pi/\omega$, *while* $x_<(t)$ *has period* π/ω.

With the graph $x_< = \bar z^2$ *the same holds with angular velocity* -2ω *and* δ *replaced by* $-(2\omega+\omega_0)$, *which never vanishes. The threshold* ε_0 *tends to* 0 *as* $\delta\to 0$.

The region $\mathcal{N}_\varepsilon$ is a positively invariant cone, not a neighbourhood of the origin: states with $|x_>| > K_0|x_<|$, in particular those with $x_< = 0\neq x_>$, are outside it, and the Theorem says nothing about them.

The proof is in Appendix C.4 below.

No resonance $j\omega=\omega_0$ with $j\neq 2$ enters: the coupling solves the pointwise equation (50), not a sequence of homological equations. The excluded value $\delta=0$ is the one at which the target

field minus the plant's own rotation, $2(\varepsilon - |x_<|)x_<$, vanishes on the whole circle $|x_<| = \varepsilon$: the cycle is then the plant's own oscillation at ω_0, and $p \sim (\varepsilon - |x_<|)^{1/2}$ is not Lipschitz there.

In the language of §2.5, with the reference state augmented to (z, η), where η is a copy of $x_>$ obeying the same equation, the coupling has coupling manifold $M = \mathrm{graph}(x = F(z,\eta))$, $F(z,\eta) = (z^2, \eta)$, and coupling $h(x,z) = Lx_> + p(z, x_>)$: by (50) the set $\{x_< = z^2\}$ is invariant for the product system, and the projection $z \mapsto z^2$ carries the reference onto $x_<$, so $\Sigma = \{z = 0\}$ and $\pi(\Sigma) = \{x_< = 0\}$. Take a ray Γ from $x_< = 0$ as the line across which the branch label is changed: the cycle encircles the branch point, meets Γ once per revolution of $x_<$, and the label changes there. The realised control $u(t)$ is smooth in time, since $z(t)$ rotates smoothly.

5.4. **Linear coupling with friction: every frequency that passes the test.** The test of §5.2 above concerns a linear coupling, and by the second Remark of §5.1 above such a coupling leaves the plant's critical mode without friction: over the reference cycle the product system carries a family of neutral tori, and the graph need not close. A cubic term of the coupling that multiplies the plant state supplies the friction. With it, every frequency that passes the test is realised, on, inside and outside the ellipse. This subsection treats the plane system (34).

For the linear part $u_0 = bz + b'\bar z$ write

$$Q_u(u_0) = 2Gu_0 + H\bar u_0 = \kappa_1 z + \kappa_0 \bar z, \qquad Q_{\bar u}(u_0) = Hu_0 + 2K\bar u_0 = \lambda_1 z + \lambda_0 \bar z,$$

that is $\kappa_0 = 2Gb' + H\bar b$, $\kappa_1 = 2Gb + H\bar b'$, $\lambda_0 = Hb' + 2K\bar b$, $\lambda_1 = Hb + 2K\bar b'$. The coupling is

$$u = h(x,z) := bz + b'\bar z + (cz + c'\bar z)\,x, \qquad c, c' \in \mathbb{C}, \tag{52}$$

and its *friction* is $\kappa := c\kappa_0 + c'\kappa_1$. With the coefficients (38) put

$$\hat F_2(\nu) := c_{20}\nu + c_{11}|\nu| + c_{02}\bar\nu, \qquad \text{so that} \qquad F_2(z) = \hat F_2(z^2).$$

Theorem (prescribed frequency, H on or outside the ellipse: friction)**.** *This Theorem is the one for H on the ellipse (41) and for H strictly outside it, where Q has folds and the Theorem of §5.3 above does not apply, for the plane system; the full plant is treated in §5.5 below. Inside the ellipse it applies as well. Consider $\dot x = i\omega_0 x + Q(u)$ with $\omega_0 > 0$. Let $\omega > 0$ with $2\omega \neq \omega_0$ and $4\omega \neq \omega_0$, let (b, b') reach ω with rotation $\sigma \in \{+2, -2\}$ (§5.2), and let c, c' satisfy*

$$\mathrm{Re}\,\kappa < -2, \qquad \text{and} \qquad c'\lambda_1 = 0 \quad \text{if} \quad \omega = \omega_0.$$

There are $\varepsilon_0 > 0$ and $C > 0$ such that for $0 < \varepsilon < \varepsilon_0$:

(i) *there is a homeomorphism Φ_ε of the disc $\{|\nu| \le 2\varepsilon\}$ onto a neighbourhood $\mathcal{N}_\varepsilon$ of $x = 0$, a C^1 diffeomorphism off $\nu = 0$, preserving orientation if $\sigma = +2$ and reversing it if $\sigma = -2$, with*

$$|\Phi_\varepsilon(\nu) - \hat F_2(\nu)| \le C\varepsilon|\nu|, \qquad \|D\Phi_\varepsilon(\nu) - D\hat F_2(\nu)\| \le C\varepsilon;$$

(ii) *for $x \in \mathcal{N}_\varepsilon$ let $\pm z$ be the two square roots of $\Phi_\varepsilon^{-1}(x)$. The coupling (52) gives the two-valued feedback $u(x,\pm) = h(x,\pm z) = \pm h(x,z)$, ramified at $x = 0$; both values give the same closed loop $\dot x = i\omega_0 x + Q(u)$, and Φ_ε conjugates*

$$\dot\nu = V(\nu) = \big[2(\varepsilon - |\nu|) + 2i\omega\big]\nu$$

to it on $\mathcal{N}_\varepsilon$;

(iii) *consequently the closed loop has a hyperbolic attracting periodic orbit $\gamma = \Phi_\varepsilon(\{|\nu| = \varepsilon\})$ of period π/ω, around which x turns once per period in the sense of σ, so at average angular velocity $\sigma\omega = \pm 2\omega$; every solution in $\mathcal{N}_\varepsilon \setminus \{0\}$ tends to γ, and the origin repels. Along γ the control has period $2\pi/\omega$. The curve γ lies within $C\varepsilon^2$ of $\hat F_2(\{|\nu| = \varepsilon\})$; in general it is not a circle.*

The hypotheses can always be met when $\omega \in \Omega$, $2\omega \neq \omega_0$ and $4\omega \neq \omega_0$. The test is an open condition on (b, b'), and $\kappa_0 = 2Gb' + H\bar{b}$ vanishes only on a real subspace of codimension two unless $G = H = 0$; so (b, b') may be taken with $\kappa_0 \neq 0$, and then $c' = 0$, $c = -3/\kappa_0$ gives $\kappa = -3$ and $c'\lambda_1 = 0$. For $Q(u) = K\bar{u}^2$ the Theorem applies to the control $\bar{u}$. Together with the classification of Ω in §5.2 above: for the plane system, a linear coupling with friction realises every frequency of Ω other than $\omega/\omega_0 = \frac{1}{4}$ and $\frac{1}{2}$. At these two values the Theorem does not apply: a component of the forcing turns at the plant's own frequency ω_0, and its response is limited only by the friction, which is of size ε. Whether these two values can be realised by other means is not decided here; for some plants nothing goes wrong at them (for $Q(u) = u^2$ and the coupling $u = z + \gamma\bar{z}x$ there is no forcing at the fourth harmonic); at $\omega = \omega_0$ a term $\vartheta^2\bar{\zeta}$ of the normal form below would be resonant, which the choice $c' = 0$ avoids.

The proof is in Appendix C.5 below.

The two constructions of frequency control differ in what is prescribed. In §5.3 above the graph is prescribed, $x_< = z^2$, and the coupling is solved for pointwise; this needs Q without folds, covers the full plant (33), and gives the exact circle $|x_<| = \varepsilon$. Here the coupling is prescribed, linear plus friction, and the graph is found; this needs only the second-order test, so it reaches the plants on and outside the ellipse, and the cycle is a curve close to $\hat{F}_2(\{|\nu| = \varepsilon\})$. The Theorem above is stated for the plane system. For the full plant the terms of $Q(u, x)$ of the form $u \cdot x$ and u^3 enter (64) at order $\sqrt{\varepsilon}$; a second averaging step then changes the effective friction κ by a constant and adds the resonances $3\omega = \omega_0$, $\omega = \omega_0$ and $\omega = 2\omega_0$. That extension is the subject of §5.5 below.

5.5. **The full plant.** The Theorem of §5.4 above is stated for the plane system. This subsection extends it to the full plant (33) of any dimension N. Two things change. The complement $x_>$ is now present, and the plant has its own terms of order two and three in (u, x) beyond $Q(u)$, such as $u \cdot x$ and u^3. In the scaled variables these terms enter at order $\sqrt{\varepsilon}$, which is larger than the friction; their average vanishes, but at second order they shift the friction by a constant and add three resonances. They also break the symmetry $z \mapsto -z$, so the cycle of $x_<$ is a closed curve winding twice around 0 in one period $2\pi/\omega$, no longer a curve run twice.

The coupling is

$$u = Lx_> + p, \qquad p = bz + b'\bar{z} + (cz + c'\bar{z})\, x_< \in \ker C_>, \tag{53}$$

with L as in §5.3 above ($A_L = A_> + C_>L$ Hurwitz), and $\kappa = c\kappa_0 + c'\kappa_1$ as in §5.4 above.

Theorem (prescribed frequency for the full plant, H on or outside the ellipse)**.** *Assume that the plant* (33) *is of class* C^5, (H1), (H2), $\omega_0 > 0$, *and let* $\omega > 0$ *with*

$$\omega/\omega_0 \notin \{\tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{2}, 1, 2\}.$$

Let (b, b') *reach* ω *with rotation* $\sigma \in \{+2, -2\}$ *for the quadratic part* $Q(u) = Gu^2 + H|u|^2 + K\bar{u}^2$ *on* $\ker C_>$ (*§5.2*). *There is a constant* $\kappa_2 \in \mathbb{C}$, *determined by the terms of order two and three of the plant, by* A_L, $A_{><}$, L *and by* (b, b'), *and independent of* c, c', *such that if*

$$\operatorname{Re}(\kappa + \kappa_2) < -2,$$

then for all small $\varepsilon > 0$*:*

(i) *the product system of the plant* (33) *under* (53) *and the reference* (35) *has an invariant surface* $\Sigma = \{x = (Y_<(z), Y_>(z)) : |z|^2 \leq 3\varepsilon\}$, *with* $Y_<(z) = F_2(z) + O(\varepsilon^{3/2})$, $DY_<(z) = DF_2(z) + O(\sqrt{\varepsilon}|z|)$, $Y_>(z) = O(\varepsilon)$. *The map* $Y_<$ *is a branched double covering ramified at* $z = 0$*: every* $x_< \neq 0$ *with* $|x_<| \leq r_\varepsilon$, r_ε *of order* ε, *has exactly two preimages, which are exchanged by continuation once around* $x_< = 0$*;*

(ii) *the branching feedback* $u(x,z) = Lx_> + p(z,x_<)$*, with branch* z *a preimage of* $x_<$ *under* $Y_<$*, is defined on the region*

$$\mathcal{N}_\varepsilon = \big\{(x,z):\ x_< = Y_<(z),\ |z|^2 \le 2\varepsilon,\ |x_> - Y_>(z)| \le K_0|z|^2\big\},$$

and solutions of the closed loop starting in $\mathcal{N}_\varepsilon$*, with the branch carried continuously, stay in the same set with* $|z|^2 \le 3\varepsilon$*; along them the branch obeys* $\dot z = w(z) + O(|x_> - Y_>(z)|)$ *and* $|x_> - Y_>(z)|$ *decays exponentially;*

(iii) *the closed loop with its branch has the hyperbolic attracting periodic orbit* $\{|z|^2 = \varepsilon\}$ *of period* $2\pi/\omega$*; its projection* $\gamma = \{(Y_<(z), Y_>(z)) : |z|^2 = \varepsilon\}$ *is periodic with that period, possibly with a smaller minimal period, and along it* $x_<$ *winds* σ *times around* 0*, i.e. turns at average angular velocity* $\sigma\omega$*; every solution in* $\mathcal{N}_\varepsilon$ *with* $z \neq 0$ *tends to* γ*. The curve* $x_<(\gamma)$ *lies within* $C\varepsilon^{3/2}$ *of* $F_2(\{|z|^2 = \varepsilon\})$*.*

As in §5.4 above, (b, b') may be taken with $\kappa_0 \neq 0$, and then $c' = 0$, $c = -(3 + \operatorname{Re}\kappa_2)/\kappa_0$ meets the condition; so every frequency of Ω outside the five resonances is realised. For the plane system $\kappa_2 = 0$, $Y_<$ is even, and the Theorem reduces to that of §5.4 above, with its smaller set of resonances.

The proof is in Appendix C.6 below.

Corollary (inside, on and outside the ellipse)**.** *Let* $\omega_0 > 0$*. The angular velocity* 2ω *(rotation* $+2$*) or* -2ω *(rotation* -2*) of* $x_<$ *is realised on an attracting cycle, according to the position of* H *relative to the ellipse* (41)*, as follows.*

(a) H strictly inside the ellipse ($D < 0$)*: by the Theorem of §5.3, for the full plant* (33) *of any dimension, every* $\omega > 0$ *with rotation* -2*, and every* $\omega > 0$ *with* $\omega/\omega_0 \neq \frac12$ *with rotation* $+2$*. The cycle is the circle* $|x_<| = \varepsilon$*.*

(b) H on the ellipse ($D = 0$)*: by the Theorem of §5.5, for the full plant of any dimension, every* $\omega > 0$ *with* $\omega/\omega_0 \notin \{\frac14, \frac13, \frac12, 1, 2\}$*, with rotation* $+2$*; for the plane system, by the Theorem of §5.4, only* $\omega/\omega_0 \in \{\frac14, \frac12\}$ *are excluded. At the two foci of a degenerate ellipse* ($|G| = |K|$*,* $H = \pm 2\sqrt{GK}$) *only* $\omega/\omega_0 < 1/\sqrt{2}$*, with the same exclusions.*

(c) H strictly outside the ellipse ($D > 0$)*: by the Theorems of §5.5 and §5.4, every* $\omega > 0$ *with* $\omega/\omega_0 < U$*,* U *as in* (45)*, with rotation* $+2$*, with the same exclusions as in* (b)*.*

In (b) *and* (c) *the cycle of* $x_<$ *is a closed curve close to* $F_2(\{|z|^2 = \varepsilon\})$*, winding twice around* 0 *per period. Not covered: rotation* -2 *in* (b) *and* (c)*, where no linear coupling passes the second-order test; frequencies at or above the ceiling; and the excluded resonances.*

Proof. By §5.3, (H3) holds exactly when H is strictly inside the ellipse, which gives (a). In (b) and (c), Ω and its rotations are as stated in §5.2 by (42), (44) and the statement on the foci, and by the paragraphs after the Theorems of §5.4 and §5.5 their hypotheses can be met for every frequency of Ω outside the stated resonances. □

A worked example, the plane system in closed form, is given in Appendix C.7.

6. Relation to prior work

The topological-obstruction and control-bifurcation directions are discussed in the Introduction. In this section we compare the method of branching QBC with optimal control, with known syntheses for the Brockett integrator, and with stabilisation methods for an uncontrollable first approximation.

6.1. **Coupling manifold in branching QBC and Lagrangian manifold.** The branching QBC construction [1] has an analogue in the geometry of optimal control; both are instances of the singularities of projections of integral manifolds. In an optimal-control problem one adjoins to the state a vector of adjoint variables and reads the synthesis off a Lagrangian manifold whose projection to state space is singular along its caustic [36], the critical value set of that projection; the optimal feedback is multivalued over the region the manifold covers several times, and the cost selects a branch, switching on the Maxwell set, where two branches have equal cost; for the Brockett integrator such syntheses are computed in [46]. The analogy here is one of roles: the reference variable z takes the place of the adjoint variable, the coupling manifold $M = \operatorname{graph}(x = F(z))$ that of the Lagrangian manifold, and the ramification locus $\pi(\Sigma)$ that of the caustic. No symplectic structure is involved, and M is not claimed to be Lagrangian. What generates the manifold differs — a cost functional there, the reference dynamics and coupling here — and here no cost need be prescribed: the singularity of F at the working point is forced by the controllability degeneracy (the Proposition of §2.4 above).

Conjecture. *For any branching feedback constructed as in §2.5 above there is a cost functional whose optimal synthesis carries the same coupling manifold, in the roles described above, $M = \operatorname{graph}(x = F(z))$. In each particular case the QBC and optimal-control constructions then read the feedback off the same manifold M through the same projection $\pi\colon M \to X$, with the same caustic.*

6.2. **The Brockett integrator: known syntheses.** The passage to a discontinuous coordinate system in which the integrator becomes tractable is Astolfi's σ-process, which [47] applies to the Brockett integrator itself; the two-phase shape — manoeuvre out, then converge — is that of the piecewise-smooth laws of Canudas de Wit and Sørdalen [48]. A distinguished surface carrying a discontinuity of the law is also the device of Bloch and Drakunov [49], but the resemblance stops there: their surface is a sliding one, reached and then followed, whereas $|\rho|$ decreases on both sides of $\{|\rho| = \rho_0\}$, which is therefore crossed once and left behind. The law of §4.3 above that ends on a horn does attach to that horn and stay on it, and there the comparison is apt.

Closest to the construction here is Zeng's time-invariant law [33]. Its discontinuities are confined to the x_3-axis; near an axis point its solutions branch out in all directions, so that no continuous extension exists, and the law is extended to the axis by selecting one branch, $u = (4|x_3|^{1/2}, 0)$; the closed loop converges uniformly exponentially. The stabilisation result of the Proposition on the estimates is therefore not new. What the present construction adds is a law in closed form, with $|u| \le c\nu$ (Zeng's law is defined through the solution of a transcendental equation), and the identification of Zeng's branches as the branches of the branching feedback: the axis is the ramification locus $\pi(\Sigma)$, and the branches over an axis point form the circle $\beta^{-1}(x)$. The size $\nu = (r^4 + x_3^2)^{1/4}$ is a homogeneous norm for the weighted dilation $(x_1, x_2, x_3) \mapsto (\lambda x_1, \lambda x_2, \lambda^2 x_3)$, under which (14) is invariant together with $u \mapsto \lambda u$; the homogeneity so introduced for stabilisation is that of Hermes [50] and Kawski [51] (for a recent account see [52]), and conclusion (iv) of the Proposition on the estimates in §4.2 above is the ρ-exponential stability of M'Closkey and Murray [53]. A different route past Brockett's obstruction is Coron's theorem [54] (see also [29] and the references therein), which supplies a feedback $u(x, t)$, periodic in t, bounded and continuous on a whole neighbourhood of the origin, and driven by a single clock common to all initial conditions. Homogeneous time-varying stabilisers of this kind are constructed for the unicycle and the slider in [15], and time-varying stabilisers built from oscillating inputs, with the Brockett integrator as an example, in [55,56]. Blow-up as a tool of feedback design is used by Jardón-Kojakhmetov, Scherpen and del Puerto-Flores [57] for slow–fast systems at non-hyperbolic points: there the controller is designed in the charts

of the blow-up; here, in addition, the exceptional set carries the branches of the feedback. A discrete variable carried by the controller that selects one of several feedbacks, as the branch does here, is also used in the hybrid controllers of Sanfelice and coauthors [30, 39]; there it is switched during the motion, here it is chosen once, at $t = 0$.

6.3. **Stabilisation with an uncontrollable first approximation.** The same hypothesis — controllability lost at the working point — opens a body of work that keeps the feedback *single-valued* and asks for equilibrium stabilisation. In the Russian literature this is the Kharkiv line: Korobov and Bebiya [58] stabilise systems *uncontrollable in the first approximation* by a power-law feedback from Korobov's controllability-function method [59], the nonlinear terms restoring the controllability the linear part lacks. In the Western literature Zaslavsky [60] treats the case of coupled oscillators, and Qian, Lin and coauthors stabilise classes of systems with uncontrollable linearisation, by sampled-data output feedback [61] and, with time delays, by linear feedback [62]. These works make the same assumption as this paper, but their feedback is single-valued and their goal is the stabilisation of an equilibrium. Here the feedback is branching (§2.5); it stabilises the whole family of plants, including the degenerate ones (the QBC ultrastabilisation problem, Definition 3.3), and besides stabilisation it installs other prescribed behaviour, such as a cycle of prescribed frequency.

6.4. **The papers of 1988–1992 and later work.** The papers [1–4] appeared in Russian, in a translated volume and in conference proceedings, and were little read in control theory. Several of the questions they treat were later taken up independently, and developed further, by other authors:

- the invariant graph of a plant coupled with a second system, used for observation [1, §6], in the theory of KKL observers [63–65];
- the design goal of an invariant manifold of the plant × target system that forces the plant to behave like the target [1, §2.3], in immersion and invariance [66];
- a controller carrying a discrete variable that is piecewise constant and continuous from the right, with finitely many jumps on each finite interval [1, §3], in hybrid feedback [30, 67, 68];
- the synthesis of a stable cycle at an uncontrollable linearisation, and the unavoidable loss of controllability in families of plants [1, §§4.3–4.11], in Hopf control and control bifurcations [25, 34, 35].

We list these connections to place the method, not to claim priority: the case of this paper rests on the new theorems above.

Appendix A. Details of the Brockett synthesis

This appendix collects the checks behind the synthesis of §4.2.

The control bound. Near the axis, moving outward is cheap and moving the height is expensive. The control A moves the state outward ($\dot r = \nu A$) and the control B moves the height ($\dot x_3 = \nu^2 B$). By (18) the two perpendicular parts of u are νA and $\nu(1+\rho^2)^{1/4}B$, so a unit of A costs ν and a unit of B costs $\nu(1+\rho^2)^{1/4}$. So near the axis the state uses $A > 0$ and sets $B = 0$: the whole control goes into the cheap outward motion, and the state moves away from the axis at constant height. This lowers $|\rho| = |x_3|/r^2$, and with it the cost of moving the height. The descent (24) below fits the bound only for $|\rho| \le \rho_{\max}$, a value given below, so it starts on a horn $|\rho| = \rho_0$ with $\rho_0 \le \rho_{\max}$. The horn $|\rho| = \rho_{\max}$ is the first one, coming from the axis, on which the descent is possible. In x: from a point at height x_3 the state must first move out to the radius $r = (|x_3|/\rho_0)^{1/2}$.

The descent formula is bounded. It is used only where $|\rho(x)| \le \rho_0$, so $|u| \le (a + b\rho_0)\, r$, and $u \to 0$ as $x \to 0$.

The outward formula is bounded. For every $r > 0$ its size is $|u| = \eta\nu(x)$; in its region $r^4 < x_3^2/\rho_0^2$, so $|u| < \eta(1 + \rho_0^{-2})^{1/4}|x_3|^{1/2}$, however small r is. On the axis the direction is supplied by the branch (the Proposition on the branching feedback in §4.2).

The control on the chart containing the axis. At a point $(0, 0, x_3)$, $x_3 \neq 0$, of the axis the feedback (26) gives a push of size $\eta|x_3|^{1/2}$ away from the axis. In Z the axis point is replaced by the circle $\{r = 0\}$ of the chart (16). The reference (23) and the coupling (25) are written in this chart and hold on its whole part $|x_3| > \rho_0 r^2$, the circles $\{r = 0\}$ included:

$$\dot r = \eta\,(r^4 + x_3^2)^{1/4}, \qquad \dot\theta = 0, \qquad \dot x_3 = 0, \qquad u = \eta\,(r^4 + x_3^2)^{1/4}(\cos\theta, \sin\theta).$$

The switch. The reference passes from the outward motion (23) to the descent (24) on the horns $|\rho| = \rho_0$, and every trajectory crosses them once, inward. At the switch the state itself is continuous; only its velocity jumps. The radial speed changes from $\dot r = \eta\nu > 0$ under (23) to $\dot r = -a\, r < 0$ under (24).

The value $\rho_{\max}$. The constant c bounds the ratio $|u|/\nu$, not the size of u, so it matters at every scale. The danger is the axis: on the descent $|u|/\nu = (a^2 + b^2\rho^2)^{1/2}(1 + \rho^2)^{-1/4} \approx b|\rho|^{1/2}$, which is unbounded as the axis is approached ($|\rho| \to \infty$), however small ν is. So with a finite c the descent can be used only on a bounded set of horns. Precisely, let $a < c$ (at $\rho = 0$ the descent has $|u|/\nu = a$) and let $\rho_{\max}$ be the positive solution of $(a^2 + b^2\rho^2)(1 + \rho^2)^{-1/2} = c^2$. Any trajectory of the descent (24) contained in $\{|\rho| \le \rho_{\max}\}$ satisfies the bound (22); since $|\rho|$ decreases along the descent, this holds for every trajectory that starts the descent on a horn $|\rho| = \rho_0$ with $\rho_0 \le \rho_{\max}$. The outward motion (23) satisfies (22) when $\eta \le c$.

The duration of the outward motion. The duration (28) of the outward motion is the same for every initial height because (23) makes $\dot r = \eta(r^4 + x_3^2)^{1/4}$ at the height x_3, which it holds constant, and the substitution $r = |x_3|^{1/2}\xi$ removes x_3 from the integral; along the way ν grows by the factor C at most. For an initial point on the axis at height x_3 the state is carried out to $r \sim |x_3|^{1/2}$, and the path length is then of order $|x_3|^{1/2}$; this is the sub-Riemannian order of magnitude for the Heisenberg metric underlying (14) [69], so nothing has been given away by moving outward first.

Remark (the switching surface can be removed). The surface $|\rho| = \rho_0$ is chosen by the designer and is not a change of branch. It can be removed: with $\chi(\rho) = \rho^2/(1 + \rho^2)$, the single pair

$$A(\rho) = \eta\,\chi - a\,(1 - \chi)(1 + \rho^2)^{-1/4}, \qquad B(\rho) = -b\,\rho\,(1 - \chi)(1 + \rho^2)^{-1/2} \tag{54}$$

agrees with the descent near $\rho = 0$ and with the outward motion near the axis, and all conclusions of the Proposition on the estimates hold for it with other constants; only the exact duration (28) of the outward motion is lost, and for that and for the choice of frequency in §4.3 below we keep the surface.

Appendix B. Proof of the Theorem on the rotating branch

In this appendix h, w and X denote the quantities defined below; they are not the coupling, the reference field or the state space.

This appendix proves the Theorem of §4.4 above: the law (32) stabilises the Brockett integrator (14) with a uniform exponential rate.

Outline. A reflection reduces the proof to the half-space $x_3 > 0$. Outside the tube the law is the field (54): there $|\rho|$ decreases, the tube is not entered again, and ν decays at a rate close to a. Near the axis we pass to a chart y in which the axis is a single regular point and the closed loop is autonomous. In that chart (a) a Dulac function rules out closed orbits, (b) a circle

around $y = 0$ is crossed only outward, (c) the trajectories that stay inside that circle converge to equilibria, at which the state shrinks along a horn, and (d) in the band where $A \le 0$ the size ν decreases. Every solution therefore converges. Uniformity follows from compactness of the sphere of shapes $\{\nu = 1\}$ times the circle of phases, together with the dilations.

Proof. Well-posedness. With $W = \nu^4 = r^4 + x_3^2$ and $L = \frac{3}{2}(\eta + a + b)$, the bound on $|u|$ gives $|\dot W| \le 4LW$, so $e^{-L(t-t_0)} \le \nu(x(t))/\nu(x(t_0)) \le e^{L(t-t_0)}$. This gives uniqueness at the origin; elsewhere the law is C^1. On the plane $x_3 = 0$ the law is $u = -a(x_1, x_2)$, so the plane is invariant, and with it each half-space. The reflection $(x_1, x_2, x_3, u_1, u_2) \mapsto (x_1, -x_2, -x_3, u_1, -u_2)$ preserves (14) and carries the law on $x_3 < 0$ to the law on $x_3 > 0$ with phase $-\varphi_0$. It suffices to treat $x_3 > 0$.

Outside the tube. There the law is (54), and (19) gives $\dot\rho = -\rho\,[(b-2a)(1-\chi) + 2\eta\chi(1+\rho^2)^{1/4}]$, $\chi = \rho^2/(1+\rho^2)$. The bracket is at least $\kappa = \min(b - 2a, 2\eta)$, so $|\rho|$ decays at rate κ, w increases, and the tube is never entered again. By (20), $\dot\nu/\nu \to -a$ as $\rho \to 0$. Hence after its entry time t_e into $\{w \ge w_1\}$ a solution satisfies $\nu(t) \le K_2 e^{-(a/2)(t-t_e)}\nu(t_e)$, with K_2 depending only on the parameters.

A chart in which the axis is a point. Let $\varphi(t) = -\omega_0 t + \varphi_0$ and

$$y = \frac{R_{-\varphi(t)}(x_1, x_2)}{\sqrt{x_3}}, \qquad h = |y| = \rho^{-1/2}, \qquad w = \frac{h}{(1+h^4)^{1/4}},$$

where R_φ is the rotation by φ. The chart (16) represents an axis point by the circle $\beta^{-1}(0, 0, x_3)$ of branches; y collapses that circle back to the single point $y = 0$, which is why a continuous law is regular here. Write $A = A(\rho)$, $J(v_1, v_2) = (-v_2, v_1)$, $e_1 = (1, 0)$, and

$$c = \frac{A\sigma}{w}, \qquad k = (1+h^4)^{1/4} A(1-\sigma), \qquad m = \frac{bh^2}{1+h^4}, \qquad \lambda = c + \tfrac{1}{2}mh^2.$$

Rotating (32) by $R_{-\varphi}$ and dividing by $\sqrt{x_3}$ gives $cy + ke_1 - mJy$; then $\dot x_3 = x_1u_2 - x_2u_1$ and the derivative of y give the autonomous system

$$\dot y = X(y) := \lambda y + (\omega_0 - m)Jy + k\big(e_1 + \tfrac{1}{2}y_2\, y\big), \qquad \frac{\dot x_3}{x_3} = -ky_2 - mh^2, \tag{55}$$

with X of class C^1 and $X(0) = \eta e_1 \ne 0$: a solution crosses the axis through a regular point. Here $\lambda = \eta(1-w^4)\sigma/w + (\frac{b}{2} - a\sigma)w^4 > 0$ for $h > 0$.

(a) No closed orbits near the axis. A decreases in w from η to $-a$; let w_A be its zero, $w_b = \min(w_A, w_1)$ and $h_b = h(w_b)$. On $|y| < h_b$, $k > 0$, and with $f = \lambda/k$,

$$\operatorname{div}\frac{X}{k} = 2f + hf' + \tfrac{3}{2}y_2.$$

Write $f = \sigma/(h(1-\sigma)) + mh^2/(2k)$. The second part is positive and increasing in h. For the first, $2f_1 + hf_1' = \sigma/(h(1-\sigma)) + \sigma_h/(1-\sigma)^2$; since $\sigma \ge (w/w_1)^2 \ge w^2$, the first term is $\ge h$, and with $\zeta = w/w_1$, $\sigma_h/(h(1-\sigma)^2) = 6(1-w^4)^{3/2}/\big(w_1^2(1-\zeta)^3(1+2\zeta)^2\big) \ge 2/(3w_1^2)$. Hence

$$\operatorname{div}\frac{X}{k} \ge \Big(\frac{2}{3w_1^2} - \frac{1}{2}\Big)h > 0 \qquad (0 < |y| < h_b),$$

and by the Bendixson–Dulac argument (Green's theorem) the disk $|y| < h_b$ contains no periodic orbit and no closed chain of orbits joining equilibria.

(b) An outward circle. The radial component of (55) gives $\dot h \ge \ell(h) := \lambda h - |k|\sqrt{1 + h^4/4}$, with $\ell(0) = -\eta < 0$ and $\ell(h_b) = \lambda(h_b)h_b > 0$. Choose $h_c < h_b$ with $\ell > 0$ on $[h_c, h_b]$. The circle $|y| = h_c$ is crossed only outward, and the collar $h_c \le |y| \le h_b$ is crossed in finite time.

(c) Trajectories that stay near the axis. An equilibrium of X in $|y| \le h_c$ has $\sin\delta = h(\omega_0 - m)/k$ and $\cos\delta = -\sigma/(1-\sigma) - \omega_0 h^3/(2k)$, δ the polar angle of y; its radius is a zero of a function

of h that is analytic on an interval containing $[0, h_c]$ and equals -1 at $h = 0$. So there are finitely many equilibria, none at $y = 0$. By the generalised Poincaré–Bendixson theorem [70, §3.7] and (a), a trajectory that stays in $|y| \le h_c$ converges to one of them, y_e. At y_e the shape is constant and $\dot\theta = \dot\varphi = -\omega_0$: the state moves on a horn with constant clock, as in §4.3, and (29), which gives $\dot\nu/\nu = \omega/(2\rho)$, with $\omega = -\omega_0$ gives $\dot\nu/\nu = -\omega_0/(2\rho) < 0$. Hence $\nu \to 0$ along such a trajectory.

(d) The band where $A \le 0$. This band, $w_A \le w < w_1$, exists only if $w_A < w_1$. Inserting (32) into (20) gives, with ψ the angle from $(x_1, x_2)/r$ to $e_+(t)$,

$$\frac{\dot\nu}{\nu} = A\sigma w^3 + A(1-\sigma)\big(w^3\cos\psi + \tfrac{1}{2}(1-w^4)^{1/2} w\sin\psi\big) - \tfrac{b}{2}(1-w^4)w^4 \le -\frac{b-a}{2}(1-w^4)w^4 < 0,$$

using $|A| \le a w^5$, $1-\sigma \le 1-w^2$ and $w^2/(1+w^2) \le \tfrac{1}{2}$. On $w = w_A$, $k = 0$ and $\dot h = \lambda h > 0$, so the band is never left towards the axis.

Every solution converges. A solution in $x_3 > 0$ either stays in $|y| \le h_c$ and converges by (c), or crosses the collar by (b) and then stays in the band (d), where ν decays, or enters $\{w \ge w_1\}$, where it decays and from which it never returns.

Uniformity. The flow commutes with the dilations and is P-periodic, so it suffices to consider initial data in the compact set $\{\nu = 1\} \times (\mathbb{R}/P\mathbb{Z})$ of states and initial phases. Each such datum reaches $\nu < \frac{1}{4}$ at some time; by continuous dependence, a neighbourhood reaches $\nu < \frac{1}{2}$ at that same time. A finite subcover gives times between $\tau_{\min} > 0$ and T after which ν has at least halved; iterating with the dilations and the growth bound gives the estimate with $K = 2e^{LT}$ and $\gamma = (\ln 2)/T$. □

Appendix C. Proofs for the control of frequency

C.1. **A linear coupling does not give an invariant graph.** This subsection proves the claims of the Remark of §5.1 above.

Outline. Along the reference trajectory that leaves the origin, the plant is a linear equation with explicit forcing. Its solution is computed in closed form, and its limit as $t \to +\infty$ differs from the periodic solution over the cycle by an explicit nonzero amount. A continuous invariant graph would force that amount to be zero. The product system then shows where the trajectory goes: to an invariant torus.

Take $\dot x = i\omega_0 x + u^2$ and $u = z$, with $\omega_0 > 0$, $\omega_0/\omega \notin \mathbb{N}$ and $\delta := 2\omega - \omega_0 \ne 0$. The reference trajectory

$$z(t) = \sqrt{s(t)}\, e^{i\omega t}, \qquad s(t) = \frac{\varepsilon}{1 + e^{-2\varepsilon t}},$$

leaves the origin and tends to the cycle. Suppose $x = F(z)$ is a continuous invariant graph over a disc containing the cycle, with $F(0) = 0$. Along the trajectory $x(t) = F(z(t)) \to 0$ as $t \to -\infty$, and with $x = e^{2i\omega t} y$ the plant becomes $\dot y = -i\delta y + s(t)$, whose solution with $y(-\infty) = 0$ satisfies

$$y(t) - \frac{s(t)}{i\delta} = -\frac{e^{-i\delta t}}{i\delta}\int_{-\infty}^{t} e^{i\delta\tau} s'(\tau)\, d\tau, \qquad \int_{-\infty}^{\infty} e^{i\delta\tau} s'(\tau)\, d\tau = \frac{\pi\delta}{2\sinh\big(\pi\delta/(2\varepsilon)\big)} \ne 0.$$

Hence $|y(t) - s(t)/(i\delta)| \to \pi/\big(2|\sinh(\pi\delta/(2\varepsilon))|\big) > 0$ as $t \to +\infty$. On the cycle, invariance makes $t \mapsto F(\sqrt{\varepsilon}e^{i\omega t})$ a $2\pi/\omega$-periodic solution of $\dot x = i\omega_0 x + \varepsilon e^{2i\omega t}$, and since $\omega_0/\omega \notin \mathbb{N}$ the only one is $(\varepsilon/i\delta)e^{2i\omega t}$; continuity of F would then give $y(t) - s(t)/(i\delta) \to 0$. So no such F exists. The defect is exponentially small in $1/\varepsilon$ and invisible at every finite order.

What happens can be seen in the product system $(z, x) \in \mathbb{C}^2$. Put $\xi := x - z^2/(i\delta)$. Along the reference, $\frac{d}{dt} z^2 = [2(\varepsilon - |z|^2) + 2i\omega] z^2$, and a direct computation gives, on all of $\mathbb{C}^2$,

$$\dot\xi = i\omega_0 \xi - 2(\varepsilon - |z|^2)\,\frac{z^2}{i\delta}.$$

Over the reference cycle the last term vanishes and $|\xi|$ is constant. So the three-dimensional set $\{|z|^2=\varepsilon\}$ is filled by the closed orbit $\Gamma_0=\{\xi=0\}$, which is the periodic solution $(\varepsilon/i\delta)e^{2i\omega t}$, and by the invariant tori $\{|\xi|=r\}$, $r>0$, nested around it; none of them attracts. The trajectory leaving the origin is pushed off Γ_0 by the last term while $|z|$ grows, and it arrives on the torus of radius $\pi/(2|\sinh(\pi\delta/(2\varepsilon))|)$, on which it winds forever with the two frequencies ω and ω_0. The same holds for every linear coupling: with $\xi=x-x_p(z)$, $x_p(z)=\sum_k d_k z^{(k)}/(i(\nu_k-\omega_0))$, where $(\nu_k,d_k,z^{(k)})$ runs over $(2\omega,d_{20},z^2)$, $(0,d_{11},|z|^2)$, $(-2\omega,d_{02},\bar z^2)$, the trajectory from the origin lands on the torus of radius $|c_*|$,

$$c_*=\frac{i\pi}{2}\sum_k\frac{d_k}{\sinh\bigl(\pi(\nu_k-\omega_0)/(2\varepsilon)\bigr)}.$$

C.2. **The second-order test.** In this subsection the letters v and w denote complex numbers that describe linear couplings, and D and E denote quantities defined below; they are not the plant field, the reference field, the linear part of the reference, or the ellipse.

This subsection proves the classification of Ω in §5.2 above: the floor (43), the case (42) inside and on the ellipse, and the ceiling (45) outside it. The quantity $\mathcal{A}$ below is the A of (39); the letter A is used here for the function $A(w)$ defined below.

Outline of proof. Up to a common factor, every coupling with $b\neq0$ is $(1,w)$ with $w=b'/\bar b$, and the couplings with $|w|>1$ reduce to $|w|<1$ for the plant (K,H,G), which has the same ellipse; so it suffices to take w in the unit disc. There (40) becomes one polynomial inequality in three invariants of the coupling, among them the ratio $|d_{02}|/|d_{20}|$. These invariants are expressed through the pseudo-hyperbolic distances from w to the roots of $q(\zeta)=G\zeta^2+H\zeta+K$, and the sign of D decides where the roots lie: one inside the unit circle and one outside if $D>0$, both on the same side if $D<0$, one on the circle if $D=0$. On the hyperbolic geodesic through the roots the inequality reduces to the positivity of a quadratic polynomial, and the ceiling U is the frequency at which the discriminant of that polynomial changes sign.

C.2.1. *Setting and result. The plane system.* Time is scaled so that $\omega_0=1$; frequencies are then measured in units of ω_0. The critical mode is

$$\dot x=ix+Q(u),\qquad Q(u)=Gu^2+H|u|^2+K\bar u^2,\qquad x,u\in\mathbb{C},$$

with $(G,H,K)\in\mathbb{C}^3\setminus\{0\}$, and the reference is the Andronov–Hopf system $\dot z=(\varepsilon+i\omega)z-|z|^2z$, $\omega>0$, $\omega\neq\frac12$. The coupling is $u=p(z)=bz+b'\bar z$ with $(b,b')\in\mathbb{C}^2\setminus\{0\}$, and

$$Q(p(z))=d_{20}z^2+d_{11}z\bar z+d_{02}\bar z^2,$$

$$d_{20}=Gb^2+Hb\bar b'+K\bar b'^2,\quad d_{11}=2Gbb'+H(|b|^2+|b'|^2)+2K\bar b\bar b',\quad d_{02}=Gb'^2+H\bar b b'+K\bar b^2.$$

The invariance equation $ix+Q(p(z))=F_z\dot z+F_{\bar z}\dot{\bar z}$ for a graph $x=F(z)$, solved order by order, determines its quadratic part $F_2=c_{20}z^2+c_{11}z\bar z+c_{02}\bar z^2$: for $\varepsilon>0$, $c_{20}=d_{20}/(2\varepsilon+i(2\omega-1))$, $c_{11}=d_{11}/(2\varepsilon-i)$, $c_{02}=d_{02}/(2\varepsilon-i(2\omega+1))$, and this appendix works with their limits at $\varepsilon=0$,

$$c_{20}=\frac{d_{20}}{i(2\omega-1)},\qquad c_{11}=i\,d_{11},\qquad c_{02}=\frac{d_{02}}{-i(2\omega+1)}$$

(§5.1 above). The solution is formal: by the Remark of §5.1 above the graph it defines at the origin need not extend to the reference cycle. With $z=re^{i\theta}$ one has $|\partial_zF_2|^2-|\partial_{\bar z}F_2|^2=r^2\varphi(\theta)$,

$$\varphi(\theta)=4\bigl(|c_{20}|^2-|c_{02}|^2\bigr)+4\,\mathrm{Re}\bigl(\mathcal{A}e^{2i\theta}\bigr),\qquad \mathcal{A}=c_{20}\bar c_{11}-c_{11}\bar c_{02},$$

so the branch point $z=0$ of F_2 is a nondegenerate double point exactly when φ has constant sign. This gives the two criteria below.

Definition. The coupling (b, b') *reaches* ω *with rotation* $+2$ if $|c_{20}|^2 - |c_{02}|^2 > |\mathcal{A}|$, and *with rotation* -2 if $|c_{02}|^2 - |c_{20}|^2 > |\mathcal{A}|$. The *reachable set* $\Omega \subset \mathbb{R}_{>0} \setminus \{\frac{1}{2}\}$ is the set of ω reached by some coupling with some rotation.

Notation.

$$\Pi = |H^2 - 4GK|, \qquad E = |H|^2 - 2\big(|G|^2 + |K|^2\big), \qquad D = E + \Pi.$$

Since $|H^2 - 4GK| \ge |H|^2 - 4|GK|$,

$$2\Pi - D = \Pi - E \ge 2\big(|G| - |K|\big)^2 \ge 0. \tag{56}$$

For $D > 0$ put

$$U = \frac{1}{2}\sqrt{\frac{D + 2\Pi}{D}} = \frac{1}{2}\sqrt{\frac{|H|^2 + 3\Pi - 2(|G|^2 + |K|^2)}{|H|^2 + \Pi - 2(|G|^2 + |K|^2)}}. \tag{57}$$

For $\omega > 0$, $\omega \ne \frac{1}{2}$, put $x = (2\omega - 1)/(2\omega + 1) \in (-1, 1) \setminus \{0\}$; then $4\omega^2 - 1 = 4x/(1 - x)^2$.

For points a, b of the open unit disk $\mathbb{D}$, $\rho(a, b) = |a - b|/|1 - \bar{b}a|$ is the pseudo-hyperbolic distance. It is invariant under the automorphisms of $\mathbb{D}$, and

$$1 - \rho(a, b)^2 = \frac{(1 - |a|^2)(1 - |b|^2)}{|1 - \bar{b}a|^2}. \tag{58}$$

If a, b, c lie in this order on one hyperbolic geodesic, then $\rho(a, c) = (\rho(a, b) + \rho(b, c))/(1 + \rho(a, b)\rho(b, c))$.

The result.

Theorem (second-order test)**.** *Let* $(G, H, K) \ne 0$.

(a) $\{H : D \le 0\}$ *is the closed ellipse with foci* $\pm 2\sqrt{GK}$ *and semi-axes* $|G| + |K|$ *and* $\big||G| - |K|\big|$*, the major axis along the direction* $\arg(GK)/2$*. If* $GK \ne 0$ *and* $|G| \ne |K|$*, then with* $\xi + i\eta := He^{-i\arg(GK)/2}$ *it is* $\xi^2/(|G| + |K|)^2 + \eta^2/(|G| - |K|)^2 \le 1$*. If* $GK = 0$*, the angle* $\arg(GK)$ *is undefined, both foci are* 0*, and the set is the disc* $|H| \le |G| + |K|$*. If* $|G| = |K| \ne 0$*, the ellipse is the segment joining the foci. At the foci* $D = -2(|G| - |K|)^2$*; the foci lie strictly inside if* $|G| \ne |K|$.

(b) *If* $D > 0$ *(strictly outside),* $\Omega = (0, U) \setminus \{\frac{1}{2}\}$*, and only rotation* $+2$ *occurs.*

(c) *If* $D < 0$ *(strictly inside),* $\Omega = \mathbb{R}_{>0} \setminus \{\frac{1}{2}\}$*, and every* $\omega \ne \frac{1}{2}$ *is reached with rotation* $+2$ *and also with rotation* -2.

(d) *If* $D = 0$ *and* $\Pi > 0$ *(on the ellipse, not at a focus),* $\Omega = \mathbb{R}_{>0} \setminus \{\frac{1}{2}\}$*, and only rotation* $+2$ *occurs.*

(e) *If* $D = 0$ *and* $\Pi = 0$*, then* $|G| = |K| \ne 0$ *and* $H = \pm 2\sqrt{GK}$ *(a focus of a degenerate ellipse),* $\Omega = (0, \frac{1}{\sqrt{2}}) \setminus \{\frac{1}{2}\}$*, and only rotation* $+2$ *occurs.*

(f) *Always* $(0, \frac{1}{\sqrt{2}}) \setminus \{\frac{1}{2}\} \subseteq \Omega$*. Equality holds exactly when* $E = \Pi$*, that is,* $|H|^2 - 2(|G|^2 + |K|^2) = |H^2 - 4GK|$.

(g) *As* H *tends to a point of the ellipse with* $\Pi > 0$ *from outside,* $U \to +\infty$.

(h) H *lies strictly inside the ellipse iff* $|G| \ne |K|$ *and* $|K\bar{H} - \bar{G}H| < \big||G|^2 - |K|^2\big|$ *(hypothesis (H3) of §5.3 above).*

(i) *A coupling with* $d_{11} = 0$ *and* $|d_{20}| \ne |d_{02}|$ *exists iff* $D < 0$*, and then one with* $|d_{02}| < |d_{20}|$ *exists. If both roots of* q *(§C.2.4) lie in the unit disk, the coupling* $(1, w)$ *with* w *their hyperbolic midpoint has* $d_{11} = 0$ *and* $|d_{02}| < |d_{20}|$.

Examples. $G = K = 1$, $H = -2$, so that $Q(u) = -4(\operatorname{Im} u)^2$: case (e). $G = K = 1$, $H = -\frac{5}{2}$: case (b) with $U = 1/\sqrt{2}$, so Ω is the floor of (f). $G = 2$, $K = i$, $H = \frac{5}{2}$: case (b) with $U \approx 1.021$, although $\xi = \frac{5}{2}$, $\eta = 0$ would satisfy the inequality of (a) without the rotation.

Remark. Ω concerns the second-order data of a linear coupling only. Such a coupling does not in general produce an invariant graph through the reference cycle (Remark of §5.1 above), so Ω is not by itself a set of frequencies realised by a synthesis. Inside the ellipse, which by (h) is the condition that Q has no folds, the Theorem of §5.3 above realises every $\omega \neq \frac{1}{2}$ with rotation $+2$ and every ω with rotation -2, by a nonlinear coupling.

C.2.2. *One test for all couplings.* For $w \in \mathbb{C}$ put

$$A(w) = G + H\bar{w} + K\bar{w}^2, \qquad B(w) = 2Gw + H(1+|w|^2) + 2K\bar{w}, \qquad C(w) = Gw^2 + Hw + K.$$

Lemma (normalisation)**.** (i) *If $b \neq 0$ and $w = b'/\bar{b}$, then*

$$(d_{20}, d_{11}, d_{02}) = \big(b^2 A(w),\ |b|^2 B(w),\ \bar{b}^2 C(w)\big).$$

(ii) *If $b = 0$, then* $(d_{20}, d_{11}, d_{02}) = (\bar{b}'^2 K, |b'|^2 H, b'^2 G)$.
(iii) *For $v \neq 0$ and $w = 1/\bar{v}$:* $A(w) = C(v)/v^2$, $B(w) = B(v)/|v|^2$, $C(w) = A(v)/\bar{v}^2$.
(iv) $(C, B, A)(w; G, H, K) = (A, B, C)(\bar{w}; K, H, G)$.

Proof. (i) $b' = w\bar{b}$ gives $bb' = |b|^2 w$, $b\bar{b}' = b^2\bar{w}$, $\bar{b}'^2 = b^2\bar{w}^2$, and similarly for d_{11}, d_{02}. (ii) is immediate. (iii): $\bar{w} = 1/v$ gives $A(w) = (Gv^2 + Hv + K)/v^2$; $w = 1/\bar{v}$ gives $C(w) = (G + H\bar{v} + K\bar{v}^2)/\bar{v}^2$ and $B(w) = (2Gv + H(|v|^2+1) + 2K\bar{v})/|v|^2$. (iv) by inspection. □

In (i) the criteria for (b, b') are those for $(1, w)$ multiplied by $|b|^4$, since c_{20}, c_{11}, c_{02} acquire the factors b^2, $|b|^2$, $\bar{b}^2$. The plant (K, H, G) has the same Π, E, D as (G, H, K), its roots (below) are the reciprocals, and it has the same ellipse and the same U.

Definition. For real $X \neq 0, \pm 1$ and $(A, B, C) \in \mathbb{C}^3$,

$$\mathrm{Crit}(X; A, B, C): \qquad |A|^2 - X^2|C|^2 > \left|\frac{2X}{1-X}\right| \big|A\bar{B} + XB\bar{C}\big|.$$

Lemma (four tests)**.** *Let $(A, B, C) = (A, B, C)(w)$ and $x = (2\omega - 1)/(2\omega + 1)$. Rotation $+2$ at $(1, w)$ is* $\mathrm{Crit}(x; A, B, C)$*, and rotation -2 at $(1, w)$ is* $\mathrm{Crit}(1/x; C, B, A)$*. Consequently every coupling and rotation is one of the four tests*

coupling	*rotation*	*test at a point of* $\mathbb{D}$
$\lvert w\rvert < 1$	$+2$	$\mathrm{Crit}(x)$ *for* (G, H, K) *at* w
$\lvert w\rvert < 1$	-2	$\mathrm{Crit}(1/x)$ *for* (K, H, G) *at* $\bar{w}$
$\lvert w\rvert > 1$ *or* $b = 0$	$+2$	$\mathrm{Crit}(x)$ *for* (K, H, G) *at* $\bar{v}$, $v = 1/\bar{w}$
$\lvert w\rvert > 1$ *or* $b = 0$	-2	$\mathrm{Crit}(1/x)$ *for* (G, H, K) *at* v

together with the couplings $|w| = 1$ (with $b = 0$ read as $v = 0$).

Proof. With $D_1 = 2\omega - 1$, $D_2 = 2\omega + 1$ (so $D_2 - D_1 = 2$): $\mathcal{A} = -(A\bar{B}D_2 + B\bar{C}D_1)/(D_1 D_2)$. Multiplying $|c_{20}|^2 - |c_{02}|^2 > |\mathcal{A}|$ by D_1^2 and using $x = D_1/D_2$, $|D_1| = |2x/(1-x)|$ gives $|A|^2 - x^2|C|^2 > |D_1|\,|A\bar{B} + xB\bar{C}|$, which is $\mathrm{Crit}(x; A, B, C)$. Multiplying $|c_{02}|^2 - |c_{20}|^2 > |\mathcal{A}|$ by D_2^2 and putting $Y = 1/x = D_2/D_1$, so that $|D_2| = |2Y/(1-Y)|$, gives $|C|^2 - Y^2|A|^2 > |D_2|\,|YA\bar{B} + B\bar{C}|$; since $|YA\bar{B} + B\bar{C}| = |C\bar{B} + YB\bar{A}|$ (complex conjugate), this is $\mathrm{Crit}(Y; C, B, A)$. The table follows from the Lemma (normalisation). □

Lemma (the quantity S)**.** *Let $A \neq 0$, $r = |C|/|A|$, $u = |B|^2/|A|^2$, $d = |B^2 - 4AC|/|A|^2$. Then*

$$\mathrm{Crit}(X; A, B, C) \iff 1 - X^2 r^2 > 0 \quad \text{and} \quad S > 0,$$

$$S := X^3 d^2 + (1-X)^2(1 - X^2 r^2)^2 - X^2\big[4u + Xu^2 + 16Xr^2 + 4X^2 r^2 u\big].$$

If $A = 0$, $\mathrm{Crit}(X; A, B, C)$ *fails.*

Proof. The quantities in Crit, and u, d, are unchanged under

$$(A, B, C) \mapsto \big(e^{i(\alpha+\beta)}A,\, e^{i\alpha}B,\, e^{i(\alpha-\beta)}C\big), \qquad \alpha, \beta \in \mathbb{R}$$

(a common output phase $e^{i\alpha}$ and a rotation of the source variable): $A\bar{B}$ and $B\bar{C}$ both acquire the factor $e^{i\beta}$, and B^2 and AC both acquire $e^{2i\alpha}$. Choose α, β with $A > 0$ and $C \geq 0$, and divide by $|A|^2$, so that $A = 1$, $C = r$. Then $|\bar{B} + XrB|^2 = u(1 + X^2r^2) + 2Xr\,\mathrm{Re}(B^2)$, and $d^2 = |B^2 - 4r|^2 = u^2 - 8r\,\mathrm{Re}(B^2) + 16r^2$, so $2Xr\,\mathrm{Re}(B^2) = \frac{X}{4}(u^2 + 16r^2 - d^2)$. The criterion says $1 - X^2r^2 > 0$ and $(1 - X)^2(1 - X^2r^2)^2 > 4X^2|\bar{B} + XrB|^2$; the difference of the two sides is S. If $A = 0$ the left side of Crit is $-X^2|C|^2 \leq 0$. □

C.2.3. *The variables t and T.*

Lemma (invariants)**.** *For every w,*

$$B^2 - 4AC = (H^2 - 4GK)(1 - |w|^2)^2, \qquad |B|^2 - 2(|A|^2 + |C|^2) = E\,(1 - |w|^2)^2.$$

Proof. $Q(p(z))$ with $p(z) = z + w\bar{z}$ is the binary form $Gu^2 + Hu\bar{u} + K\bar{u}^2$ after the substitution $(u, \bar{u}) = (z + w\bar{z},\ \bar{w}z + \bar{z})$, of determinant $1 - |w|^2$; the discriminant of a binary quadratic form is multiplied by the square of the determinant. The second identity is checked by expanding both sides as polynomials in $G, \bar{G}, H, \bar{H}, K, \bar{K}, w, \bar{w}$. □

For $|w| < 1$ with $A(w) \neq 0$ put $\lambda = (1 - |w|^2)^2/|A|^2 > 0$ and

$$t = \tfrac{1}{2}(\Pi - E)\lambda, \qquad T = \tfrac{1}{2}D\lambda.$$

By the Lemma (invariants), $d = \Pi\lambda = t + T$ and $u = 2(1 + r^2) + E\lambda = 2(1 + r^2) - t + T$. The inequality $|u - 4r| \leq d \leq u + 4r$ (valid for any B, r) becomes

$$(1 - r)^2 \leq t \leq (1 + r)^2, \qquad T \geq -(1 - r)^2. \tag{59}$$

If $2\Pi \neq D$, put

$$\mu = \frac{D}{2\Pi - D}; \qquad \text{then} \qquad T = \mu t.$$

Thus $\mu > 0$ outside the ellipse, $\mu = 0$ on it, and $-1 \leq \mu < 0$ inside it (by (56)). In the variables (r, t, T), by direct expansion,

$$S = (1 + X)P_0(r, t) + T\,S_1(r, t), \qquad S_1 = -4X^2\big[(1 + X)(1 + Xr^2) - Xt\big], \tag{60}$$

$$P_0 = X^5r^4 - 3X^4r^4 - 4X^3r^4 + 4X^3r^2t - 10X^3r^2 - 10X^2r^2 + 4X^2t - 4X^2 - 3X + 1,$$

and at the ends of the band

$$P_0\big|_{t=(1+r)^2} = (Xr-1)^3(X^2r-3Xr+3X-1), \qquad P_0\big|_{t=(1-r)^2} = (Xr+1)^3(X^2r-3Xr-3X+1). \tag{61}$$

Lemma (sign of S_1)**.** *If $|X|r < 1$ and (59) holds, then $S_1 \leq 0$.*

Proof. The bracket in S_1 is affine in t with slope $-X$. For $X > 0$ its minimum on the band is at $t = (1 + r)^2$, where it equals $(1 - Xr)^2$. For $X < 0$ the minimum is at $t = (1 - r)^2$, where it equals $(1 + Xr)^2$. □

C.2.4. *Roots, and points on a geodesic.* For $G \neq 0$ write $C(\zeta) = G(\zeta - r_1)(\zeta - r_2)$, the polynomial $q(\zeta) = G\zeta^2 + H\zeta + K$. From $|G|^2|r_1 - r_2|^2 = \Pi$, $|G|^2|r_1 + r_2|^2 = |H|^2$, $|G|^2|r_1 r_2|^2 = |K|^2$:

$$|G|^2(1 - |r_1|^2)(1 - |r_2|^2) = -\tfrac{1}{2}D, \qquad |G|^2|1 - r_1\bar{r}_2|^2 = \Pi - \tfrac{1}{2}D. \tag{62}$$

Hence: if $D > 0$, exactly one root lies in $\mathbb{D}$; if $D < 0$, both roots lie in $\mathbb{D}$ or both outside $\overline{\mathbb{D}}$ (and then the roots $1/r_1, 1/r_2$ of the plant (K, H, G) lie in $\mathbb{D}$); if $D = 0$, a root lies on the unit circle. Also $A(w) = G(1 - r_1\bar{w})(1 - r_2\bar{w})$ and $C(w) = G(w - r_1)(w - r_2)$. If $G = 0$ we take $r_2 = \infty$, i.e. $C(w) = H(w - r_1)$, $A(w) = H\bar{w}(1 - r_1\bar{w})$, $r_1 = -K/H$.

Lemma (outside)**.** *Let $D > 0$, $|r_1| < 1 < |r_2|$, $a = 1/\bar{r}_2$ ($a = 0$ if $G = 0$), $\delta = \rho(r_1, a)$. Then $\mu = (1 - \delta^2)/\delta^2$ (with $\mu = \infty$ when $\delta = 0$), and for $w \in \mathbb{D} \setminus \{a\}$, with $p = \rho(w, r_1)$, $q = \rho(w, a)$:*

$$r = \frac{p}{q}, \qquad t = \frac{(1 - p^2)(1 - q^2)}{\mu\, q^2}.$$

For $\delta > 0$, on the geodesic through r_1 and a: between r_1 and a, $t = (1 + r)^2$ and r takes every value in $[0, \infty)$; beyond r_1, $t = (1 - r)^2$ and r takes every value in $(0, 1)$; beyond a, $t = (1 - r)^2$ and r takes every value in $(1, \infty)$.

Proof. $\delta = |1 - r_1\bar{r}_2|/|r_1 - r_2|$, so by (62) $\Pi(1 - \delta^2) = \frac{1}{2}D$, i.e. $2\Pi - D = 2\Pi\delta^2$ and $\mu = (1 - \delta^2)/\delta^2$. Next $|w - r_1| = p|1 - r_1\bar{w}|$, $|1 - r_2\bar{w}| = |r_2||w - a|$ and $|w - r_2| = |r_2||1 - \bar{a}w|$, so $|C|/|A| = p/q$. By (58) applied to $|1 - r_1\bar{w}|$ and $|1 - \bar{a}w|$, and by (62), $\lambda = 2(1 - p^2)(1 - q^2)/(Dq^2)$; then $t = \frac{1}{2}(2\Pi - D)\lambda = \frac{D}{2\mu}\lambda$. On the geodesic, $\delta = (p + q)/(1 + pq)$ between the two points and $\delta = |p - q|/(1 - pq)$ beyond them, so $1 - \delta^2 = (1 - p^2)(1 - q^2)/(1 \pm pq)^2$, and $t = (1 \pm r)^2$ follows. The ranges follow by continuity and the end values. For $G = 0$: $q = |w|$, $|H|^2(1 - |r_1|^2) = |H|^2 - |K|^2 = \frac{1}{2}D$, and the same computation applies. □

Lemma (inside)**.** *Let $D < 0$ and $r_1, r_2 \in \mathbb{D}$, $\delta = \rho(r_1, r_2)$, $s = 1/\sqrt{1 - \delta^2} \geq 1$. Then $\mu = -1/s^2$, and with $p_j = \rho(w, r_j)$:*

$$r = p_1 p_2, \qquad t = s^2(1 - p_1^2)(1 - p_2^2).$$

On the geodesic through r_1, r_2: between them, $t = (1 + r)^2$ and r takes every value in $[0, r_{\max}]$, $r_{\max} = (s - 1)/(s + 1)$, the maximum being attained at the hyperbolic midpoint; beyond r_1, $t = (1 - r)^2$ and r takes every value in $[0, 1)$.

Proof. $|C|/|A| = p_1 p_2$. By (58) and (62), $\lambda = -2(1 - p_1^2)(1 - p_2^2)/D$, and $1 - \delta^2 = (1 - |r_1|^2)(1 - |r_2|^2)/|1 - r_1\bar{r}_2|^2 = -D/(2\Pi - D)$, so $\mu = -(1 - \delta^2)$ and $t = \frac{1}{2}(2\Pi - D)\lambda = s^2(1 - p_1^2)(1 - p_2^2)$. The geodesic relations are as in the Lemma (outside). At the midpoint $p_1 = p_2 = p$ with $2p/(1 + p^2) = \delta$, so $(1 + p^2)/(1 - p^2) = s$ and $p^2 = (s - 1)/(s + 1)$. If $\delta = 0$ then $s = 1$, $r_{\max} = 0$, and every ray from r_1 serves as the second arc. □

Lemma (on the ellipse)**.** *Let $D = 0$, $\Pi > 0$.* (i) *If $r_1 \in \mathbb{D}$ and $|r_2| = 1$, then on the geodesic from r_1 to r_2 one has $r = \rho(w, r_1) \in [0, 1)$ and $t = (1 + r)^2$.* (ii) *If $|r_1| = |r_2| = 1$, then $r = 1$ for every w, and $t = 4$ on the geodesic joining r_1 and r_2.*

Proof. Here $t = \Pi\lambda$. Let $P(z, \zeta) = (1 - |z|^2)/|\zeta - z|^2$ be the Poisson kernel; for an automorphism φ of $\mathbb{D}$, $P(\varphi z, \varphi\zeta) = P(z, \zeta)/|\varphi'(\zeta)|$ and $|\varphi\zeta_1 - \varphi\zeta_2|^2 = |\varphi'(\zeta_1)||\varphi'(\zeta_2)||\zeta_1 - \zeta_2|^2$ on the circle. (i) $|w - r_2| = |1 - r_2\bar{w}|$ gives $r = \rho(w, r_1)$, and (58) gives $t = (1 - r^2)P(w, r_2)/P(r_1, r_2)$. The automorphism with $\varphi(r_1) = 0$ maps the geodesic to a radius towards $\zeta = \varphi(r_2)$ and w to $r\zeta$, leaving the ratio of Poisson kernels unchanged, so $t = (1 - r^2)(1 + r)/(1 - r) = (1 + r)^2$. (ii) $r = 1$ as in (i), and $t = |r_1 - r_2|^2 P(w, r_1)P(w, r_2)$ is invariant under automorphisms. Sending r_1, r_2 to $1, -1$, the geodesic becomes $(-1, 1)$, where $t = 4(1 - y^2)^2/((1 - y)^2(1 + y)^2) = 4$. □

C.2.5. *The margin on a geodesic.*

Lemma (margin)**.** *If $T=\mu t$, $t=(1+\varrho)^2$ and $r=|\varrho|$, then $S=(1-X\varrho)^2R(\varrho)$ with*

$$R(\varrho)=X^2\big[(X-3)(X+1)-4\mu\big]\varrho^2+2X\big[(1+X)^2-4\mu X\big]\varrho+(1+X)(1-3X)-4\mu X^2.$$

Its discriminant is $16X^2(1+X)^2\big[(1-X)^2-\mu(6X-1-X^2)\big]$, and

$$R(1/X)=-4\mu(1+X)^2,\qquad R(-1)=(1+X)^2(X^2-6X+1),\qquad R(1)\big|_{\mu=0}=(1+X)^2(1-X)^2,$$

$$R\Big(\frac{s-1}{s+1}\Big)\Big|_{\mu=-1/s^2}=\frac{(1-X)^2\big(s(1+X)+1-X\big)^2}{(s+1)^2}.$$

Proof. S depends on r only through r^2; substitute into (60) and factor. □

By the Lemmas (outside), (inside) and (on the ellipse), $t=(1+\varrho)^2$ holds on the geodesics there, with $\varrho=r$ on the arc between the two points and $\varrho=-r$ on the arcs beyond them.

C.2.6. *Proof of the Theorem. No rotation -2 on or outside the ellipse.* Let $D\ge 0$ and $|X|>1$, $|X|r<1$. Then $T\ge 0$, $S_1\le 0$ (the Lemma (sign of S_1)), and P_0 is affine in t with slope $4X^2(1+Xr^2)>0$. For $X>1$, $P_0\le P_0|_{t=(1+r)^2}<0$ by (61), since $X^2r-3Xr+3X-1=Xr(X-3)+3X-1>0$ for $0\le Xr<1$; and $1+X>0$. For $X<-1$, $P_0\ge P_0|_{t=(1-r)^2}>0$, since with $X=-\sigma$ both factors are positive; and $1+X<0$. So $S<0$ by (60). By the Lemmas (the quantity S) and (four tests) no coupling with $|w|\ne 1$ reaches any ω with rotation -2.

The couplings $|w|=1$. Then $C(w)=w^2A(w)$, $r=1$, and $\lambda=0$, so $u=4$, $d=0$. If $A(w)=0$ both criteria fail. Otherwise the test needs $|X|<1$ and $S=(1+X)^4(X^2-6X+1)$, which is positive iff $x<3-2\sqrt{2}$, i.e. $\omega<1/\sqrt{2}$. So these couplings reach exactly $(0,\frac{1}{\sqrt{2}})\setminus\{\frac12\}$, with rotation $+2$.

Statement (b): the ceiling. Let $D>0$, $\omega>\frac12$ (so $0<x<1$), $|w|<1$, and suppose $\mathrm{Crit}(x)$ holds; then $S>0$ and $|x|r<1$. If $\delta=0$, then $2\Pi=D$, $t=0$, $r=1$ and $S=(1+x)^4(x^2-6x+1)+TS_1$ with $S_1\le 0$, so $\omega<1/\sqrt{2}=U$. Let $\delta>0$; then $T=\mu t>0$. Put $h=6x-1-x^2$ and $L=hT-(1-x)^2t$. We show $L<0$. Suppose $L\ge 0$. If $h<0$, then $L\le hT<0$. If $h=0$, then $t=0$, so $r=1$ by (59), and $S=(1+x)P_0(1,0)+TS_1=(1+x)^4(x^2-6x+1)+TS_1=TS_1\le 0$. If $h>0$, then $T\ge T_*=(1-x)^2t/h$ and, S being nonincreasing in T, $hS\le\Phi(r,t):=h(1+x)P_0+(1-x)^2tS_1$. The coefficient of t^2 in Φ is $4x^3(1-x)^2>0$, so Φ is convex in t, and at the ends of the band

$$\Phi\big|_{t=(1+r)^2}=-(rx-1)^2W_+^2,\qquad \Phi\big|_{t=(1-r)^2}=-(rx+1)^2W_-^2,$$

$W_\pm=rx^3-4rx^2-rx\pm(x^2+4x-1)$. So $\Phi\le 0$ and $S\le 0$. In every case $S>0$ fails; hence $L<0$. Since $T/t=\mu=D/(2\Pi-D)$, $L<0$ reads $hD<(1-x)^2(2\Pi-D)$, i.e. $2xD<(1-x)^2\Pi$, i.e. $4\omega^2-1<2\Pi/D$, i.e. $\omega<U$. By the Lemma (four tests) the same holds for the plant (K,H,G), hence for $|w|>1$ and $b=0$, and the couplings $|w|=1$ give only $\omega<1/\sqrt{2}\le U$ (by (56), $2\Pi/D\ge 1$). So $\Omega\subseteq(0,U)$, and only rotation $+2$ occurs.

Statement (b): every $\omega<U$ is reached. Let $D>0$, $X=x$, and first $\delta>0$, so $0<\mu<\infty$. The leading coefficient of R is negative since $(x-3)(x+1)<0$. The vertex ϱ_* satisfies

$$1-x\varrho_*=\frac{(1+x)\big(2(1-x)+4\mu\big)}{(3-x)(1+x)+4\mu}>0,\qquad 1+x\varrho_*=\frac{4(1+x)+4\mu(1-x)}{(3-x)(1+x)+4\mu}>0.$$

If $x<0$, then $6x-1-x^2<0$ and the discriminant of R is positive. If $x>0$ and $\omega<U$, then $\mu(6x-1-x^2)<(1-x)^2$ by the computation above, and again the discriminant is positive. Then $R(\varrho_*)>0$, and $R>0$ on an interval around ϱ_*; choose ϱ there with $\varrho\ne-1$ and $|x\varrho|<1$. By

the Lemma (outside) there is $w \in \mathbb{D}$ on the geodesic with this ϱ, and by the Lemmas (margin) and (the quantity S) it reaches ω with rotation $+2$.

If $\delta = 0$, then $2\Pi = D$, $t \equiv 0$, $r \equiv 1$, and $T = \frac{1}{2}D\lambda$ takes all small positive values as $|w| \to 1$. Then $S = (1+x)^4(x^2-6x+1) - 4x^2(1+x)^2T > 0$ for small T when $\omega < 1/\sqrt{2} = U$.

Statement (c). Let $D < 0$; by (62) and the Lemma (four tests) we may assume both roots in $\mathbb{D}$ (for $G = 0$, $D < 0$ means $|H| < |K|$, and $(K, H, 0)$ has the roots 0, $-H/K$). Use the Lemmas (inside) and (margin) with $\mu = -1/s^2$.

Rotation $+2$: take $w =$ the hyperbolic midpoint of r_1, r_2. Then $R(r_{\max}) > 0$ since $s(1+x) + 1 - x > 0$, and $|x| r_{\max} < 1$; so $S > 0$ and w reaches ω.

Rotation -2: by the table, use the coupling $w = 1/\bar{v}$ with $\mathrm{Crit}(1/x)$ for (G, H, K) at v; put $X = 1/x$, $|X| > 1$. Note $R(1/X) = 4(1+X)^2/s^2 > 0$. If $X > 1$ and $X r_{\max} < 1$, take $v =$ the midpoint ($R(r_{\max}) > 0$, as $X \neq 1$). If $X > 1$ and $X r_{\max} \geq 1$, take v between r_1 and r_2 with $\varrho = r$ slightly below $1/X$. If $X < -1$, take v beyond r_1 with $\varrho = -r$ slightly above $1/X$. In each case $R(\varrho) > 0$ and $|X\varrho| < 1$, so $S > 0$.

Statements (d) and (e). Let $D = 0$. Then $T = 0$, and rotation -2 is excluded above.

(d) $\Pi > 0$, so $\mu = 0$. If one root lies in $\mathbb{D}$ and one on the circle (after passing to (K, H, G) if the other root lies outside; for $G = 0$ or $K = 0$ the roots are 0 and a point of the circle), take w on the geodesic of the Lemma (on the ellipse)(i): $S = (1-xr)^2 R(r) \to (1-x)^4(1+x)^2 > 0$ as $r \to 1$. If both roots lie on the circle, take w as in the Lemma (on the ellipse)(ii): $\varrho = 1$ and $S = (1-x)^2 R(1) = (1+x)^2(1-x)^4 > 0$. Every $\omega \neq \frac{1}{2}$ is reached.

(e) $\Pi = 0$: $q = G(\zeta - r_0)^2$, and $D = -2|G|^2(|r_0|^2 - 1)^2$ by (62), so $|r_0| = 1$, $|K| = |G||r_0|^2 = |G|$, $H = -2Gr_0 = \pm 2\sqrt{GK}$. (For $G = 0$, $\Pi = 0$ forces $H = 0$ and then $D = 0$ forces $K = 0$.) For every w, $A = G(1 - r_0\bar{w})^2$, $C = G(w - r_0)^2$, $B = 2G(w - r_0)(1 - r_0\bar{w})$, so $r = 1$, $d = 0$, $u = 4$, and $S = (1+X)^4(X^2 - 6X + 1)$; the same holds for (K, H, G) and for $|w| = 1$. So $\Omega = (0, \frac{1}{\sqrt{2}}) \setminus \{\frac{1}{2}\}$.

Statement (f). For $D > 0$, $U^2 = \frac{1}{4}(1 + 2\Pi/D) \geq \frac{1}{2}$ by (56), with equality iff $D = 2\Pi$, i.e. $E = \Pi$. With (c)–(e) this gives the floor in all cases. Ω equals the floor in case (b) with $E = \Pi$ and in case (e), where $E = \Pi = 0$; if $E = \Pi$ and $D \leq 0$ then $D = 2\Pi \leq 0$ forces case (e).

Statements (a) and (g). (a) With $f = 2\sqrt{GK}$: $(|H - f| + |H + f|)^2 = |H - f|^2 + |H + f|^2 + 2|H^2 - f^2| = 2|H|^2 + 8|GK| + 2\Pi$, so $|H - f| + |H + f| \leq 2(|G| + |K|)$ iff $D \leq 0$. This is the ellipse with foci $\pm f$ and major semi-axis $|G| + |K|$; its minor semi-axis is $\sqrt{(|G| + |K|)^2 - 4|GK|} = \big||G| - |K|\big|$, and if $f \neq 0$ its major axis lies along f, i.e. along $\arg(GK)/2$. If $f = 0$ (that is, $GK = 0$) the two semi-axes are equal and the set is the disc $|H| \leq |G| + |K|$; if the minor semi-axis is 0 (that is, $|G| = |K|$) the set is the segment $[-f, f]$. At $H = \pm f$, $\Pi = 0$ and $D = 4|GK| - 2|G|^2 - 2|K|^2 = -2(|G| - |K|)^2$. (g) $U^2 = \frac{1}{4}(D + 2\Pi)/D \to +\infty$ as $D \to 0^+$ with Π bounded away from 0.

Statement (h). Expanding both sides as polynomials in $G, \bar{G}, H, \bar{H}, K, \bar{K}$ gives the identity

$$\big(|G|^2 - |K|^2\big)^2 - |K\bar{H} - \bar{G}H|^2 = \tfrac{1}{4}\big(E^2 - \Pi^2\big) = \tfrac{1}{4}D(D - 2\Pi).$$

By (56), $D - 2\Pi \leq 0$, with equality only if $D = 2\Pi \geq 0$. If $D < 0$ the right side is positive, so $|G| \neq |K|$ and the inequality of (H3) holds. Conversely, (H3) makes the left side positive, hence $D(D - 2\Pi) > 0$, which with $D - 2\Pi \leq 0$ forces $D < 0$.

Statement (i). Necessity. For a coupling $(1, w)$ with $|w| = 1$: $C = w^2 A$, so $|d_{20}| = |d_{02}|$. For $|w| \neq 1$ and $B = 0$, the Lemma (invariants) gives $-4AC = (H^2 - 4GK)(1 - |w|^2)^2$ and

$-2(|A|^2+|C|^2) = E(1-|w|^2)^2$, hence

$$2\big(|A|-|C|\big)^2 = -(E+\Pi)(1-|w|^2)^2 = -D\,(1-|w|^2)^2,$$

so $|A| \neq |C|$ forces $D<0$. By the Lemma (normalisation)(i) this covers every coupling with $b \neq 0$. For $b=0$: $(d_{20}, d_{11}, d_{02}) = (\bar{b}'^2 K, |b'|^2 H, b'^2 G)$, so $d_{11}=0$ means $H=0$, where $D = 4|GK| - 2|G|^2 - 2|K|^2 = -2(|G|-|K|)^2$, and $|d_{20}| \neq |d_{02}|$ means $|G| \neq |K|$, i.e. $D<0$.

Sufficiency. Let $D<0$. If both roots lie in $\mathbb{D}$, let w be their hyperbolic midpoint. By the Lemma (inside), $r = r_{\max} = (s-1)/(s+1) < 1$, $t = (1+r)^2 = 4s^2/(s+1)^2$ and $T = \mu t = -t/s^2$, so

$$u = 2(1+r^2) - t + T = \frac{4(s^2+1)}{(s+1)^2} - \frac{4s^2}{(s+1)^2} - \frac{4}{(s+1)^2} = 0,$$

that is $B=0$, and $|C|/|A| = r < 1$. If both roots lie outside $\overline{\mathbb{D}}$, the plant (K,H,G) has both roots in $\mathbb{D}$; its midpoint v gives $B=0$ and $|C|<|A|$ for (K,H,G) at v, hence, by the Lemma (normalisation)(iv), $B=0$ and $|A|<|C|$ for (G,H,K) at $\bar{v}$, and then, by the Lemma (normalisation)(iii), the coupling $(1, 1/v)$ for (G,H,K) (or $(0,1)$ if $v=0$) has $d_{11}=0$ and $|d_{02}|<|d_{20}|$. If $G=0$, $D<0$ means $|H|<|K|$, and $(K,H,0)$ has the roots 0, $-H/K$ in $\mathbb{D}$, so the same argument applies. □

C.3. **Proof of the Lemma on quadratics without folds.** This subsection proves the Lemma of §5.3 above.

Outline. On the unit circle the Jacobian of Q is a positive multiple of the derivative of $\arg Q$, so (H3) makes $\arg Q(e^{i\theta})$ strictly monotone. A root count bounds the winding number, monotonicity makes it ± 2, and homogeneity extends the conclusions from the circle to the plane.

Proof. Q is even and homogeneous of degree two. At $u = e^{i\theta}$, Euler's identity $dQ(u)\,u = 2Q(u)$ gives Jacobian $= 2|Q(u)|^2\,\partial_\theta \arg Q(e^{i\theta})$, so under (H3) $Q(e^{i\theta}) \neq 0$ and $\arg Q(e^{i\theta})$ is strictly monotone. Since $Q(e^{i\theta}) = e^{-2i\theta}\big(Ge^{4i\theta} + He^{2i\theta} + K\big)$ and $G\zeta^2 + H\zeta + K$ has at most two roots in the unit disc, the winding number of $\theta \mapsto Q(e^{i\theta})$, $0 \le \theta \le 2\pi$, lies in $\{-2, 0, 2\}$; it is nonzero by monotonicity. So $\theta \mapsto \arg Q(e^{i\theta})$ covers the circle twice, each direction being attained at two antipodal points, and homogeneity gives the first two assertions. The bound on dQ^{-1} follows by homogeneity from compactness of the unit circle. □

C.4. **Proof of the Theorem for H strictly inside the ellipse.** This subsection proves the Theorem of §5.3 above.

Outline. (i) Writing $x_< = s^2 e^{i\phi}$ and $p = sq$ turns the pointwise equation (50) into $Q(q) = i\delta e^{i\phi}$ plus a small term; the Lemma and the implicit function theorem give exactly two roots, exchanged once around $x_< = 0$. (ii) Labelling the branch by z with $z^2 = x_<$ turns the closed loop into a skew product over the reference, in which the complement $x_>$ is driven by a forcing of size $|z|^2$ and contracts; this makes the cone $\mathcal{N}_\varepsilon$ positively invariant. (iii) Over the reference cycle the period map of $x_>$ is a contraction, and its fixed point is the cycle γ. (iv) The periods are read off from $z(t)$.

Proof. (i) The pointwise equation. Write $x_< = s^2 e^{i\phi}$ with $s = |x_<|^{1/2}$, and $p = sq$. Dividing (50) by s^2 gives

$$Q(q) = i\delta e^{i\phi} + 2(\varepsilon - s^2)e^{i\phi} - s^{-2}T(sq, x). \tag{63}$$

On $\mathcal{N}_\varepsilon$, $|x| \le (1+K_0)s^2$ and $s^2 \le 2\varepsilon$, so for $|q| \le R_0$ the bounds (49) give $|s^{-2}T(sq,x)| \le Cs$ and $|\partial_q(s^{-2}T)| \le Cs$, with C depending on K_0 and R_0. The right side of (63) is therefore $i\delta e^{i\phi} + O(\sqrt{\varepsilon})$, uniformly in ϕ, $x_>/s^2$ and $|q| \le R_0$. By the Lemma of §5.3, $Q(q) = i\delta e^{i\phi}$ has the two solutions $\pm q_*(\phi)$, with $|q_*|$ bounded above and below by multiples of $|\delta|^{1/2}$ and $dQ(\pm q_*)$ uniformly invertible. Take $R_0 = 2\max|q_*|$. The implicit function theorem, uniform in

the parameters by compactness, gives for small ε exactly two solutions of (63) in $|q| \le R_0$, one near each of $\pm q_*(\phi)$, C^k in x for $x_< \neq 0$. Following ϕ once around the circle turns q_* into $-q_*$, because Q winds twice around 0 on the unit circle while $i\delta e^{i\phi}$ winds once; so the two solutions are exchanged. Finally $|p| = s|q| \le R_0|x_<|^{1/2}$.

(ii) The skew product. By the Lemma the loop $\theta \mapsto i\delta e^{2i\theta}$ lifts through Q to a closed loop $q_1(\theta)$; for $z \neq 0$ let $p(z, x_>) = |z|\, q$ with q the solution of (i) near $q_1(\arg z)$. Then $p(z, x_>)$ and $p(-z, x_>)$ are the two branches at $x_< = z^2$, and along a solution with $x_< \neq 0$ choosing $z(t)$ continuously with $z^2 = x_<$ selects a branch continuously. By (50), $\dot{x}_< = V(x_<)$, i.e. $2z\dot{z} = 2z\, w(z)$, so $\dot{z} = w(z)$; the second equation of (51) is the complement equation of (33) under the feedback. The disc $|z|^2 \le 2\varepsilon$ is positively invariant for w, and $\frac{d}{dt}|z|^2 = 2|z|^2(\varepsilon - |z|^2) \ge -2\varepsilon|z|^2$ on it. Fix a norm with $\langle x_>, A_L x_>\rangle \le -a|x_>|^2$, $a > 0$. On $\tilde{\mathcal{N}}_\varepsilon$, $|u| \le |L|K_0|z|^2 + R_0|z|$, and $R(u, x)$ has order ≥ 2, so the forcing in (51) is at most $C_2|z|^2$ with C_2 independent of K_0 once ε is small. Then $\frac{d}{dt}|x_>| \le -a|x_>| + C_2|z|^2$, and with $K_0 = 2C_2/a$ and $\varepsilon < a/4$,

$$\frac{d}{dt}\big(|x_>| - K_0|z|^2\big) \le \big(-aK_0 + C_2 + 2K_0\varepsilon\big)|z|^2 < 0$$

where $|x_>| = K_0|z|^2$, $z \neq 0$: $\tilde{\mathcal{N}}_\varepsilon$ is positively invariant. By (49) and the Lemma, $\partial_{x_>} p = -(dQ(p) + \partial_p T)^{-1}\partial_{x_>} T = O(1)$, so the forcing is Lipschitz in $x_>$ with constant $\ell = O(\sqrt{\varepsilon})$, and (51) has unique solutions for $z \neq 0$. If $x(0) = 0$, then $x_>(0) = 0$; were $x_<(t_1) \neq 0$ for some $t_1 > 0$, the lift $z(t)$ on $(t_0, t_1]$, t_0 the last time before t_1 with $x_<(t_0) = 0$, would solve $\dot{z} = w(z)$ with $z(t) \to 0$ as $t \to t_0^+$, which is impossible since w is Lipschitz and $w(0) = 0$.

(iii) The cycle. On $|z|^2 = \varepsilon$, $z(t) = \sqrt{\varepsilon}\, e^{i(\omega t + \theta_0)}$, and the second equation of (51) is $2\pi/\omega$-periodic in t, contracting at rate a up to the Lipschitz perturbation $\ell < a/2$. Its period map is a contraction of the positively invariant ball $|x_>| \le K_0\varepsilon$, so it has a unique periodic solution $x_>^*(t)$, and from $\frac{d}{dt}|x_>| \le -a|x_>| + C_2\varepsilon$ one gets $|x_>^*| \le C_2\varepsilon/a = \frac{1}{2}K_0\varepsilon$: γ lies in the interior of $\mathcal{N}_\varepsilon$. Different θ_0 give time shifts of one orbit. For any solution in $\tilde{\mathcal{N}}_\varepsilon$ with $z(0) \neq 0$, $|z(t)|^2 \to \varepsilon$ and $\arg z(t) - \omega t$ converges, both exponentially; the difference between $x_>(t)$ and the periodic solution with the limiting phase then obeys a contracting equation with exponentially decaying forcing, and tends to 0. The closed loop is C^k near γ, where $x_< \neq 0$; its Floquet multipliers are 1, $e^{-4\pi\varepsilon/\omega}$ (the radial multiplier of the reference, since $\frac{d}{dt}|z|^2 = 2|z|^2(\varepsilon - |z|^2)$ has derivative -2ε at $|z|^2 = \varepsilon$) and the multipliers of the period map of $x_>$, of modulus at most $e^{-(a-\ell)2\pi/\omega}$. So γ is hyperbolic and attracting, with asymptotic phase.

(iv) Along γ, $x_<(t) = z(t)^2$ and $u(t) = Lx_>^*(t) + p\big(z(t), x_>^*(t)\big)$, with $z(t)$ of period $2\pi/\omega$.

For the graph $x_< = \bar{z}^2$, $\frac{d}{dt}\bar{z}^2 = 2(\varepsilon - |z|^2 - i\omega)\bar{z}^2$, and (50) becomes $Q(p) + T = [2(\varepsilon - |x_<|) - i(2\omega + \omega_0)]x_<$; the proof is the same. The implicit function theorem in (i) needs the $O(\sqrt{\varepsilon})$ perturbation small compared with $|\delta|$, whence $\varepsilon_0 \to 0$ as $\delta \to 0$. □

C.5. **Proof of the Theorem with friction.** This subsection proves the Theorem of §5.4 above.

Outline. After scaling, the product system is a skew product over the reference, with a two-dimensional fibre (Step 1). A normal-form substitution removes all terms of order ε in the fibre equation except the friction term $\varepsilon\kappa|\vartheta|^2\zeta$; this is where $2\omega \neq \omega_0$ and $4\omega \neq \omega_0$ are used (Step 2). An invariant section of the fibre equation is the fixed point of a contraction, over the disc (Step 3) and over a surrounding annulus (Step 4). The condition $\operatorname{Re}\kappa < -2$ makes the fibre contract faster than the reference contracts towards its cycle, and this makes the section C^1. Written as a function of $\nu = z^2$, the section gives the map Φ_ε (Step 5), which conjugates the field V to the closed loop (Step 6).

Proof. Step 1: scaling. Put $z = \sqrt{\varepsilon}\,\vartheta$, $x = \varepsilon y$, $s = |\vartheta|^2$, $\psi(\vartheta) = b\vartheta + b'\bar{\vartheta}$, $\ell(\vartheta) = c\vartheta + c'\bar{\vartheta}$. Then $u = \sqrt{\varepsilon}\,(\psi + \varepsilon\ell y)$, and from $Q(\psi + k) = Q(\psi) + Q_u(\psi)k + Q_{\bar{u}}(\psi)\bar{k} + Q(k)$ the product system

is exactly

$$\dot\vartheta = w(\vartheta) := i\omega\vartheta + \varepsilon(1-s)\vartheta, \qquad \dot y = i\omega_0 y + Q(\psi) + \varepsilon\big(\alpha(\vartheta)y + \beta(\vartheta)\bar y\big) + \varepsilon^2 Q(\ell y), \tag{64}$$

$$\alpha = c\kappa_1\vartheta^2 + \kappa s + c'\kappa_0\bar\vartheta^2, \qquad \beta = \bar c'\lambda_1\vartheta^2 + (\bar c\lambda_1 + \bar c'\lambda_0)s + \bar c\lambda_0\bar\vartheta^2.$$

It is a skew product over the reference, and it is invariant under $(\vartheta, y) \mapsto (-\vartheta, y)$.

Step 2: normal form. Let L be the derivative along $i\omega\vartheta$, so that $L(\vartheta^j\bar\vartheta^m) = i(j-m)\omega\,\vartheta^j\bar\vartheta^m$; call $j-m$ the harmonic of the monomial. By (38), $LF_2 = i\omega_0 F_2 + Q(\psi)$, and F_2 is homogeneous of degree two. Hence $\tilde y = y - F_2$ obeys $\dot{\tilde y} = i\omega_0\tilde y + \varepsilon\varphi_1 + \varepsilon(\alpha\tilde y + \beta\bar{\tilde y}) + \varepsilon^2 Q(\ell y)$ with $\varphi_1 = \alpha F_2 + \beta\bar F_2 - 2(1-s)F_2$, a polynomial whose harmonics lie in $\{0, \pm2, \pm4\}$. Solve $(L - i\omega_0)F_3 = \varphi_1$ monomial by monomial; the divisors $i(k\omega - \omega_0)$, $k \in \{0, \pm2, \pm4\}$, are nonzero because $\omega_0 > 0$, $2\omega \neq \omega_0$, $4\omega \neq \omega_0$. Solve $L\Xi = \alpha - \kappa s$ (harmonics ±2, divisors $\pm2i\omega$) and $(L - 2i\omega_0)\Theta = \beta$ (harmonics $2, 0, -2$, divisors $i(2\omega - 2\omega_0)$, $-2i\omega_0$, $-i(2\omega + 2\omega_0)$; the first occurs only if $c'\lambda_1 \neq 0$). The substitution $y = F_2 + \varepsilon F_3 + \zeta + \varepsilon(\Xi\zeta + \Theta\bar\zeta)$ turns (64) into

$$\dot\zeta = \big(i\omega_0 + \varepsilon\kappa s\big)\zeta + \varepsilon^2 f(\vartheta, \zeta), \tag{65}$$

since the $\varepsilon(1-s)\vartheta$ part of w acting on F_3, Ξ, Θ, and the $O(\varepsilon)$ part of $\dot{\tilde\zeta}$, contribute only at order ε^2. Here f need not be a polynomial, since solving the substitution for ζ inverts the real-linear map $\zeta \mapsto \zeta + \varepsilon(\Xi\zeta + \Theta\bar\zeta)$; but it is smooth in (ϑ, ζ) with derivatives bounded uniformly for small ε, and it is even in ϑ. Moreover $f(0,0) = 0$: at $\vartheta = 0$ the functions $F_2, F_3, \Xi, \Theta, \alpha, \beta$ vanish, so $y = \zeta$ and $\zeta = 0$ is an equilibrium. Evenness gives $\partial_\vartheta f(0, \zeta) = 0$ for every ζ. Hence on $\{s \le 2,\ |\zeta| \le 1\}$, with a constant C_f,

$$|f(\vartheta, 0)| \le C_f s, \qquad |f(\vartheta, \zeta_1) - f(\vartheta, \zeta_2)| \le C_f|\zeta_1 - \zeta_2|, \qquad \|\partial_\vartheta f(\vartheta, \zeta)\| \le C_f|\vartheta|.$$

Step 3: the invariant section over the disc $s \le 1$. The flow ϕ_τ of w preserves $\{s \le 1\}$ and the circle $s = 1$, and $\dot s = 2\varepsilon s(1-s)$. Along the backward orbit of ϑ, let $I(\tau) = \int_\tau^0 s(\phi_\sigma\vartheta)\,d\sigma$ and $E(\tau) = \exp\int_\tau^0 (i\omega_0 + \varepsilon\kappa s(\phi_\sigma\vartheta))\,d\sigma$, so $|E(\tau)| = e^{-\varepsilon|\operatorname{Re}\kappa| I(\tau)}$. A bounded invariant section $\zeta = Z(\vartheta)$ of (65) is a fixed point of

$$(\mathcal T Z)(\vartheta) = \varepsilon^2 \int_{-\infty}^0 E(\tau)\, f\big(\phi_\tau\vartheta, Z(\phi_\tau\vartheta)\big)\, d\tau.$$

Since $dI = -s\,d\tau$, and since $I(-\infty) = \int_0^{s_0} \frac{ds}{2\varepsilon(1-s)} = -\frac{\ln(1-s_0)}{2\varepsilon}$ for $s_0 = s(\vartheta) < 1$ (and $I(-\infty) = \infty$ on the circle),

$$\int_{-\infty}^0 |E(\tau)|\, s(\phi_\tau\vartheta)\, d\tau = \int_0^{I(-\infty)} e^{-\varepsilon|\operatorname{Re}\kappa| I}\, dI \le \min\Big\{\frac{-\ln(1-s_0)}{2\varepsilon}, \frac{1}{\varepsilon|\operatorname{Re}\kappa|}\Big\} \le \frac{2s_0}{\varepsilon},$$

using $-\ln(1-s_0) \le 2s_0$ for $s_0 \le \frac12$ and $|\operatorname{Re}\kappa| > 2$. In the norm $\|Z\| = \sup_{0<s\le1} |Z(\vartheta)|/s$, $\mathcal T$ therefore maps the ball $\|Z\| \le 4C_f\varepsilon$ into itself and is a contraction with constant $2C_f\varepsilon$, for small ε. Its fixed point Z is continuous, even in ϑ, and $|Z(\vartheta)| \le 4C_f\varepsilon s$.

For the derivative, $P(t) = DZ(\phi_t\vartheta)$, a real-linear map of $\mathbb C$, satisfies along orbits

$$\dot P = (i\omega_0 + \varepsilon\kappa s)P - P\,Dw(\vartheta) + \varepsilon^2\,\partial_\zeta f \cdot P + \varepsilon\kappa\, Z\, Ds + \varepsilon^2\partial_\vartheta f,$$

where $\partial_\zeta f \cdot P$ denotes the real derivative of f in ζ composed with P. The symmetric part of $Dw = \omega J + \varepsilon[(1-s)I - 2\vartheta\vartheta^\top]$ (with ϑ read as a vector of $\mathbb R^2$ and J the rotation by $\pi/2$) has the eigenvalues $\varepsilon(1-s)$ and $\varepsilon(1-3s)$, so the logarithmic norm of the homogeneous part is at most $\varepsilon\operatorname{Re}\kappa\, s + \varepsilon(3s-1) + C_f\varepsilon^2 = \varepsilon[(\operatorname{Re}\kappa + 3)s - 1] + C_f\varepsilon^2 \le -\varepsilon m/2$ on $s \le 1$, with $m = \min\{1, -(\operatorname{Re}\kappa + 2)\} > 0$, for small ε. The inhomogeneous terms are $O(\varepsilon^2|\vartheta|)$, because $|Z| \le C\varepsilon s$ and $|Ds| = 2|\vartheta|$. Hence the derivative equation has a unique bounded solution along every orbit in the closed disc, of size $O(\varepsilon)$. To see that it is DZ: the iterates $Z_k = \mathcal T^k(0)$ are C^1 and

converge uniformly to Z; their derivatives obey the derivative equation with the inhomogeneity evaluated at Z_{k-1}, and by the rate bound the map from DZ_{k-1} to DZ_k is a contraction in the uniform norm. So the DZ_k converge uniformly, Z is C^1, and $DZ = P$. The same argument for $P/|\vartheta|$, using $\frac{d}{dt}|\vartheta| = \varepsilon(1-s)|\vartheta|$, has the rate $\varepsilon[(\operatorname{Re}\kappa + 4)s - 2] \le -\varepsilon \min\{2, -(\operatorname{Re}\kappa + 2)\}$ and bounded inhomogeneity, so $\|DZ(\vartheta)\| \le C\varepsilon|\vartheta|$. It is here, and only here, that $\operatorname{Re}\kappa < -2$ is used: on the circle the derivative of the reference contracts in the radial direction at the rate 2ε, and the fibre must contract faster.

Step 4: the annulus $1 \le s \le 2$. Every ϑ with $1 < s \le 2$ lies on the forward orbit of a unique point of the circle $s = 2$. Define Z there by $Z = 0$ on $s = 2$ and transport by (65). Along forward orbits $\frac{d}{dt}|Z| \le -\varepsilon|\operatorname{Re}\kappa||Z| + C_f\varepsilon^2(s + |Z|)$, so $|Z| \le C\varepsilon$. For the derivative, choose $\eta > 0$ with $(\operatorname{Re}\kappa + 3)(1+\eta) < 1$: on $1+\eta \le s \le 2$ the rate is at most 5ε, during a time at most $\frac{1}{2\varepsilon}\ln\frac{1+\eta}{2\eta}$, so the growth is by a bounded factor; on $s \le 1+\eta$ the rate is negative. So $\|DZ\| \le C\varepsilon$ on the annulus. As $s \to 1$, Z and DZ converge to the values of Step 3 on the circle, because both equations contract there and their bounded solutions over the circle are unique. Hence Z is C^1 and even on $\{s \le 2\}$, with $|Z| \le C\varepsilon s$ and $\|DZ\| \le C\varepsilon|\vartheta|$.

Step 5: the map Φ_ε. Undoing Step 2, the surface $\{y = Y(\vartheta)\}$, $Y = F_2 + \varepsilon F_3 + Z + \varepsilon(\Xi Z + \Theta\bar{Z})$, is invariant for (64), $Y(-\vartheta) = Y(\vartheta)$, and $|Y - F_2| \le C\varepsilon s$, $\|DY - DF_2\| \le C\varepsilon|\vartheta|$ on $s \le 2$. Write $Y(\vartheta) = \hat{Y}(\vartheta^2)$ and put $\Phi_\varepsilon(\nu) = \varepsilon\hat{Y}(\nu/\varepsilon)$. Since $\hat{F}_2$ is homogeneous of degree one and $\|D(\vartheta^2)^{-1}\| = 1/(2|\vartheta|)$, this gives the two estimates of (i). Next, $\hat{F}_2$ is a homeomorphism of $\mathbb{C}$: by $\det D(z^2) = 4|z|^2 > 0$ its Jacobian has the constant sign of that of F_2; and since F_2 is homogeneous of degree two with sign-definite Jacobian, $\arg F_2(e^{i\theta})$ is strictly monotone (as in the proof of the Lemma of §5.3, Appendix C.3 above), $F_2(e^{i\theta}) = e^{2i\theta}(c_{20} + c_{11}e^{-2i\theta} + c_{02}e^{-4i\theta})$ winds $\sigma = \pm 2$ times, and $\hat{F}_2$ restricted to the unit circle is a simple closed curve winding once. Being homogeneous, $\hat{F}_2$ is then bi-Lipschitz, with $|\hat{F}_2(\mu_1) - \hat{F}_2(\mu_2)| \ge m_0|\mu_1 - \mu_2|$. As $\hat{Y} - \hat{F}_2$ has Lipschitz constant at most $C\varepsilon < m_0$, $\hat{Y}$ is injective, and it is a local diffeomorphism off 0 with the orientation of $\hat{F}_2$. So Φ_ε is a homeomorphism onto its image $\mathcal{N}_\varepsilon$, which is a neighbourhood of 0.

Step 6: the closed loop. For $x = \Phi_\varepsilon(\nu)$ and $z^2 = \nu$, the coupling gives $h(x, -z) = -h(x, z)$, and Q is even, so both branches give the same vector field $X_\varepsilon(x) = i\omega_0 x + Q(h(x, z))$ on $\mathcal{N}_\varepsilon$. On the invariant surface the reference moves $\nu = z^2$ by $\dot{\nu} = 2z\dot{z} = 2(\varepsilon + i\omega - |\nu|)\nu = V(\nu)$, so $X_\varepsilon(\Phi_\varepsilon(\nu)) = D\Phi_\varepsilon(\nu)V(\nu)$ for $\nu \ne 0$, and both sides vanish at 0: Φ_ε conjugates V to the closed loop, which proves (ii). The field V has the repelling origin and the attracting hyperbolic cycle $|\nu| = \varepsilon$, traversed in time π/ω, which attracts every $\nu \ne 0$ of the disc. Its image under Φ_ε, a C^1 diffeomorphism near the cycle, is hyperbolic and attracting, and x turns once around 0 along it, in the sense of σ. Solutions are unique, since they correspond to solutions of the Lipschitz field V. The control $h(x(t), z(t))$ follows the reference $z(t)$ and has period $2\pi/\omega$. Finally $|\Phi_\varepsilon(\nu) - \hat{F}_2(\nu)| \le C\varepsilon|\nu| = C\varepsilon^2$ on $|\nu| = \varepsilon$. □

C.6. **Proof of the Theorem for the full plant.** This subsection proves the Theorem of §5.5 above.

Outline. A centre manifold of the product system reduces it to the critical plane, as a skew product over the reference (Step 1). The plant's own terms of order two and three now enter the scaled critical equation at order $\sqrt{\varepsilon}$ (Step 2). A normal form in two steps removes them, except for a constant shift κ_2 of the friction, at the price of the three further resonances $3\omega = \omega_0$, $\omega = \omega_0$ and $\omega = 2\omega_0$ (Step 3). The construction of the invariant section in Appendix C.5 above then applies with $\kappa + \kappa_2$ in place of κ (Step 4). The section is a branched double covering (Step 5), and the closed loop near it is analysed in the coordinates $(z, x_> - Y_>(z))$ (Step 6).

Proof. Step 1: reduction to the critical plane. Regard ε as a parameter. At $\varepsilon = 0$ the product system has at the origin the eigenvalues $\pm i\omega$ (reference) and $\pm i\omega_0$ (critical mode) on the imaginary axis, and those of A_L, with real parts $\le -a < 0$. By the center manifold theorem for families depending on parameters, in the form of the reduction principle [71, 72], the family has near the origin a C^5 locally invariant center manifold $S_\varepsilon = \{x_> = \mathcal{H}(z, x_<, \varepsilon)\}$, defined on a neighbourhood of fixed size for all small ε, tangent at the origin to the eigenspace $x_> = Mx_<$ of the critical eigenvalues, and attracting the nearby orbits (a center manifold of any finite class C^k exists for a C^k family, on a neighbourhood depending on k; class C^5 is what Step 3 uses); we write S and $\mathcal{H}(z, x_<)$ for fixed ε; the region $|z|^2 \le 2\varepsilon$, $|x_<| \le C\varepsilon$ lies inside it for small ε. Its Taylor expansion is fixed by invariance: with $z = \sqrt{\varepsilon}\vartheta$, $x_< = \varepsilon y$, $\mathcal{H} = \varepsilon\big(\Lambda y + H_2(\vartheta)\big) + O(\varepsilon^{3/2})$, where Λ solves $A_L\Lambda - \Lambda J_0 = -A_{><}$ (J_0 the multiplication by $i\omega_0$) and H_2 solves $L_\omega H_2 = A_L H_2 + R_{uu}(\psi) - \Lambda Q(\psi)$ with L_ω the operator L of Step 2 in Appendix C.5 above; both equations are uniquely solvable because A_L is Hurwitz. On S the product system is a skew product over the reference, with the two-dimensional fibre y.

Step 2: the scaled critical equation. Expanding $Q(u,x)/\varepsilon$ on S with $u = \sqrt{\varepsilon}\,\psi(\vartheta) + \varepsilon L(\Lambda y + H_2) + \varepsilon^{3/2}\ell(\vartheta)y + O(\varepsilon^{3/2})$ gives

$$\dot y = i\omega_0 y + Q(\psi) + \sqrt{\varepsilon}\big(a_1(\vartheta)y + a_1'(\vartheta)\bar y + q_1(\vartheta)\big) + \varepsilon\big(\tilde\alpha(\vartheta)y + \tilde\beta(\vartheta)\bar y + q_2(\vartheta) + \varpi(y)\big) + O(\varepsilon^{3/2}),$$

where a_1, a_1' are linear in $(\vartheta, \bar\vartheta)$ (harmonics ±1), q_1 is cubic (harmonics $\pm1, \pm3$), $\tilde\alpha, \tilde\beta$ are quadratic (harmonics $0, \pm2$) with $\tilde\alpha = \alpha + \alpha_2$, α as in Step 1 of Appendix C.5, q_2 has degree four (harmonics $0, \pm2, \pm4$), and ϖ is a quadratic form in $(y, \bar y)$ with constant coefficients. The $\sqrt{\varepsilon}$ terms come from $2B_{uu}(\psi, L(\Lambda y + H_2))$, $Q_{ux}(\psi, (y, \Lambda y + H_2))$ and the cubic part $Q_3(\psi)$; the coefficients c, c' enter only through α, via $2B_{uu}(\psi, \ell y) = \alpha y + \beta\bar y$.

Step 3: normal form. Substitute $y = F_2 + \sqrt{\varepsilon}F_{5/2} + \varepsilon F_3 + \zeta + \sqrt{\varepsilon}(\Xi_1\zeta + \Theta_1\bar\zeta) + \varepsilon(\Xi_2\zeta + \Theta_2\bar\zeta)$ and solve the homological equations order by order, as in Step 2 of Appendix C.5. At order $\sqrt{\varepsilon}$ the forcing has harmonics $\pm1, \pm3$ (divisors $i(k\omega - \omega_0)$, nonzero since $\omega \ne \omega_0$, $3\omega \ne \omega_0$), the ζ-terms have harmonics ±1 (divisors $\pm i\omega$), and the $\bar\zeta$-terms harmonics ±1 (divisors $i(\pm\omega - 2\omega_0)$, nonzero since $\omega \ne 2\omega_0$). At order ε, after the second-order terms produced by the first step are added, the forcing has harmonics $0, \pm2, \pm4$ (divisors nonzero since $2\omega, 4\omega \ne \omega_0$), the ζ-terms harmonics $0, \pm2$, and the $\bar\zeta$-terms harmonics $0, \pm2$ (divisors $i(k\omega - 2\omega_0)$, nonzero since $\omega \ne \omega_0$). Every term is removed except the stationary ζ-term, which has a coefficient of the form $(\kappa + \kappa_2)s$, $s = |\vartheta|^2$: it is of degree two and harmonic 0. This defines κ_2; it collects the stationary parts of α_2 and of the products of the $\sqrt{\varepsilon}$-coefficients, none of which involves c, c'. The result is

$$\dot\zeta = \big(i\omega_0 + \varepsilon(\kappa + \kappa_2)s\big)\zeta + \varepsilon\varpi(\zeta) + \varepsilon^{3/2} f(\vartheta, \zeta, \sqrt{\varepsilon}),$$

where ϖ is the quadratic form of Step 2, kept apart from f. On $\{s \le 3,\ |\zeta| \le 1\}$, with a constant C_f,

$$|f(\vartheta, 0)| \le C_f s, \qquad |f(\vartheta, \zeta_1) - f(\vartheta, \zeta_2)| \le C_f|\zeta_1 - \zeta_2|, \qquad \|\partial_\vartheta f(\vartheta, \zeta)\| \le C_f\big(|\vartheta| + |\zeta|\big). \quad (66)$$

The last bound cannot be improved to $C_f|\vartheta|$ as in Appendix C.5: a plant term $u\,x_<^2$ in $Q(u,x)$ contributes $\psi(\vartheta)\zeta^2$ to f, whose ϑ-derivative at $\vartheta = 0$ is the real-linear map $\delta \mapsto \psi(\delta)\zeta^2$, nonzero for $\zeta \ne 0$. To prove (66), go back to the original variables. There $z = \sqrt{\varepsilon}\vartheta$ and $x_< = \mathcal{P}(z, \varepsilon) + (1 + \Xi)\xi + \Theta\bar\xi$ with $\xi = \varepsilon\zeta$, where $\mathcal{P}$, Ξ, Θ collect the terms of the substitution above; the parity of the harmonics makes every power of ε in them an integer power once $\vartheta = z/\sqrt{\varepsilon}$ is inserted. The ξ-equation is $\dot\xi = (i\omega_0 + (\kappa + \kappa_2)|z|^2)\xi + \varpi(\xi) + \mathcal{R}(z, \xi, \varepsilon)$ with $\mathcal{R}$ of class C^5, and $f(\vartheta, \zeta) = \varepsilon^{-5/2}\mathcal{R}(\sqrt{\varepsilon}\vartheta, \varepsilon\zeta, \varepsilon)$. Give z the weight 1 and ε, ξ the weight 2; a monomial $z^j\varepsilon^k\xi^l$ of weight $j + 2k + 2l$ becomes $\varepsilon^{(j+2k+2l-5)/2}\vartheta^j\zeta^l$ in f. The normal form has removed every term of $\mathcal{R}(z, 0, \varepsilon)$ of weight ≤ 4 and every term of $\partial_\xi\mathcal{R}(z, 0, \varepsilon)$ of weight ≤ 2, and the only part of $\mathcal{R}$ quadratic in $(\xi, \bar\xi)$ with constant coefficients is ϖ, which has been taken out; by

Taylor's theorem for the C^5 function $\mathcal{R}$, the remainders have the corresponding weighted orders. Moreover $\mathcal{R}(z,0,\varepsilon)=O(|z|^2)$ for every ε: every term of $\mathcal{P}$ has degree ≥ 2 in z, the reference has the linear part $(i\omega+\varepsilon)z$, and at $\xi=0$ the plant's critical equation contributes, besides $i\omega_0 x_<$, only terms of order ≥ 2 in (u,x), all of whose factors vanish at $z=0$ (the complement $x_>$ on the center manifold vanishes at $z=0$, and receives no term linear in z because $C_>p=0$). Hence on $s\leq 3$: every term of $f(\vartheta,0)$ is bounded and of degree ≥ 2 in ϑ, which gives $|f(\vartheta,0)|\leq C_f s$ and $\|\partial_\vartheta f(\vartheta,0)\|\leq C_f|\vartheta|$; $\partial_\zeta f=\varepsilon^{-3/2}\partial_\xi\mathcal{R}$ is bounded, which gives the Lipschitz bound; and $\partial_\zeta\partial_\vartheta f=\varepsilon^{-1}\partial_\xi\partial_z\mathcal{R}$ is bounded, so that $\|\partial_\vartheta f(\vartheta,\zeta)\|\leq\|\partial_\vartheta f(\vartheta,0)\|+C|\zeta|$.

Step 4: the invariant surface. Steps 3–4 of Appendix C.5 apply, except for the evenness in ϑ, with κ replaced by $\kappa+\kappa_2$ and ε^2 by $\varepsilon^{3/2}$, and with two additions. First, the ball of the contraction is now $|Z|\leq C\sqrt{\varepsilon}s$, and on it the term $\varepsilon\varpi(Z)$ is at most $C\varepsilon^2 s$, with Lipschitz constant $C\varepsilon^{3/2}$; it therefore enters the contraction and the derivative equation like the terms of $\varepsilon^{3/2}f$, and its derivative $\varepsilon D\varpi(Z)$, of size $C\varepsilon^{3/2}$, is absorbed into the rate. Second, the derivative estimate uses $\partial_\vartheta f$ only along the section and along the iterates Z_k, which lie in the same ball; there $|\zeta|\leq C\sqrt{\varepsilon}s\leq|\vartheta|$ for small ε, since $s\leq 3$, so (66) gives $\|\partial_\vartheta f(\vartheta,Z_k(\vartheta))\|\leq 2C_f|\vartheta|$, which is the bound the argument for $P/|\vartheta|$ needs. On the annulus $1\leq s\leq 3$ a bounded $\partial_\vartheta f$ suffices. They give an invariant section $\zeta=Z(\vartheta)$ over $s\leq 3$ (the annulus of Step 4 now being $1\leq s\leq 3$, with the rate bound 8ε in place of 5ε), C^1, with $|Z|\leq C\sqrt{\varepsilon}s$ and $\|DZ\|\leq C\sqrt{\varepsilon}|\vartheta|$. Undoing Steps 3 and 1 gives Σ, with $Y_<=\varepsilon\, y(z/\sqrt{\varepsilon})$ satisfying the estimates of (i) and $Y_>=\mathcal{H}(z,Y_<(z))=O(\varepsilon)$.

Step 5: $Y_<$ is a branched double covering. In scaled form $Y_<=\varepsilon(F_2+R)$ with $|R|\leq C\sqrt{\varepsilon}s$, $\|DR\|\leq C\sqrt{\varepsilon}|\vartheta|$. Since F_2 is homogeneous of degree two with sign-definite Jacobian, on each circle $|\vartheta|=r$ one has $|\partial_\theta F_2|\geq mr^2$ and $\partial_r|F_2|=2|F_2|/r\geq mr$, while the derivatives of R are $O(\sqrt{\varepsilon}r^2)$ and $O(\sqrt{\varepsilon}r)$. So for small ε, $\arg Y_<(re^{i\theta})$ is strictly monotone in θ with total increase $2\pi\sigma$, and $|Y_<|$ is strictly increasing along each curve on which $\arg Y_<$ is constant. Hence each $x_<\neq 0$ in the disc of radius $r_\varepsilon=\min_{|z|^2=3\varepsilon}|Y_<(z)|$ has exactly two preimages, one on each of the two curves on which $\arg Y_<=\arg x_<$, and going once around $x_<=0$ exchanges them.

Step 6: the closed loop. For $(x,z)\in\mathcal{N}_\varepsilon$ put $e=x_>-Y_>(z)$. Along the closed loop, $DY_<(z)\dot z=\dot x_<=i\omega_0 x_<+Q(u,x)$, and on Σ $(e=0)$ this equals $DY_<(z)w(z)$; the difference is $O\big((|u|+|x|)|e|\big)=O(|z||e|)$, and $\|DY_<(z)^{-1}\|\leq C/|z|$, so $\dot z=w(z)+O(|e|)$. Similarly $\dot e=\big(A_L+O(|z|)\big)e$, so $|e(t)|\leq|e(0)|e^{-at/2}$ for small ε. On the boundary $|e|=K_0|z|^2$, using $\frac{d}{dt}|z|^2\geq -2\varepsilon|z|^2-C|z||e|$,

$$\frac{d}{dt}\big(|e|-K_0|z|^2\big)\leq\big(-\tfrac{a}{2}+2\varepsilon+CK_0|z|\big)K_0|z|^2<0,$$

so the bound $|e|\leq K_0|z|^2$ persists, and z never reaches 0 from $z\neq 0$. The perturbation changes $|z|^2$ by at most $\int C|z||e|\,dt\leq C'\varepsilon^{3/2}$ beyond what the reference allows, so $|z|^2\leq 2\varepsilon$ initially gives $|z|^2\leq 3\varepsilon$ for all $t\geq 0$ when ε is small. The perturbation of $\dot z=w(z)$ decays exponentially, and w has the hyperbolic attracting cycle $|z|^2=\varepsilon$, whose basin is the punctured disc; so $z(t)$ tends to the cycle and $e(t)\to 0$, that is $x(t)\to\gamma$. In the coordinates (z,e) the Floquet multipliers of γ are 1, $e^{-4\pi\varepsilon/\omega}$ and those of the contracting e-equation, so γ is hyperbolic and attracting. By Step 5 and σ, $x_<$ winds σ times around 0 along γ; and $|Y_<-F_2|\leq C\varepsilon\cdot\sqrt{\varepsilon}s$ gives the last estimate. □

C.7. **A worked example: the plane system in closed form.** For the plane system (34) ($x_>$ absent, $T=0$), (50) reads $Q(p)=g(x)$, $g(x):=[2(\varepsilon-|x|)+i\delta]x$, and it is solved in closed form. With $\eta:=p^2$ one has $Q(p)=G\eta+H|\eta|+K\bar\eta$. Put $\Delta:=|G|^2-|K|^2$ and $\mathcal{A}:=G\bar H-H\bar K$; (H3) says $|\Delta|>|\mathcal{A}|$. Together with its conjugate, $G\eta+H\rho+K\bar\eta=g$ is a real-linear system in

$\eta, \bar{\eta}$ once $\rho := |\eta|$ is held fixed, with solution

$$\eta = P - \rho\,\frac{\bar{\mathcal{A}}}{\Delta}, \qquad P := \frac{\bar{G}g - K\bar{g}}{\Delta}.$$

Taking the modulus of both sides gives one real quadratic equation for ρ alone:

$$\rho^2\,\frac{\Delta^2 - |\mathcal{A}|^2}{\Delta^2} + 2\rho\,\mathrm{Re}\Big(\frac{P\mathcal{A}}{\Delta}\Big) - |P|^2 = 0. \tag{67}$$

Since $|\Delta| > |\mathcal{A}|$, the leading coefficient is positive and the constant term is ≤ 0, so exactly one root is nonnegative (both vanish exactly at $x = 0$). Taking that root $\rho(x) \ge 0$ gives $\eta(x)$, and the exact branching feedback is

$$u(x, j) = (-1)^j \sqrt{\eta(x)}, \qquad j \in \{0, 1\}, \tag{68}$$

which closes the loop to $\dot{x} = V(x) = [2(\varepsilon - |x|) + 2i\omega]x$ exactly: the circle $|x| = \varepsilon$ is an attracting cycle traversed at angular velocity 2ω, and the origin is repelling.

For $Q(u) = Gu^2$ ($H = K = 0$) everything is explicit:

$$u(x, j) = (-1)^j \sqrt{\big[2(\varepsilon - |x|) + i(2\omega - \omega_0)\big]x/G}.$$

At fixed $\varepsilon > 0$ its leading part near the origin is $(-1)^j\sqrt{[2\varepsilon + i(2\omega - \omega_0)]x/G}$, which closes the loop to $\dot{x} = (2\varepsilon + 2i\omega)x$, an unstable focus. At $\varepsilon = 0$ this leading part reduces to $(-1)^j\sqrt{i(2\omega - \omega_0)x/G}$ and closes the loop to $\dot{x} = 2i\omega x$, a centre with no attracting cycle; the term $-2|x|$ supplies the attraction to the circle $|x| = \varepsilon$.

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